

Creating the Bottleneck: Vertical Integration in Platform Markets*

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Abstract

We study vertical integration between a platform (an operating system and its application store) and one of two competing device manufacturers, when applications port costlessly across systems. In our benchmark, separation leaves the platform with no upstream market power, because each manufacturer can run an equally good system of its own. Integration creates market power over the developer's access to the buyers of the integrated device, and none over the rival manufacturer. How this bottleneck is priced depends on the strength and the asymmetry of network effects: harm to the rival moves with the developer fee and is not sufficient evidence of consumer harm, and the merger can benefit every party absent any efficiency gain. Richer contracts let separation copy the merger's tax on the developer, but only inside part of the region where the merged firm taxes, and never where integration is pro-competitive. Merger-specific efficiency gains, porting costs and pre-existing dominance all give the platform's system an edge, and what integration should be credited with then depends on whether the merger created that edge or merely prices one that was already there.

KEYWORDS: Vertical integration; two-sided markets; indirect network effects.

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1 Introduction

MOTIVATION. Connected televisions are sold by manufacturers that do not write the software they run. Google, Amazon and Roku supply an operating system together with the application store attached to it; Samsung, LG and Hisense either license one of those systems or run one of their own. The system monetizes the buyer of the set it equips, chiefly through advertising on the home screen, and the platform hands part of that revenue back to the manufacturer carrying it. The main streaming applications are available on every one of these systems. Over the last decade several of the platforms started selling televisions themselves while continuing to license their systems to rival manufacturers, Amazon in 2021 and Roku in 2023; the smartphone industry had seen the same structural change earlier, by acquisition, with Google/Motorola and Microsoft/Nokia. Competition authorities have taken notice: the European Commission cleared Google/Fitbit only against ten-year commitments preserving rivals’ access to the Android ecosystem, and whether the standard assessment of vertical mergers applies to platform markets remains an open question.¹

THE MODEL AND THE MECHANISM. Two manufacturers sell competing devices. A platform supplies an operating system and runs the application store attached to it; a developer pays a fee per application published on the store; buyers value the applications running on their device, and the developer values the buyers its applications can reach. Each manufacturer also has an outside option, an in-house system or an open-source fork that is functionally equivalent but carries no store, and applications port costlessly across systems. Revenue sharing follows industry practice: the platform shares with an adopting manufacturer a fraction $\beta \leq 1$ of the per-user revenue its system generates, as licensors do; it also posts a fee per published application, which is not made contingent on which manufacturers adopt. The operating-system layer is thus *contestable* by construction, which is the polar case we want: under separation, self-supply leaves the platform no upstream market power, so whatever market power appears after the merger has been created by it rather than assumed into the benchmark.

Integration creates market power in exactly one place. It creates none over the rival manufacturer, which remains protected by self-supply: there is no ‘raise the rival’s cost’ channel in the baseline. It creates market power over the *developer*, since the merged firm owns the only access to the buyers of its own device and prices that access. The developer fee then trades off a one-sided logic, extracting surplus from the developer, against a two-sided one, subsidizing one side of the market to feed participation on the other. When buyers’ valuation for applications dominates, the integrated firm subsidizes publication and recoups on its device price; in the opposite, developer-skewed configuration, it taxes publication and sells a cheaper device. A single frontier in the space of network effects separates the two regimes, and the sign of the developer fee organizes the welfare analysis.

WHAT THE MERGER DOES. Three findings organize the welfare analysis, and Figure 1 in Section 3 draws them. First, harm to the rival moves with the fee rather than with foreclosure as usually understood: the rival loses exactly when the fee rises, and it gains when the integrated firm subsidizes the application commons. Second, harm to the rival is not sufficient evidence of harm to buyers. Where network effects are strong and developer-skewed, the integrated firm taxes the developer and passes the proceeds into cheaper devices, so that buyers gain while the rival loses; how often the two come

¹See OECD [2018] and Yun [2020].

apart depends on the intensity of product-market competition (Section 3.3). Third, where network effects are sufficiently buyer-skewed the merger subsidizes publication and improves on separation for every party at once, with no efficiency gain in sight, while where they are weak and balanced the fee is extraction and nothing else, and every other party loses.

WHY A MERGER, RATHER THAN A CONTRACT? Any theory of vertical integration owes an answer to the Coasian question: whatever the merged firm implements, why could contracts not implement it without a merger? We solve the game in which the cap on revenue sharing is lifted, so that the platform may pay a manufacturer more per user than the revenue that user generates. Strictly inside the developer-skewed region, separation can then tax the developer, by hosting a single manufacturer on a tax-and-rebate package that leaves it exactly indifferent while the outsider, offered nothing, is not consulted, a contracting externality in the sense of Segal [1999]. That package copies the merger, exactly so on one curve of the parameter plane. Taxing under separation always requires paying for adoption, and it can take more than one form in the space we study, but subsidizing never can: a subsidy to the developer is collected whether or not anyone adopts, so no rebate recoups it. The argument rests on developer terms that are posted and not contingent on the manufacturers' adoption, the practice of application stores (Section 4); a fee schedule written on adoption would reopen the financing of subsidies. Where integration is pro-competitive, no contract in this space could have done the same.

SCOPE. Section 5 relaxes the assumptions that carry the analysis: the platform's system may hold an advantage over self-supply, whether created by the merger or pre-existing, buyers may enjoy a direct network effect among themselves, porting an application may be costly, devices may be complements, there may be more than two of them, the rival may own a system of its own, and the store may charge a commission rather than a fee per application. The case that matters most for policy is a pre-existing advantage, since the mergers that authorities examine are by platforms that are already dominant; there the pre-merger benchmark is itself distorted, and we ask how much of what we attribute to integration should be attributed to the contestability of the platform layer instead.

RELATED LITERATURE. Our framework belongs to the strategic approach to vertical integration initiated by Ordover et al. [1990] and developed by Chen [2001]: upstream suppliers compete to serve downstream rivals, and the question is whether integration creates market power and forecloses [see Rey and Tirole, 2007, Riordan, 2008, for surveys]. These models have no room for the indirect network effects that link device buyers to application developers, and adding them is not a marginal amendment: with a contestable upstream layer the merger creates no market power over rivals at all. Closest in spirit, Carlton and Waldman [2002] and Farrell and Katz [2000] study how a firm creates market power in a complementary market or in systems markets; there the complementarity is a technological one between components, whereas ours runs through a third party whose output is a public good across systems, which is what makes self-supply a discipline in the first place. From the literature on two-sided markets [Caillaud and Jullien, 2003, Rochet and Tirole, 2006, Armstrong, 2006, Weyl, 2010, Jullien et al., 2021], which has not addressed vertical mergers, we borrow the insight that the structure of prices across sides, not their level, carries the welfare weight, and we push it into merger analysis. The papers on content exclusivity [Doganoglu and Wright, 2010, Hagiu and Lee, 2011, Lee, 2013] and on integration with one side of the market [Church and Gandal, 2000, Hagiu and Spulber, 2013] share our concern but not our question, which is the merger between

the platform and one of the manufacturers whose devices connect the two sides.

A recent literature studies platforms that already hold market power over their sellers or developers: hybrid modes and self-preferencing [Hagiu et al., 2022, Etro, 2021b, Anderson and Bedre-Defolie, 2024, Madsen and Vellodi, 2025], gatekeeping and its regulation [Bisceglia and Tirole, 2023, Kang and Muir, 2022], commissions and seller entry [Etro, 2023], commission setting under platform competition [Jeon and Rey, 2024], spillovers across competing bottlenecks [Teh and Wright, 2026], and the screening of digital mergers under network effects [Johnen and Shekhar, 2026]. Those papers start where our benchmark ends: their platforms hold market power over developers with or without vertical integration, whereas ours has to create it, and our comparative statics in the platform’s pre-existing advantage (Section 5.2) are the bridge between the two settings. Closest to us, Padilla et al. [2022] study integration by an upstream platform monopoly under network externalities and also find that network effects can dampen foreclosure, from the opposite premise: their bottleneck is assumed and its contract space is rich, so it is the richness of contracting that delivers neutrality. Their neutrality is a statement about public two-part tariffs, under which integration leaves the aggregate outcome unchanged; under linear contracts they find that integration raises consumer surplus and welfare. Our platform, by contrast, faces self-supply, and it is the *poverty* of the contract space, one instrument capped by industry practice, that gives integration its bite. Section 4 is where the two meet: remove our cap and neutrality returns on one curve of the parameter plane, though not on a band along the taxing frontier, where integration is strictly more powerful than any contract we study (see also Bisceglia et al., 2022, on integration and innovation with competing ecosystems). Finally, the contract-space analysis connects to the literature on contracting with externalities [Segal, 1999, Jullien, 2011], and the run-away of two-host packages beyond the admissible set echoes the non-existence result of Schutz [2024] for two-part tariffs in vertically related markets.

ORGANIZATION. Section 2 presents the model, Section 3 the benchmark, the integrated firm’s fee and the welfare map, Section 4 the choice between integration and separation once the cap on revenue sharing is lifted, and Section 5 the scope of the analysis; Section 6 concludes. Proofs are in the Appendix, computations and numerical verifications in an Online Appendix. We work with the linear specification throughout, the level of generality at which the effects of integration can be signed; Appendix A.4 states which building blocks do not depend on it.

2 The Model

We consider a two-sided market in which a *platform* supplies an operating system to two competing device manufacturers and runs the application store attached to it; developers pay the platform to publish their applications; buyers purchase devices. Indirect network effects link the two sides: buyers value the number of applications running on their device, and the developer values the number of buyers its applications can reach.

2.1 Environment

PLATFORM, MANUFACTURERS, AND THE OUTSIDE OPTION. There are two symmetric manufacturers, M_1 and M_2 , and one platform, denoted by I . The platform owns an operating system and its application store. If manufacturer M_k equips its devices with

the platform’s operating system, each buyer of M_k ’s device generates a per-user benefit $r > 0$, the monetization of user-generated data and ancillary services through advertising, and the contract between the platform and M_k specifies how this benefit is shared: M_k keeps $\beta^k r$ per user and the platform keeps $(1 - \beta^k)r$, with $\beta^k \in [0, 1]$.² The platform also charges the developer a fee a per application published on its store.³ The fee can be negative, in which case the platform subsidizes publication.

Each manufacturer also has an *outside option*: it can costlessly equip its devices with a functionally equivalent operating system that it sources in-house or from an open-source project. The outside option provides the operating-system functionality but *no store*: no fee can be charged to, and no subsidy paid to, the developer for publishing on it, and a manufacturer using it keeps the entire per-user benefit r .⁴ The outside option is thus summarized by the permanent terms $(\beta = 1, a = 0)$. It is what disciplines the platform: as in the strategic approach to vertical integration [Ordover et al., 1990, Chen, 2001], upstream market power is competed away in the benchmark, and any market power the platform enjoys will have to be *created*, by integration or by the contracts we rule out in Section 4.

The restriction $\beta^k \leq 1$ states that the platform cannot pay a manufacturer more, per user, than the revenue this user generates: revenue sharing is allowed, pure adoption subsidies are not.⁵ We impose it throughout, and we characterize exactly when and how it bites in Section 4: the constraint turns out to be slack in the region of network effects where integration will be found to be pro-competitive, and to rule out, elsewhere, a contractual replication of integration that rests on a contracting externality between manufacturers.

PAYOFFS. Let n_B^k denote the number of buyers of M_k ’s device and q_S the number of applications developed. If M_k adopts the platform’s operating system it earns $(p_k + \beta^k r) n_B^k$, where p_k is the price of its device (marginal production costs are normalized to

²Revenue-sharing agreements of this kind are pervasive in platform industries. In the *Google Android* case the European Commission documented portfolio-based agreements under which Google shared search advertising revenue with device makers and mobile network operators (Commission Decision C(2018) 4761 final of 18 July 2018); Roku and Amazon share home-screen advertising revenue with the television manufacturers that license their systems. The per-user benefit does not depend on the stock of applications, so that the platform’s profit is additively separable across the two sides. If monetization instead rose with the number of applications, an additional motive to subsidize publication would enter the integrated firm’s first-order condition (7); we have not characterized the resulting map.

³Stores charge developers account fees, fees per published application and commissions on transactions. We work with a fee per application, which is the membership-based instrument of the two-sided literature; Section 5.7 discusses commissions.

⁴Device makers routinely maintain such alternatives: forks of the Android Open Source Project (Amazon’s FireOS, Huawei’s HarmonyOS), in-house systems in the smart-TV industry (Samsung’s Tizen, LG’s webOS, Hisense’s VIDAA). That these alternatives are as good as the platform’s system, and that their storefronts do not monetize developers, are the model’s assumptions rather than documented facts; Section 5.2 relaxes the first and Section 5.6 the second. Formally, the outside option is equivalent to a non-strategic competitive fringe of interchangeable upstream platforms: both deliver, off the equilibrium path, an operating system on the terms $(\beta = 1, a = 0)$.

⁵Arrangements in which the net flow runs the other way are the exception. The clearest instance is the Microsoft-Nokia partnership of 2011, under which Nokia disclosed quarterly platform support payments from Microsoft of \$250 million alongside the software royalties it owed in return, the former expected over the life of the agreement to exceed the latter. We read the episode as suggestive rather than probative: the published figures compare two contractual flows, not a payment against the revenue a manufacturer’s users generate, which is what $\beta \leq 1$ bounds.

zero); if it uses its outside option it earns $(p_k + r) n_B^k$. Writing $n_B^I \equiv \sum_k \mathbb{1}_{\{M_k \text{ adopts } I\}} n_B^k$ for the platform's installed base, the platform's profit is

$$\Pi = \sum_{k=1,2} \mathbb{1}_{\{M_k \text{ adopts } I\}} (1 - \beta^k) r n_B^k + a q_S \mathbb{1}_{\{\text{the developer publishes on } I\}}.$$

BUYERS. The demand for M_k 's device depends on both prices and on the numbers of applications available on each device, n_S^k and n_S^ℓ :

$$Q_B^k(p_k, p_\ell, n_S^k, n_S^\ell) = \left(v - p_k - \gamma \left(p_k - \frac{p_k + p_\ell}{2} \right) \right) + \left(u_B n_S^k + \frac{\gamma}{2} u_B (n_S^k - n_S^\ell) \right), \quad (1)$$

with $v > 0$, $\gamma \geq 0$, and $u_B > 0$. The first bracket is a standard linear demand for imperfectly substitutable products, γ measuring the intensity of product-market competition; the second describes how applications shift demand, u_B measuring how much buyers value them. We take (1) as a primitive.⁶

THE DEVELOPER. A representative developer creates q_S applications at cost $C_S(q_S) = q_S^2/2$ and ports them costlessly across operating systems.⁷ Each application earns its developer $u_S \geq 0$ per buyer reached. Publication is decided system by system: applications are published on the outside operating systems in any case, there being nothing to pay there, and on the platform's store if and only if the per-application gain from reaching the platform's installed base covers the fee, $u_S n_B^I \geq a$, a condition we impose throughout: a fee above $u_S n_B^I$ collects nothing and strips the devices on the platform's system of every application, and we restrict the platform to fees at which the developer publishes on its store, in every adoption configuration.⁸ The number of applications available on a device then equals q_S regardless of its operating system, $n_S^1 = n_S^2 = q_S$, and the developer's investment solves $\max_{q_S} q_S [u_S(n_B^1 + n_B^2) - a \mathbb{1}_{\{u_S n_B^I \geq a\}}] - q_S^2/2$, so that

$$q_S = u_S (n_B^1 + n_B^2) - a \mathbb{1}_{\{u_S n_B^I \geq a\}}. \quad (2)$$

The indicator is the publication rule itself, and it handles both directions of the fee: a tax enters only where publication survives it, while a subsidy ($a < 0$) is collected on the platform's store regardless of adoption. Two features of (2) drive much of the analysis. First, applications are a *public good across operating systems*: because porting is costless, any application called into existence by one side of the market is enjoyed on every device. Second, the fee a operates on the *total* stock of applications, so that taxing access to the platform's buyers destroys applications on the rival's device too. Both features are relaxed in Section 5.3, where porting is costly, and the unconditionality of the subsidy drives the free-riding argument below.

⁶It is the aggregate of a population of single-homing buyers, with a captive segment and a contested one for each manufacturer, on the range where both segments are interior; that construction shows the system is not ad hoc. Online Appendix B.10 gives it, and gives the calibration that keeps both segments interior at the prices of the model.

⁷The representative developer is equivalent to a continuum of atomistic developers, each supplying one application at a development cost distributed with unit density: an application is created whenever its cost lies below the per-application revenue, which yields (2) exactly, with the fee operating on the extensive margin of entry, and the same aggregate surplus.

⁸The admissible sets of Sections 3 to 5 include the condition at the integrated optimum; it holds with slack at every point of the reference grid there, and at the contractual optima of Section 4 and the benchmarks of Section 5 where they are computed (Online Appendix B.1).

TIMING. The game has four stages. (1) The platform publicly posts (β^1, β^2, a) .⁹ (2) Manufacturers simultaneously choose between the platform's operating system and their outside option. (3) Manufacturers simultaneously set device prices. (4) Buyers purchase and the developer invests and publishes. All choices are publicly observed. We study subgame-perfect equilibria, with two standard conventions and one selection rule spelled out in Section 2.3.

2.2 Demands, Network Effects and Rivalry

Consider stage 4, in a configuration where all applications are published on all systems in use. Given prices (p_1, p_2) and a per-application fee a , buyer participation and developer investment are simultaneous, and the numbers (n_B^1, n_B^2, q_S) solve the fixed point formed by (1) (with $n_S^1 = n_S^2 = q_S$) and (2). Provided

$$2 u_B u_S < 1, \quad (3)$$

this system has a unique solution *within a publication regime*, obtained by substitution (Appendix A.1); we use the affine system it yields in every configuration, on and off the equilibrium path, and verify interiority where it is used:

$$D_k(p_k, p_\ell, a) = \frac{2v - (2 + \sigma) p_k + \sigma p_\ell - 2u_B a}{2(1 - 2u_B u_S)}, \quad (4)$$

$$D_S(p_1, p_2, a) = \frac{u_S(2v - p_1 - p_2) - a}{1 - 2u_B u_S}, \quad (5)$$

where

$$\sigma \equiv \gamma - 2u_B u_S(1 + \gamma). \quad (6)$$

Condition (3) bounds the feedback loop between the two sides (more buyers attract more applications, which attract more buyers), and $1/(1 - 2u_B u_S)$ is the resulting *network multiplier*: any shock to participation on either side is amplified by this factor.

Network effects do not only shift demands; they change the nature of competition between manufacturers. From (4), $\partial D_k / \partial p_\ell = \sigma / [2(1 - 2u_B u_S)]$, so devices are demand substitutes or demand complements according to the sign of σ . The decomposition behind (6) is general and holds for any demand system: a higher p_ℓ steals business from the rival, which is the standard effect, but it also shrinks total participation, hence the stock of applications, hence the demand for *both* devices. When network effects are strong relative to product-market competition ($2u_B u_S(1 + \gamma) > \gamma$), this second channel dominates and devices become complements. As in the bulk of the literature on strategic vertical integration, we focus on the substitutes case; Section 5.4 studies what changes under complements.

Assumption 1. *Network effects are not too strong relative to product-market competition: $\sigma \geq 0$.*

Under Assumption 1 and condition (3), the pricing subgame of stage 3 is well behaved: profits are strictly concave in own prices, best responses are increasing in the rival's price with slope less than one, and stage 3 admits a unique Nash equilibrium in which prices decrease in the developer fee and in own perceived per-user benefit (Appendix A.1).

⁹Contracts are publicly posted, as in most of this literature. We therefore leave aside secret contracting and the opportunism problem it raises [Hart and Tirole, 1990].

2.3 Equilibrium

We look for subgame-perfect equilibria and adopt two tie-breaking conventions: the developer publishes when indifferent, and a manufacturer indifferent between the platform's contract and its outside option accepts the contract. The adoption subgame of stage 2 may admit several Nash equilibria, sustained by the manufacturers' beliefs about each other's choice rather than by the platform's terms. As is standard in the platform literature, we do not let the platform profit from such miscoordination:

SELECTION RULE. *If the adoption subgame admits several Nash equilibria, the one maximizing the manufacturers' joint profit is played, ties going to adoption.*

This is the analogue, in our setting, of the coordination refinements routinely used in two-sided markets [Caillaud and Jullien, 2003, Jullien, 2011]: it removes market power that would rest on a pure failure of the manufacturers to coordinate their adoption choices, and nothing else. The separation benchmark of the next section uses it, and not only the presence of an outside option.

3 Separation and Integration

3.1 Separation: the Platform Has No Market Power

Consider first the situation of *separation*, in which the platform is independent of both manufacturers. The outside option and the public-good nature of applications leave the platform empty-handed:

Proposition 1 (Separation benchmark). *Under separation, every subgame-perfect equilibrium of the game of Section 2, selection rule included, yields the same outcome: the platform makes no profit, no fee is collected from the developer, and market outcomes are as if both manufacturers operated on the terms of the outside option, ($\beta = 1, a = 0$).*

Proof. See Appendix A.2. □

The economics of Proposition 1 is a free-riding argument that recurs throughout the paper. The platform cannot *tax*: if it posts $a > 0$, each manufacturer individually may tolerate the fee, but abandoning the platform jointly restores $a = 0$ at no cost, and the selection rule picks that equilibrium. The platform cannot *subsidize* either: a posted subsidy $a < 0$ is paid to the developer whether or not any manufacturer adopts the platform, so a manufacturer can decline whatever share $\beta < 1$ was meant to finance the subsidy and still enjoy the subsidized applications on its outside operating system. Applications being a public good across operating systems, generosity cannot be recouped and toughness cannot be enforced. The zero-profit benchmark should be read as a limit: with any positive cost of operating the store, the separated platform would exit, and the benchmark captures a contestable operating-system layer in its polar form.

3.2 Integration Creates Market Power over the Developer, and over No One Else

Suppose now that the platform and M_1 integrate. The merged entity chooses the price of device 1 and the developer fee.

NO MARKET POWER OVER THE RIVAL. The outside option protects M_2 exactly as it protected both manufacturers before the merger: to attract M_2 onto its operating system, the integrated platform must offer $\beta^2 \geq 1$, and it is indifferent about doing so.¹⁰ Absent efficiency gains, integration creates no market power over the non-integrated manufacturer: there is no ‘raise the rival’s cost’ channel here, by construction, since the rival’s outside option is self-supply at no loss of applications, so that the integrated firm’s withdrawal from the licensing market leaves the rival’s terms untouched. This is the polar case of the mechanism in [Ordover et al. \[1990\]](#) and [Chen \[2001\]](#), where the remaining independent supplier has market power; the channel returns as soon as the outside option is degraded (Section 5).

MARKET POWER OVER THE DEVELOPER. What integration does create is a bottleneck. The developer can reach the buyers of device 1 only through the integrated platform’s store: those buyers are now captive, and the fee a prices the access to them; the merger thus creates a competitive bottleneck in the sense of [Armstrong and Wright \[2007\]](#). Under separation this bottleneck existed potentially, since each manufacturer’s buyers are reachable only through the store of the system it uses, but competition kept it unpriced. Integration is what makes it exploitable, because the integrated firm internalizes M_1 ’s profit and no longer needs to be compensated through β for the harm the fee causes on the device side.

At stage 3, the integrated firm chooses p_1 to maximize $(p_1 + r) D_1 + a D_S$, internalizing the effect of its device price on its store revenues, while M_2 maximizes $(p_2 + r) D_2(p_2, p_1, a)$. The pricing subgame has a unique equilibrium $(p_1^I(a), p_2^I(a))$, with $dp_k^I/da \leq 0$ ([Appendix A.1](#)): a higher fee depresses the stock of applications, hence device demands, hence prices. At stage 1, the fee solves $\max_a \Pi^I(a)$, where $\Pi^I(a) \equiv (p_1^I + r) D_1 + a D_S$, with first-order condition

$$\underbrace{D_S + a \left(\frac{\partial D_S}{\partial a} + \frac{\partial D_S}{\partial p_2} \frac{dp_2^I}{da} \right)}_{\text{rent extraction from the developer}} + \underbrace{(p_1^I + r) \left(\frac{\partial D_1}{\partial a} + \frac{\partial D_1}{\partial p_2} \frac{dp_2^I}{da} \right)}_{\text{depreciation of the device business}} = 0. \quad (7)$$

The first bracket collects what a higher fee extracts from the developer; the second, that fewer applications depress demand for the integrated firm’s own device, and it is negative at $a = 0$. Which force dominates there is a race between the two network-effect parameters: extraction is powerful when the developer values buyers (u_S high), costly when buyers value applications (u_B high). In the linear model, the comparison has an explicit answer. Some restrictions on parameters are needed first, implied neither by [Assumption 1](#) nor by the feedback condition (3): the integrated firm’s problem must be concave, the rival must still serve a positive demand at the integrated optimum, and the developer must still publish on the platform’s store there. We call *admissible* the set $\mathcal{A}$ of parameters that meet them, defined in [Appendix A.1](#), and state the propositions of this section on it. [Sections 4 and 5](#), where the platform chooses a share alongside the fee, work on a slightly smaller set $\mathcal{A}^*$, on which that two-instrument problem is concave as well.

¹⁰With $\beta^2 = 1$ the platform earns nothing from licensing, and M_2 ’s profit, demand and pricing are identical whether it adopts the platform’s system or its outside option: only the developer fee affects M_2 , and it does so through the total stock of applications regardless of M_2 ’s own choice of operating system. Without loss of generality we let M_2 use its outside option. The merged firm therefore does not necessarily win the rival’s market, and does not necessarily want to.

Proposition 2 (The integrated platform’s fee). *On $\mathcal{A}$, the integrated firm leaves $\beta^2 = 1$ to the rival, and its fee a^{**} is the unique solution of (7). There is a function $\underline{h}$, with $\underline{h}(u_S) > u_S$ for all $u_S > 0$, such that $a^{**} > 0$ if and only if $u_B < \underline{h}(u_S)$: the integrated platform taxes the developer when network effects are balanced or developer-skewed, and subsidizes it when they are sufficiently buyer-skewed.*

Proof. See Appendix A.2. □

In Appendix A.2 the frontier takes the form $u_B/u_S < R_a(\sigma)$, with R_a given in (A.10): a condition on the ratio of the two network effects, compared with a threshold that itself depends on their strength and on product-market competition through σ . Arbitrarily weak effects still call for a subsidy when they are skewed enough toward buyers, and R_a increases in σ from a value of 2 at $\sigma = 0$, so that subsidizing always requires buyers to value an extra application at least twice as much as the developer earns from an extra buyer. The comparative statics run through σ : the threshold rises with effective downstream rivalry, so that as the network channel becomes stronger relative to product-market competition, σ falls and a smaller buyer-side asymmetry suffices for the platform to subsidize. This is a statement about the sign of the fee, not about the size of the subsidy. Integration is strictly profitable whenever $a^{**} \neq 0$: the merged entity can always replicate separation by setting $a = 0$, so $\Pi^I(a^{**}) > \Pi^I(0)$, which equals the joint profit of the platform and M_1 under separation.

3.3 The Impact of Integration

Since the integrated outcome coincides with separation when $a = 0$, the competitive assessment of the merger reduces to signing the effect of moving the fee from 0 to a^{**} on each constituency.

THE NON-INTEGRATED MANUFACTURER. Differentiating $\pi_2(a) = (p_2^I + r) D_2$ and using the envelope theorem,

$$\frac{d\pi_2}{da} = (p_2^I + r) \left(\frac{\partial D_2}{\partial a} + \frac{\partial D_2}{\partial p_1} \frac{dp_1^I}{da} \right) < 0 : \quad (8)$$

at given device prices a higher fee reduces the supply of applications on *every* device ((2)), which is the first term, and the integrated firm’s price cut then takes buyers from the rival, which is the second. Both lower the rival’s demand, whatever the equilibrium stock of applications turns out to be.

Proposition 3 (Harm to the rival). *On $\mathcal{A}$, the merger reduces the non-integrated manufacturer’s profit if and only if the developer fee rises above its pre-merger level: harm occurs exactly when $u_B < \underline{h}(u_S)$, and M_2 benefits from the merger when network effects are sufficiently buyer-skewed.*

Two features distinguish this effect from foreclosure as usually understood. It is not the outcome of a ‘raise the rival’s cost’ strategy: the integrated firm prices the developer side for its own sake, and the impact on M_2 is collateral, transiting through a public good. And it is not exclusionary in the input-market sense: whatever the fee does to the stock of applications it does to both devices alike, and M_2 ’s access to applications is never degraded relative to M_1 ’s. We accordingly speak of *harm to the non-integrated*

manufacturer rather than of foreclosure. Beyond the linear model, (8) and Proposition 3 require only the pass-through property $dp_k^I/da \leq 0$ together with $\partial D_2/\partial p_1 \geq 0$.

BUYERS AND THE DEVELOPER. The surpluses of the two sides are intertwined by the network multiplier. Taxing the developer lowers both device prices, which benefits buyers and, by drawing in more of them, replenishes the developer's incentive to create applications. The direct effect of the fee on the application stock is thus offset in equilibrium: partially where the induced price cuts fall short of the fee's direct bite, and *more than fully* in the strongly developer-skewed range, where they outweigh it and the stock of applications rises. Subsidizing has the mirror effects. Buyers therefore face a conflict between the price of their device and the availability of applications, and the developer faces one between the fee it pays and the size of the market it reaches. In the linear model, both are settled by explicit frontiers.

Proposition 4 (Buyers and developer). *On $\mathcal{A}$, there are two frontiers, $\bar{h}_B$ and $\bar{h}_S$, such that the merger (i) raises buyer surplus if $u_B > \underline{h}(u_S)$ or $u_S \geq \bar{h}_B(u_B)$;¹¹ (ii) raises the developer's surplus if and only if $u_B > \underline{h}(u_S)$ or $u_S > \bar{h}_S(u_B)$. On the frontier $u_B = \underline{h}(u_S)$ itself the fee is zero and the merger reproduces separation, so no surplus moves.*

Proof. See Appendix A.2, where $\bar{h}_B$ and $\bar{h}_S$ are given in (A.12). $\square$

Proposition 5 (Total welfare). *On $\mathcal{A}$: (i) Whenever the integrated platform subsidizes the developer ($a^{**} < 0$), the merger raises total welfare: the merged entity's profit, the rival's profit, buyer surplus and developer surplus all rise. (ii) Whenever it taxes the developer ($a^{**} > 0$), the merger reduces total welfare in a neighborhood of the taxing frontier $\underline{h}$, for every γ .*

Proof. See Appendix A.2. $\square$

READING THE MAP. Figure 1 assembles Propositions 2–4, and three messages emerge. First, when network effects are sufficiently *buyer-skewed* (green region), integration is Pareto-improving without any efficiency gains: the integrated firm subsidizes applications, recoups the subsidy on its device business, the rival free-rides on the richer application stock, and buyers pay more for devices that carry more applications. What competition could not deliver, because no one could recoup a subsidy under separation, the merger delivers, because it ties the store to a device business. Second, when network effects are strong and *developer-skewed* (orange region, and the upper part of the blue one), the integrated firm taxes the developer hard and passes the proceeds into cheaper devices: buyers gain while the rival and possibly the developer lose. Where exactly buyers stop gaining, just below $\bar{h}_B$ inside the red region, is the dividing curve of Proposition 4(i) rather than the curve $\bar{h}_B$ drawn in the figure. Harm to the rival is thus *not sufficient* evidence of harm to buyers, though it is necessary: by Propositions 3 and 4, a rival that gains from the merger faces a subsidizing platform, and buyers then gain too. How often the two harms come apart is governed by product-market competition: on the reference grid, the configurations in which the rival loses while buyers gain become rare as γ grows

¹¹These two conditions leave a region open, and Online Appendix B.2 settles it: there buyer surplus rises on the developer-skewed side of a single dividing curve, given in parametric form, its uniqueness following from endpoint signs that hold as polynomial identities. That curve runs close to $\bar{h}_B$, and the Online Appendix reports how little of the open region lies beyond it.

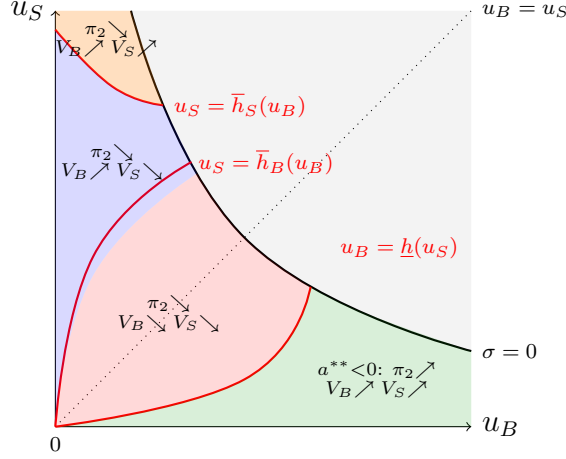


Figure 1: The regimes of Propositions 2 to 4 (drawn as a schematic; $\bar{h}_B$ is a sufficient condition, and the buyers' exact frontier, just below it, is not drawn). The grey area violates Assumption 1, and other restrictions of the admissible set are not represented. Online Appendix B.11 draws the computed maps on the admissible set, welfare included.

and are frequent when γ is small (Online Appendix B.1).¹² An antitrust screen that infers consumer harm from the rival's complaint is therefore unreliable exactly where product-market competition is soft. Third, it is only when network effects are weak and balanced (red region, around the origin) that the merger harms every other constituency at once: there little value can be created by re-balancing the price structure, and the fee is pure extraction. Proposition 5 signs total welfare at the two ends of the map, throughout the subsidizing region and just inside the taxing frontier; in between, welfare can go either way, and the map is computed (Online Appendix B.2).

4 Integration or Separation?

Two restrictions define our contract space: the platform cannot pay a manufacturer more per user than the revenue this user generates ($\beta^k \leq 1$), and the developer fee cannot be made contingent on manufacturers' adoption decisions.¹³ Proposition 1 rests on the first: a tax is refused jointly as long as no manufacturer is offered more than one. This section drops the cap and asks whether the platform can buy the acceptance of a tax with $\beta > 1$, and, where it can, whether it still wants the merger. It works on the set $\mathcal{A}^*$: the platform now chooses shares alongside the fee, and the concavity of its two-instrument program is the stronger requirement (Appendix A.1).

A subsidy cannot be part of the answer. It is paid to the developer whether or not

¹²Numerical statements in what follows are read on the *reference grid*, a step-0.02 grid in (u_B, u_S) over $[0.02, 1.40]^2$ screened on the admissible set, and at the *reference calibration* $v = 2$, $r = 1/2$ and $\gamma \in \{1/2, 1, 2, 4, 8, 16\}$, with $\gamma = 4$ where a single value is used. Departures from either are stated where they occur, and the Online Appendix names the script behind each figure.

¹³Developer fee schedules are posted and uniform: a store publishes one schedule for all developers, and does not renegotiate it with the developer side each time a manufacturer adopts or leaves its system. Per-download incentive programs, which would tie payments to the realized installed base, are usage-based instruments; they belong to the proportional-fee family discussed in Section 5.7.

anyone hosts, so every hosted manufacturer must be offered $\beta^k \geq 1$, on which the platform earns nothing while paying the subsidy; that family is empty on the whole admissible set. The asymmetry matters: a tax bites only if the platform hosts buyers, so its proceeds arrive tied to adoption and can be handed back through $\beta > 1$, whereas a subsidy has no such tie. Contractual replication of the merger, where it exists, is available in one direction only.

THE TAX-AND-REBATE PACKAGE. Suppose the platform hosts M_1 alone on a package (β, a) , M_2 remaining outside. Write $\pi_1(\beta, a)$ for the host's equilibrium profit in that configuration, and π^{bench} for what a manufacturer earns in the benchmark of Proposition 1, which is what the host obtains by walking away. The host's participation constraint binds at the optimum, and defines the *compensating share* $\bar{\beta}(a)$ by $\pi_1(\bar{\beta}(a), a) = \pi^{bench}$, with $\bar{\beta}(0) = 1$ and $\bar{\beta}'(0) > 0$ everywhere on the admissible set: the fee depresses the shared application stock, and the host must be compensated through its share for carrying the system that levies it. The rebate above one is therefore not assumed but required. The platform's profit along the constraint, $\Phi(a)$, is a strictly concave quadratic with $\Phi(0) = 0$, so a profitable package exists if and only if $\Phi'(0) > 0$, and the best one is the fee $\hat{a}$ that maximizes Φ , in closed form in Appendix A.3. Writing $\Lambda \equiv 8 + \sigma(8 + \sigma)$,

$$\text{sign } \Phi'(0) = \text{sign } [\Lambda u_S - (4 + 3\sigma) u_B], \quad (9)$$

so that the family is profitable below a frontier $h_\Phi(u_S)$, an object of the same kind as $\underline{h}$, whose ratio form is $u_B/u_S < R_\Phi(\sigma) \equiv \Lambda/(4 + 3\sigma)$. The deal it supports is a *tax-and-rebate* one: tax the developer, hand the proceeds to the single host. A package hosting both manufacturers can do better than this one, and does so only where network effects are strictly developer-skewed: its interior optimum calls for a tax only where $u_S > u_B$ (Lemma B.1 of the Online Appendix), where it wins is a grid finding, and where it wins it does so through two rebates so generous that each manufacturer accepts whatever the other does.

Proposition 6 (Unrestricted contracts). *On $\mathcal{A}^*$, consider the game of Section 2 with the restriction $\beta^k \leq 1$ removed.*

- (i) *If $u_B \geq h_\Phi(u_S)$, no single-host package, no capped package and no subsidy earns the platform anything, and on the reference grid no package hosting both manufacturers does either: the outcome is that of Proposition 1. This region contains the no-tax region $u_B \geq \underline{h}(u_S)$, since $h_\Phi \leq \underline{h}$ with equality only at $\sigma = 0$; on the band $h_\Phi(u_S) \leq u_B < \underline{h}(u_S)$ that separates them, integration would strictly profit from taxing the developer and no contract does.¹⁴*
- (ii) *If $u_B < h_\Phi(u_S)$, the platform earns a strictly positive profit, and the best single-host package is $\beta^1 = \bar{\beta}(\hat{a}) > 1$, $a = \hat{a} > 0$. It is the optimal contract wherever $u_B \geq u_S$, on the reference grid; where $u_S > u_B$ a package hosting both manufacturers earns more on the region where condition (A.23) of Appendix A.3 holds, a grid finding reported in Online Appendix B.7.*

Proof. See Appendix A.3. □

¹⁴The first statement is analytic (Appendix A.3). The second is a grid finding within bounds on the fee and the shares that Online Appendix B.7 certifies: beyond them no package hosting both manufacturers, selected by the rule or not, earns as much as the single-host package (`rep33_bothhost_grid.py`).

Part (ii) rests on a contracting externality in the sense of Segal [1999]. Both manufacturers walking away would restore the benchmark; but held exactly at its benchmark, the host accepts whatever the other does, and the outsider, offered nothing beyond its share of one, is not consulted. Joint refusal is simply not an equilibrium, and the selection rule has nothing left to choose from.

THE COPY AND ITS PRICE. On the region of part (ii), which we call the *unraveling region*, the platform has two ways of taxing the developer, and two questions follow: which one it prefers, and which one costs the market more. Both gains are in closed form, the separation gain $\Phi(\hat{a})$ and the integration gain $\Pi^I(a^{**}) - \Pi^I(0)$ of Section 3, and their comparison is a single polynomial inequality. Write

$$K(u_B, u_S) \equiv \Lambda u_S - (2 + \sigma)(4 + 3\sigma) u_B. \quad (10)$$

At any common positive fee, the single-host package and integration differ in the two device prices and in the application stock by one common factor, proportional to aK ; this is an identity. Where $K = 0$ the two arrangements coincide fee by fee: the compensating package reproduces the integrated outcome, $\hat{a} = a^{**}$, and the platform is exactly indifferent between them. Where $K > 0$ the compensating host prices above the integrated level at any common fee, the rival follows, and the application stock is smaller; where $K < 0$ the reverse. The two arrangements are evaluated at different fees away from that curve, and comparing them there is a computation, reported in Online Appendix B.7 as two propositions. The platform prefers integration to its best single-host package when network effects are strong, balanced or tilted toward buyers, and separation on a small region around the curve $K = 0$, essentially where they are weak in level and tilted toward the developer, by a thin margin. On the reference grid, and where both arrangements tax, the single-host package leaves total welfare, buyer surplus and developer surplus below their integrated values where $K > 0$ and above them where $K < 0$; the region on which the platform prefers separation lies mostly on the $K > 0$ side.

The band of Proposition 6(i) is a rivalry phenomenon: it closes at $\sigma = 0$ and widens as the two devices become closer substitutes. Online Appendix B.7 gives the mechanism.

WHAT THE CAP DOES, AND WHAT IT DOES NOT. On the unraveling region the zero-profit benchmark of Proposition 1 rests on the cap, and without it the separated platform taxes the developer. What the cap does not do is protect the merger's pro-competitive case: in the buyer-skewed region, part (i) shows that no restriction on the size of per-user rebates is needed, because the missing instrument there is a recoupable subsidy to the developer, missing for free-riding reasons that no transfer can fix. Integration thus matters where per-user rebates beyond generated revenue are unavailable *and* developer pricing cannot condition on the device market's structure; and where the first restriction is lifted, what separation gains is a copy of the merger, exact on one curve. Replication is replication within the contract space we study: a developer fee contingent on adoption would reopen the financing of subsidies.

5 Scope and Extensions

The analysis so far rests on five choices: the platform's system has no advantage over the outside option, devices are substitutes, there are two of them, the rival owns no system of its own, and the developer pays per published application. This section relaxes each

of them. The first three subsections relax the first, in three ways, efficiency gains, a dominant platform, and coordination motives or porting costs; they share one structure, and it is worth laying it out once, because it organizes how the results of each should be read. The remaining four change the structure of the baseline rather than adding to it, and are taken up afterwards.

ONE PROBLEM. Each of the three departures gives the platform's operating system an edge over self-supply: the merged entity monetizes users better than a standalone platform, or forks monetize users less well than the platform's system, or a manufacturer values sharing a system with its rival. Call this edge the *adoption premium* and write it Δ , measured in the units of the per-user benefit r . Its effect on the game is always the same: at equal terms a manufacturer now prefers the platform's system, unilaterally under an efficiency gain or dominance and jointly under a coordination motive or a porting cost (Section 5.3), so the platform can keep part of the per-user benefit and still be adopted. The rival's participation constraint takes the form

$$\beta^2 \geq 1 - \frac{\Delta}{r} \theta(\sigma), \quad (11)$$

a floor on the share the rival must be left, equal to one at $\Delta = 0$ and falling as the premium grows, where θ is a market-structure factor taking two values, according to whether the premium is measured against the per-user benefit of the better system, r_0 then replacing r in (11), or against r .¹⁵ Section 3 is the case $\Delta = 0$, in which the constraint reduces to $\beta^2 \geq 1$ and the fee is the only instrument.

With $\Delta > 0$ the integrated platform solves a two-instrument problem, choosing the share β^2 it leaves the rival, between the floor (11) and one, and the fee a it charges the developer, and the structure of network effects decides which constraint binds. When effects are strongly developer-skewed, the platform wants buyers on both devices, because the developer values them and the fee taxes what the developer earns from them; the cheapest way to attract them is to leave the rival the whole per-user benefit, so that the rival prices low, and to tax the developer: the cap $\beta^2 \leq 1$ binds, and the premium is passed through. Call this the *cap regime*. When effects are buyer-skewed or moderate, the floor (11) binds instead: the platform extracts the whole premium from the rival, and may at the same time subsidize the developer. Call this the *participation regime*. In between neither constraint binds. The three regimes are drawn, for the efficiency-gains case, in Online Appendix B.4. (For premia large enough that the floor (11) falls below zero, the share is exhausted at $\beta^2 = 0$, a corner Online Appendix B.5 records; the floor reaches zero at $\delta = 8r(1 + \sigma)/\Lambda$, which is 0.357 at $\sigma = \gamma = 4$.)

THREE CHANNELS. The welfare effect of the merger can then be read as the sum of three channels, each of which the reader has already met or will meet in one of the subsections below.

The first is the *fee channel* of Section 3: the merger creates a bottleneck over the buyers of the integrated device, and the sign of the fee that prices it is set by the structure of network effects. This channel is present in every extension, and it is the only one at $\Delta = 0$.

¹⁵Efficiency gains and a dominant platform share $\theta = \Lambda/(8(1 + \sigma))$, with Λ as in Section 4; the ratio Δ/r is then δ/r under dominance and $(r_0 - r)/r_0$ with efficiency gains, the premium being measured against the per-user benefit of the better system in each case. Coordination motives and porting costs share $\theta = (4 + 3\sigma)/(4(1 + \sigma))$, with the premium measured against r . The floor is written $\underline{\beta}$ throughout, and its closed form in each environment is in Online Appendices B.4, B.5 and B.6.

The second is the *premium channel*: whatever the integrated platform extracts from the rival through $\beta^2 < 1$. This is a transfer, but not a neutral one, because both device prices rise. The rival keeps less per user and prices higher, and the merged firm, now earning a margin on the rival's sales, competes less hard against it. When the platform passes the premium through instead, in the cap regime, this channel is shut, and what the rival faces is the fee channel together with whatever value the premium it is left carries.

The third is the *value of the premium* itself, and here the three sources part company. An efficiency gain is value the merger creates: a higher per-user benefit on the integrated system, which the integrated firm passes into a lower device price, and which the rival receives when it is left the share. A coordination benefit and a saved porting cost are value as well, a higher demand for both devices or more applications, but value that does not depend on the merger, since both manufacturers could enjoy it on the platform's system under separation. And the shortfall under dominance is no value at all: it is the outside option being worse, and nothing is created by pricing it.

WHICH COUNTERFACTUAL. The last distinction decides what the merger is compared with. When the premium is created by the merger, the counterfactual is Proposition 1: the platform earns nothing under separation, and the merger's welfare effect is the sum of the three channels in full, the value of the premium included. When the premium exists before the merger, the separated platform may already price it, in which case the counterfactual already contains the premium channel, and part of the fee channel too, and the merger adds only the difference. Which of the two applies depends on the adoption game. Under dominance and under coordination motives alike the separated platform prices part of the premium, so the merger is measured against a distorted benchmark in both cases, but the constraint that limits the platform differs: the outside option of a manufacturer that adopts alone in Section 5.2, that of a manufacturer that walks alone in Section 5.3.

READING THE SECTION. Section 5.1 activates the fee channel, the premium channel and the value of a merger-created premium, against the undistorted benchmark. Section 5.2 activates the fee and the premium channels against a distorted benchmark, and the value channel is empty. Section 5.3 activates the fee channel and the premium channel against a benchmark in which the separated platform already prices part of a pre-existing premium, and the value channel, present on both sides of the comparison, cancels. The four subsections that follow, complements, more manufacturers, an integrated rival and proportional fees, add no premium: they change the rivalry between devices, their number, who owns a store, or the instrument, and revisit the fee channel alone. The Online Appendix carries the formal treatment of each; what follows reports the mechanism and what it does to the map of Section 3.

5.1 Efficiency Gains

Suppose the merger creates synergies: on the integrated system the per-user benefit becomes $r_0 > r$, the merged entity monetizing user data better than a standalone platform, while the outside option still yields r . The premium is $r_0 - r$, and it is created by the merger. Licensing becomes strictly valuable to both parties: a rival adopting the integrated system earns $\beta^2 r_0$ per user and faces a softer competitor, since the integrated firm keeps $(1 - \beta^2)r_0$ on every unit the rival sells and internalizing this licensing revenue shifts its own price upward. The rival therefore accepts shares strictly below one, which

is where the factor θ of (11) comes from. Efficiency gains thus resurrect the channel that Section 3 had shut down, through the value the integrated system brings to the rival rather than through a degradation of the rival’s alternatives.

The three channels then read as follows. In the cap regime the premium channel is shut: the rival keeps the whole per-user benefit, efficiency gain included, and the platform finances its buyer subsidy with a higher fee. The gain is passed through in full, and the rival is nevertheless worse off on the grids of Online Appendix B.4, because the fee channel outweighs it. In the participation regime the premium channel is fully open and the platform may at the same time subsidize the developer. The rival is then held at its outside value, so its fate is the fate of self-supply: the subsidized application commons raises that value and the better-monetizing competitor lowers it, and the rival gains when the first outweighs the second. Passing the gain through and gaining on net are thus two different things. The qualitative map of Figure 1 carries over, on those same grids: buyers and the developer still gain where network effects are strong and skewed, on regions that widen with the gain, as the value channel adds to the fee channel, and the region where every party but the merged entity loses remains the neighborhood of weak and balanced effects, where extraction, from the developer through the fee and from the rival through the share, dominates whatever the gain adds. What does not carry over is the exact equivalence of Proposition 3: with two instruments the sign of the developer fee is no longer a sufficient statistic for harm to the rival, since the share moves too.

5.2 A Dominant Platform

MOTIVATION. Our benchmark rests on the operating-system layer being contestable: a manufacturer that walks away keeps the same per-user benefit. In many platform markets it does not. Forks and in-house systems work, but they monetize users less well. We therefore let the outside option deliver a per-user benefit of $r - \delta$ rather than r , with $\delta \in (0, r)$, and ask what survives. Two reasons make this worth doing. First, the vertical mergers that competition authorities examine are mergers by platforms that are already dominant, so a benchmark in which the pre-merger platform earns nothing is precisely the assumption a practitioner would challenge. Second, δ plays for the non-integrated platform exactly the role that efficiency gains play for the integrated one. Online Appendix B.5 characterizes the separation benchmark and the integrated policy in closed form, for every δ , and compares them.

DOMINANCE CREATES MARKET POWER UNDER SEPARATION. The premium is now δ , and it exists before any merger: the participation constraint is the one of efficiency gains with δ/r in the place of the relative gain, but it is the *separated* platform that faces it. Two things then change with respect to Proposition 1. The platform earns a strictly positive profit without charging the developer at all, by setting the shares just high enough that adoption beats walking away and keeping δ per user on both devices: the premium channel is open under separation. And the free-riding argument no longer protects the developer: a manufacturer that refuses in isolation now compares hosting with a strictly worse outside option, so the platform can make joint refusal unprofitable without the rebates above the cap that Section 4 required, and it needs to buy one manufacturer’s adoption only, the other, held at its outside value, following once its rival is on the system. That package is the single-host deal of Section 4 with the premium in the role of the rebate, and it taxes the developer exactly where that deal was profitable, on

$u_B < h_\Phi(u_S)$: the fee channel is open under separation on all but the band of Proposition 6(i). How the separated platform uses that market power depends, as everywhere in our analysis, on the structure of network effects. When they are developer-skewed it gives up most or all of the per-user benefit to one manufacturer, all of it at the cap, holds the other at its outside value, and taxes the developer; when they are buyer-skewed a developer fee would depress device demand more than it would raise, and the platform leaves the developer alone, or subsidizes it when δ is large. This is the same arbitration between the one-sided and the two-sided logic as in Section 3, now carried out by a platform that has not merged with anyone.

THE INCENTIVE TO INTEGRATE, AND WELFARE. Integration remains profitable at every admissible point of the reference grid, at every γ and every δ up to 0.45 (Online Appendix B.5). Its value, however, does not move monotonically with the platform’s advantage on those computations, and the two sides of the separated platform’s own taxing frontier, which is the frontier h_Φ of Proposition 6 (Online Appendix B.5), move in opposite directions. Where the dominant platform already taxes the developer, integration tends to add less, since the fee it brings is one the platform was collecting anyway, and at balanced effects one it was collecting in larger amount. Where the dominant platform does not tax, δ instead creates a problem that integration is well placed to solve: the platform keeps δ per user from manufacturers whose price-cost markups are already positive, which is a double margin, and integration eliminates it on the device it owns. There integration becomes more valuable as δ grows, at every point of that region on our grids; the first tendency holds at most points of its own, and Online Appendix B.5 reports the exceptions cell by cell.

That double margin is also why dominance is costly. Under separation, the first units of δ strictly reduce welfare at every admissible configuration, a result we prove at $\delta = 0$: the manufacturers’ markups are already positive there, so the output loss caused by the platform’s own margin is of first order. At that margin the platform’s gain exceeds the manufacturers’ loss as soon as devices are substitutes, so the industry as a whole is not made worse off; buyers and the developer pay for dominance, and they pay more than the industry gains. Whether the welfare effect of the merger itself improves with δ is a comparison of two moving outcomes: on our admissible set at the reference calibration, the share of configurations at which integration reduces welfare falls from about two thirds at $\delta = 0$ to about one seventh at $\delta = 0.3$, and to zero where rivalry is weaker (Online Appendix B.5), and it falls largely because the counterfactual deteriorates: both outcomes move, and at large premia separation welfare can even rise where the platform subsidizes (Online Appendix B.5), while the private gain, as we have just seen, does not follow the same path. The incentive to merge and the welfare effect of merging come apart, and they come apart further the more dominant the platform already is.

5.3 Coordination Motives and Porting Costs

Two further sources of an adoption premium are endemic to platform markets, and both make the platform’s system more valuable to a manufacturer precisely when the rival manufacturer uses it too. The first is a direct network effect among buyers, the paper’s own effects being indirect: operating systems carry services whose value to a buyer grows when the other device runs the same system, which we capture by an extra gross benefit α_B accruing to every buyer when manufacturers share an operating system. The second

is porting: if the developer bears a cost of porting an application onto each additional system, at a unit cost $c > 0$, joining the system where applications already are saves resources and yields more applications, so that self-supply loses its equivalence. Both premia are value, and value that does not depend on the merger; the thresholds and the pricing policies are in Online Appendix B.6.

Coordination motives and porting costs resurrect the licensing channel that Section 3 had shut down, as efficiency gains and dominance do, and the two-instrument problem of (11) applies unchanged. What is particular to them is that the value they bring exists only when both manufacturers are on the same system, and the rest follows. The value is not created by the merger: both manufacturers could enjoy it on the platform's system under separation, and the independent platform can price it, so that the benchmark is distorted, as under dominance. But the adoption game limits what it prices. A unilateral adopter enjoys neither the coordination benefit nor the saved porting cost, so joint refusal survives as an equilibrium of the adoption game at every capped package, and the platform cannot divide and conquer as it does under dominance; and a unilateral walker keeps the whole per-user benefit, so joint adoption is an equilibrium only above a floor on the share, at which the independent platform collects part of the premium, the whole of it only at $\sigma = 0$.¹⁶ The merger adds two things to that benchmark (Online Appendix B.6). It lowers the floor: facing a rival that walks, the integrated firm internalizes the fee it collects on its own buyers and prices lower than a hosted manufacturer would, so the rival's outside option is worse, and the integrated firm extracts more of the premium than the independent platform could, by an amount that vanishes at $\sigma = 0$ and grows with σ . And it changes the fee, which is now on both sides of the comparison: the independent platform taxes the developer at most points, converting the premium into taxing capacity along the compensated locus of Online Appendix B.5, and subsidizes it at the strongly buyer-skewed points, where the free-riding argument no longer bites because a manufacturer that walked to enjoy the subsidized commons would lose the premium. Against this benchmark the maps of Online Appendix B.11 read as follows. Where effects are developer-skewed the independent platform had converted the premium into taxing capacity, and the integrated firm taxes more at every point and takes back part of the rival's share at many: the rival loses everywhere, the developer and total welfare at shares that rise with γ . Where effects are buyer-skewed or moderate the benchmark sits at the floor, the integrated firm taxes less than the independent platform did at most points or subsidizes more, and the developer loses nowhere at $\gamma \leq 4$ and at a minority of points beyond, while the rival still loses at most taxing points to the lower floor. And near the origin, where the fee is worth little to either firm, the merger is the lower floor alone, both device prices rise, and the rival, the buyers and welfare lose. At $\gamma = 4$ and $\alpha_B = 0.2$ the rival loses at 71% of the admissible set, buyers at 31%, the developer at 46% and welfare falls at 38%; the developer's share is where the corrected benchmark matters most, since against the undistorted benchmark it would read 68%, the pre-merger fee being then counted as an effect of the merger.

¹⁶Porting costs break the benchmark's immunity to free-riding for any positive cost, since hosting both manufacturers saves the duplicated porting cost, provided the developer still finds it worthwhile to port to a second system when one is in use, that is $usn_B^2 \geq c$; where this fails, catalogues differ across systems and the analysis does not apply.

5.4 When Devices Become Complements

Assumption 1 confined the analysis to markets where product-market rivalry dominates the network channel. Yet the industries that motivate the scrutiny of platform mergers are often those where network effects are strongest, and strong network effects can turn devices into demand complements: a cheaper rival device then means more buyers, more applications, and a *higher* demand for one's own device. Online Appendix B.3 relaxes Assumption 1 and studies $\sigma \in (-1, 0)$, the lower bound being where the two-sided feedback diverges. Everything below is stated on the interior admissible set defined there, which requires positive margins on both devices at the integrated optimum and is stricter than $\mathcal{A}$: it excludes the configurations in which the fee finances below-cost device pricing, where the alignment described next can fail.

On that set the architecture of Section 3 survives intact: the separation benchmark extends, the absence of market power over the rival extends, and the frontier $\underline{h}$ continues smoothly across the boundary $\sigma = 0$. What changes is the welfare map, and it changes by becoming simpler. Under substitutes the merger's fee redistributes across constituencies, taxing the developer to finance cheaper devices, and that is the disconnect at the heart of Figure 1. Under complements the rivalry margin fades: all participations move together, a fee shrinks the whole system and a subsidy expands it, and the rival, the buyers and the developer rise or sink together. On the subsidizing side this is proved, and total welfare rises with them, exactly as in Proposition 5(i). On the taxing side it is verified on the grid of the reference calibration, with one exception: the alignment of buyers requires product-market competition to be strong enough, and where it is soft a losing rival can coexist with gaining buyers. Because the rival, the buyers and the developer now move together, complaints from non-integrated manufacturers regain much of their evidentiary value where network effects are strongest, subject to that exception. Total welfare, which adds the merged entity's gain to the others' losses, falls at almost every taxing point of those grids, the exceptions lying where buyers gain.

5.5 More Manufacturers

The second manufacturer is the model's only source of downstream rivalry, and the duopoly is a modelling convenience rather than a feature of the industries. Let there be $N \geq 2$ of them, symmetric. The demand system rewrites itself as it stands, with Nu_{BUS} in the place of $2u_{BUS}$, but the equilibrium is not a relabeling, since the integrated firm now faces $N - 1$ rivals rather than one; Online Appendix B.8 recomputes it and gives the two polynomials in σ that replace $4 + \sigma$ and $4 + 3\sigma$ in the closed forms of the Appendix.

Two things move with N . The feedback loop tightens, the common application stock being fed by N buyer bases, so that the stability condition (3) becomes $Nu_{BUS} < 1$. And the frontier moves, for a reason that is the paper's own free-riding argument turned on the merged firm. The integrated firm subsidizes the developer to enlarge the application stock and recoups the subsidy on its device margin; with N manufacturers it pays the whole subsidy and captures roughly $1/N$ of what it buys, the other $N - 1$ free-riding on a common they did not finance. The ratio of network effects at which the integrated firm switches from taxing to subsidizing, $R_a^N(\sigma)$, equals N exactly at $\sigma = 0$ and is strictly increasing in N at fixed σ , as a polynomial identity; since σ itself falls with N at fixed primitives, the statement about regions is a grid finding: on the reference grid at the

reference calibration, the subsidizing region shrinks as manufacturers are added. Harm to a non-integrated manufacturer still coincides exactly with a rise in the fee, at every N , by the envelope argument of Proposition 3.

On those grids the share of the admissible set on which the merger raises total welfare falls both as the downstream market fragments and as product-market competition intensifies, so the two comparative statics of the paper survive (Online Appendix B.8 reports the shares). The reading to resist is that the result evaporates: the pro-competitive case for integration is a phenomenon of concentrated downstream markets, and its receding as the device market fragments is a prediction of the model rather than a caveat.

5.6 An Integrated Rival

A real market often contains one manufacturer that owns its operating system outright rather than licensing one. Let M_2 own its system *and* its store, so that it keeps the whole per-user benefit and posts its own developer fee a_2 , while the platform posts a_1 on its own store. Applications are still written once and published wherever the developer is better off publishing, so the demand system is the paper’s at $a = a_1 + a_2$, and the developer keeps multi-homing.¹⁷ The two fees are perfect substitutes in the developer’s cost, which makes the two stores Cournot complements, and everything follows from that. Before the merger the platform licenses to M_1 and earns nothing, by the free-riding argument of Proposition 1 run with a single licensee. After the merger there are two symmetric integrated firms.

Two things are analytic. First, the taxing frontier does not move: the same bracket signs the integrated fee of Section 3, M_2 ’s pre-merger fee and each firm’s post-merger fee, so all three change sign on the same ratio $R_a(\sigma)$, and the taxing and subsidizing regions of Figure 1 carry over unchanged. Second, the merger raises the total fee exactly where the region is taxing, the change factoring into that same bracket times a polynomial that is positive on the whole admissible set (Online Appendix B.9). The merger amplifies what the benchmark was already doing, more tax where M_2 taxed and more subsidy where it subsidized.

The mechanism is the mirror of the paper’s. There, the merger *creates* market power over the developer where none existed. Here M_2 already had it, and the merger *duplicates* it: two stores tax the same developer, and neither internalizes the damage its own fee does to the other’s application base. That is a double margin on the developer side, and it can be measured against the counterfactual in which the two firms coordinate their fees while still competing in prices. On the reference grid the uncoordinated total fee exceeds the coordinated one at almost every admissible taxing point, by about half as much again: two independent stores tax the developer more than a single owner of both would. One reading of the map also changes, again as a grid finding. The rival and the developer now move together, both being hurt by a higher total fee, so that a rival’s complaint, insufficient evidence of consumer harm in the paper, is here a reliable signal of harm to the developer. After the merger the market is the one the smartphone

¹⁷Under single-homing the developer would have to be allocated between the two stores, which we do not model. Cross-platform development is the object of Jeon and Rey [2024], where developers may single- or multi-home and economies of scope in porting an application link the two platforms, and of Teh and Wright [2026], where competing bottlenecks generate spillovers across platforms; both work with commissions, whereas our stores post a fee per published application.

industry displays: two integrated systems, each with its own store and its own market power over the developer, competing in devices. What the extension does not add is a third manufacturer, since our platform has a single licensee and the merger absorbs it; Section 5.5 is where their number varies.

5.7 Proportional Fees

We have modeled the developer fee as a per-application charge; many stores charge instead a commission proportional to developer revenues. A proportional fee τ on the revenues the developer earns from the platform’s buyers changes three things. First, the platform’s store revenue scales with its own installed base, so the integrated firm gains an additional reason to price its device low, at given decisions on the other side; this is the channel Etro [2021a] isolates for device-funded platforms, where what is raised on one side finances cheaper hardware on the other. Second, a proportional fee applies only to interactions on the platform’s system, so a manufacturer that leaves escapes its *direct* incidence. It does not escape the mechanism: what harms a departing manufacturer in this paper is the shrunken application commons, which is common to every system, and a proportional fee bears on application incentives in the same way; whether it shrinks the equilibrium stock depends on the device-price response, which we have not characterized. What weakens is the extraction of Section 4, whose grip comes from the contracting externality, since the outside option is a better refuge against a fee one can leave behind. One tool would have to be rebuilt rather than transposed: because adoption now changes the commission base, a hosted manufacturer at $\beta = 1$ is no longer in the position of a walker, so the equivalence that organizes Sections 3 and 4 does not survive.

Third, tractability. With the commission multiplying the network multiplier, the integrated firm’s problem loses concavity in the region of strong and skewed network effects where the interesting results live. This is the familiar divide between membership-based models in the tradition of Armstrong [2006] and usage-based ones in the tradition of Rochet and Tirole [2006], and the economics of proportional fees in platform competition is the subject of Jeon and Rey [2024] and Bisceglia and Tirole [2023]. Our per-application fee with a strictly convex development cost is what delivers interior fees and closed forms here; the loss of concavity is a property of the formulation we explored, not a claim about commissions in general. What would survive is a question for that model, not a result of this one.

A charge levied per transaction rather than per unit of developer revenue raises the same issues without being the same instrument. Both depend on usage rather than on membership, and they coincide only if the value of a transaction is fixed and the two bases are the same. Where that value or the intensity of use responds to the fee, or to the device price, the two give the developer different incentives at the margin, and we claim no equivalence between them.

6 Concluding Remarks

This paper asked how indirect network effects change the competitive assessment of vertical integration. In markets where the operating-system layer is contestable, the merger creates a bottleneck over the buyers of the integrated device and nowhere else, and the merged firm prices the developer’s access to them. How it prices that access is disciplined

by the structure of network effects: when they are sufficiently buyer-skewed, the integrated firm subsidizes publication and the merger can benefit every constituency with no efficiency gain in sight; when they are weak and balanced, the fee is pure extraction and every other party loses. Without the industry’s cap on per-user rebates, separation can copy the merger’s tax on the developer through a tax-and-rebate deal with one manufacturer, on part of the taxing region and exactly on one curve, but it can never subsidize.

For merger control, the lesson is not that platform mergers are benign, but that harm to a non-integrated rival is not sufficient evidence of harm to buyers: the rival can lose while buyers gain, and on the reference grid this happens most often where product-market competition is soft. The complaint recovers its evidentiary value where network effects turn devices into complements, up to the same qualification about competition. What screens the cases is the strength and the skew of network effects together with the contestability of the platform layer, quantities about which the agencies’ growing practice on digital ecosystems already generates evidence.

Two boundaries of the exercise are worth naming. The benchmark is a polar case, and Section 5 measures how fast the merger’s distinctive effects erode as the platform layer becomes entrenched; how much a manufacturer loses by leaving the platform’s system, in applications or in monetization, is the empirical quantity that decides which of the two literatures applies. And our contract space is deliberately lean: richer instruments in the hands of *competing* strategic platforms raise existence questions we do not settle, and costly or endogenous multi-homing by the developer, and developer terms contingent on adoption, are the first assumptions we would relax. And every application earns the same u_S per buyer, which makes publication on the platform’s store all or nothing; with heterogeneous returns the marginal applications could stay on the free system and skip the taxed store, and an input-foreclosure channel absent here would reopen.

A Appendix

A.1 Preliminaries: Pricing, Welfare, Admissible Set

DEMANDS. Solving the stage-4 fixed point under condition (3) yields (4)–(5). Direct computation gives $\partial D_k/\partial p_k = -(2+\sigma)/[2(1-2u_B u_S)] < 0$, $\partial D_k/\partial p_\ell = \sigma/[2(1-2u_B u_S)]$, $\partial D_k/\partial a = -u_B/(1-2u_B u_S) < 0$, $\partial D_S/\partial p_k = -u_S/(1-2u_B u_S) < 0$ and $\partial D_S/\partial a = -1/(1-2u_B u_S) < 0$.

STAGE 3. Let b_k denote M_k 's perceived per-user benefit ($b_k = \beta^k r$ on the platform, $b_k = r$ outside). M_k 's profit $(p_k + b_k)D_k$ is strictly concave in p_k , with best response

$$R_k(p_\ell; b_k, a) = \frac{2(v - b_k) + \sigma(p_\ell - b_k) - 2u_B a}{2(2 + \sigma)}, \quad (\text{A.1})$$

so that $\partial R_k/\partial p_\ell = \sigma/[2(2 + \sigma)]$: prices are strategic complements for $\sigma > 0$ and strategic substitutes for $\sigma \in (-1, 0)$, with $|\partial R_k/\partial p_\ell| < 1$ in both cases; the pricing subgame has a unique equilibrium. When both firms are non-integrated the equilibrium prices are

$$\hat{p}_k = \frac{-2u_B a(4 + 3\sigma) + 2v(4 + 3\sigma) - (2 + \sigma)(\sigma b_\ell + 2(2 + \sigma)b_k)}{(4 + \sigma)(4 + 3\sigma)}, \quad (\text{A.2})$$

and when the integrated firm faces a *hosted* M_2 at share $b_2 = \beta^2 r$ (the merged firm's objective being $(p_1 + r)D_1 + aD_S + (r - b_2)D_2$, where the last term is the licensing margin, which vanishes at $b_2 = r$), the equilibrium prices are

$$p_1^I = -b_2 + \frac{2v - r - a(2u_B + u_S) + 3b_2}{4 + \sigma} + \frac{b_2 - r - au_S}{4 + 3\sigma}, \quad (\text{A.3})$$

$$p_2^I = -b_2 + \frac{2v - r - a(2u_B + u_S) + 3b_2}{4 + \sigma} + \frac{r + au_S - b_2}{4 + 3\sigma}. \quad (\text{A.4})$$

From (A.2)–(A.4): for $\sigma \geq 0$ equilibrium prices decrease in a (e.g. $\partial \hat{p}_k/\partial a = -2u_B/(4 + \sigma) \leq 0$, $\partial p_1^I/\partial a = -(2u_B + u_S)/(4 + \sigma) - u_S/(4 + 3\sigma) \leq 0$); in the complements range the rival's response satisfies $\partial p_2^I/\partial a < 0$ only where $u_B/u_S > |\sigma|/(4 - 3|\sigma|)$, which the admissible complements set of Proposition B.1 satisfies throughout. Prices decrease in own perceived benefit with a pass-through smaller than one, and, setting $a = 0$ and $b_k = r$, deliver the benchmark values (A.8).

WELFARE OBJECTS. Buyer surplus is computed from the quadratic representative utility behind (1), $V_B = v(n_B^1 + n_B^2) + u_B \sum_k n_S^k n_B^k - \frac{1}{4(1+\gamma)}(2 \sum_k (n_B^k)^2 + \gamma(\sum_k n_B^k)^2) - \sum_k p_k n_B^k$, which shares its price and application gradients with the aggregate surplus of the single-homing population ($\partial V_B/\partial p_k = -n_B^k$, $\partial V_B/\partial q_S = u_B \sum_k n_B^k$ when all systems carry the applications). Developer surplus is $V_S = q_S^2/2$.

ADMISSIBLE SET. The integrated platform's reduced profit $\Pi^I(a)$ is a quadratic in a ; strict concavity holds if and only if

$$H \equiv (\sigma + 4)^2(3\sigma + 4)^2 - 2(\sigma + 2)(3\sigma + 4)^2 u_B^2 - 2(3\sigma + 4)(\sigma(7\sigma + 32) + 32)u_B u_S - 2(\sigma(\sigma(7\sigma + 40) + 64) + 32)u_S^2 > 0. \quad (\text{A.5})$$

Two admissible sets are used, and it is worth separating them. Propositions 2–5 concern the *one-instrument* problem, and their admissible set is

$$\mathcal{A} = \{(u_B, u_S) : 2u_B u_S < 1, H > 0, D_2(a^{**}) > 0, u_S n_B^I(a^{**}) \geq a^{**}\}, \quad (\text{A.6})$$

concavity together with the requirement that the rival still serve a positive demand at the integrated optimum, and that the developer publish on the platform's store there (Section 2). The rival-demand condition is not a technicality: where it fails the fee would drive the rival out of the market altogether, and a proposition about harm to a rival that remains active has nothing to say there. On the reference grid it is the only interiority condition that ever binds, the publication condition never binding there, and it binds only at strongly developer-skewed points (Online Appendix B.1). The *two-instrument* problem of Section 5, in which the integrated firm chooses β and a jointly, requires in addition $N_2 < 0$, a strictly stronger condition ($N_2 < 0$ implies $H > 0$), characterized in Online Appendix B.1; a simple sufficient condition for it is $(u_B + u_S)^2 < 2$. Write

$$\mathcal{A}^* = \{(u_B, u_S) : 2u_B u_S < 1, \ N_2 < 0, \ D_2(a^{**}) > 0, \ u_S n_B^I(a^{**}) \geq a^{**}\} \subset \mathcal{A}. \quad (\text{A.7})$$

Section 4 is stated on $\mathcal{A}^*$ as well. The reason is not the integrated firm's problem, whose benchmark there is the one-instrument optimum, but the separated platform's own two-host program, whose second-order condition is exactly $N_2 < 0$ in disguise (Lemma B.1 and Corollary B.3 of the Online Appendix). For $\sigma \in (-1, 0)$, $\mathcal{A}$ additionally imposes positive markups, as Section 5.4 states.

A.2 Proofs of Propositions 1 to 5

TWO MONOTONICITY FACTS. Let $\hat{\pi}_k(b_k, b_\ell, a)$ denote M_k 's stage-3 equilibrium profit. From (A.1)–(A.2), for all $\sigma \in (-1, \gamma]$: (i) $\hat{\pi}_k$ is strictly increasing in b_k ; (ii) $\hat{\pi}_k$ is strictly decreasing in a ; the derivative is proportional to $-u_B(4 + 2\sigma)/(4 + \sigma) < 0$.

Proof of Proposition 1. Existence. Let the platform post $(1, 1, 0)$; all adoption profiles yield the benchmark outcome and zero platform profit; manufacturers are indifferent. *Uniqueness of the outcome.* Consider any posted (β^1, β^2, a) with $\beta^k \leq 1$. Suppose $a > 0$. By fact (i), a manufacturer offered $\beta^k < 1$ prefers its outside option whatever the other does, since the fee it faces is unchanged by its own adoption when the other is hosted, and dies when neither is; so hosting requires $\beta^k = 1$, in which case each hosted manufacturer is exactly indifferent to walking alone. ‘Both walk’ is then also a Nash equilibrium of the adoption subgame, with joint profit $2\hat{\pi}(r, r, 0) > 2\hat{\pi}(r, r, a)$ by fact (ii); the selection rule picks it, and the platform earns nothing. Suppose $a < 0$. The developer then publishes on the platform's store regardless of adoption, so the subsidy is paid unconditionally and the application stock is the same function of prices in every configuration; a manufacturer's comparison therefore reduces to its own perceived benefit, and fact (i) again makes $\beta^k < 1$ refused whatever the other does. Hosting requires $\beta^k = 1$, and the platform's profit is $a q_S < 0$, dominated by posting $(1, 1, 0)$. Hence in every equilibrium the platform earns zero and the outcome is the benchmark. Evaluating (A.2) at $b_k = r$, $a = 0$ gives the benchmark prices, participation and application stock,

$$p^S = \frac{2(v - r)}{4 + \sigma} - \frac{r\sigma}{4 + \sigma}, \quad n_B^S = \frac{(2 + \sigma)(v + r)}{(4 + \sigma)(1 - 2u_B u_S)}, \quad q_S^S = 2u_S n_B^S. \quad (\text{A.8})$$

The selection rule never overrides an individual preference. The rule chooses only where the adoption subgame admits several equilibria, and under $\beta^k \leq 1$ that happens only at $\beta^k = 1$. For $\beta^k < 1$ refusal strictly dominates acceptance for M_k , as the two branches above establish: with $a > 0$ the fee it faces is unchanged by its own adoption

when the other is hosted and dies when neither is, and with $a < 0$ the subsidy is collected in every configuration, so in both cases the comparison reduces to the perceived benefit and fact (i) settles it. Every profile in which some manufacturer accepts at $\beta^k < 1$ is therefore not an equilibrium, joint refusal is the unique one, and the rule has nothing to select. At $\beta^k = 1$ with $a > 0$, joint acceptance and joint refusal are both equilibria, and each manufacturer strictly prefers the second: perceived benefits are r in both, while the fee is live at joint acceptance and dead at joint refusal, so fact (ii) applies. At $\beta^k = 1$ with $a < 0$ the subsidy is collected whatever the adoption profile and all four profiles yield the same payoffs, so the selection is immaterial. The rule therefore never retains an outcome that a manufacturer ranks below another equilibrium of the subgame, asymmetric profiles included: what it supplies is not the preference but the belief. The argument is analytic and rests on the two monotonicity facts and the publication rule; `rep8_adoption.py` enumerates the subgame and reproduces the three cases, but nothing above depends on it. $\square$

Proof of Proposition 2. $\beta^2 \geq 1$ follows from (A.4): $p_2^I + b_2$ is increasing in b_2 , its derivative being $8(1+\sigma)/[(4+\sigma)(4+3\sigma)]$, and hosting at $b_2 = r$ is outcome-equivalent to self-supply; the platform is indifferent to licensing at $\beta^2 = 1$. The fee is restricted to the publication regime (Section 2), on which the reduced profit $\Pi^I(a) = (p_1^I + r)D_1 + aD_S$ evaluated at (A.3)–(A.4) (with $b_2 = r$) is quadratic in a with second derivative $-2H/[(\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)]$; on the admissible set the first-order condition gives

$$a^{**} = \frac{(3\sigma + 4)(v + r) [(3\sigma(\sigma + 7) + 16) + 32]u_S - 2(\sigma + 2)(3\sigma + 4)u_B}{H}, \quad (\text{A.9})$$

with $H > 0$ on $\mathcal{A}$. Thus $a^{**} > 0$ if and only if $u_B/u_S < R_a(\sigma)$, where

$$R_a(\sigma) \equiv \frac{3\sigma(\sigma + 7) + 16}{2(\sigma + 2)(3\sigma + 4)}, \quad (\text{A.10})$$

evaluated at $\sigma = \gamma - 2u_B u_S(1 + \gamma)$; R_a is strictly increasing, with $R_a(0) = 2$. Along the locus $a^{**} = 0$ one has $u_B u_S$ fixed by σ , and eliminating u_B yields a curve that lies strictly below the 45° line and crosses each horizontal line $\{u_S = \text{const}\}$ at most once within the admissible set, since along such a line u_B/u_S rises while σ , and with it $R_a(\sigma)$, falls; this defines $\underline{h}$ with $\underline{h}(u_S) > u_S$, with the convention $\underline{h} = +\infty$ on the lines it does not meet, which are exactly those with $u_S > \frac{1}{2}\sqrt{\gamma/(\gamma + 1)}$: since $R_a \geq 2$, a subsidy requires $u_B > 2u_S$, hence $2u_S^2 < u_B u_S$, while $\sigma \geq 0$ caps $u_B u_S$ at $\gamma/(2(1 + \gamma))$, so that

$$a^{**} < 0 \implies u_S < \frac{1}{2}\sqrt{\gamma/(\gamma + 1)} < \frac{1}{2}, \quad (\text{A.11})$$

the bound being reached at $\sigma = 0$, where the locus $a^{**} = 0$ attains its highest point (details and the curve's geometry in Online Appendix B.2). $\square$

Proof of Proposition 3. Immediate from (8): both terms of the bracket are negative for $\sigma \geq 0$ ($\partial D_2/\partial a < 0$; $\partial D_2/\partial p_1 \geq 0$ and $dp_1^I/da \leq 0$), so π_2 strictly decreases in a , and the merger moves the fee from 0 to a^{**} . $\square$

Proof of Proposition 4. The two frontiers of the statement are

$$\bar{h}_B(u_B) = \frac{(\gamma + 2)u_B}{2(\gamma + 1)u_B^2 + 1}, \quad \bar{h}_S(u_B) = \frac{1}{2} \left(\sqrt{2(\gamma + 4) + (\gamma + 3)^2 u_B^2} - (\gamma + 3)u_B \right). \quad (\text{A.12})$$

Around the pre-merger fee, and evaluated at the symmetric pre-merger equilibrium, buyer surplus and the application stock move according to

$$\left. \frac{dV_B}{da} \right|_{a=0} = \underbrace{-Q_B \left(\frac{dp_1^I}{da} + \frac{dp_2^I}{da} \right)}_{\text{price effect (+)}} + \underbrace{\frac{\partial V_B}{\partial n_S} \frac{dD_S}{da}}_{\text{application effect}}, \quad (\text{A.13})$$

$$\left. \frac{dD_S}{da} \right|_{a=0} = \frac{1}{1 - 2u_B u_S} \left(\underbrace{\frac{\partial q_S}{\partial a}}_{(-)} + \underbrace{\frac{\partial q_S}{\partial n_B} \left(\frac{\partial Q_B^1}{\partial p_1} + \frac{\partial Q_B^1}{\partial p_2} \right) \left(\frac{dp_1^I}{da} + \frac{dp_2^I}{da} \right)}_{(+)} \right), \quad (\text{A.14})$$

where Q_B is the common device demand of (1), the two price derivatives of Q_B^1 sum to -1 , and $\partial q_S / \partial a = -1$ and $\partial q_S / \partial n_B = u_S$ are read on (2) before the participation fixed point is imposed. Taxing the developer lowers both device prices, which benefits buyers and, by drawing in more of them, replenishes the developer's incentive to create applications: partially where the net application effect in (A.13) is negative, more than fully in the strongly developer-skewed range, where the stock rises. Direct computation yields the factorizations

$$\Delta V_B(a) = a \left(a - \frac{2(\sigma+2)(3\sigma+4)^2(v+r)[(\sigma+2)u_B - u_S]}{K_B} \right) \frac{K_B}{(\sigma+4)^2(3\sigma+4)^2(1-2u_B u_S)^2}, \quad (\text{A.15})$$

$$\Delta V_S(a) = \frac{(\sigma - 2u_S(2u_B + u_S) + 4)^2}{2(\sigma+4)^2(1-2u_B u_S)^2} a \left(a - \frac{4(\sigma+2)u_S(v+r)}{\sigma - 2u_S(2u_B + u_S) + 4} \right), \quad (\text{A.16})$$

where $K_B = (\sigma+2)^2(3\sigma+4)^2 u_B^2 + [(\sigma+2)(\sigma(\sigma+16)+16) - 2u_B u_S(\sigma+1)(\sigma+4)^2] u_S^2 - 2u_B u_S(\sigma+2)(3\sigma+4)^2$. Since $2u_B u_S < 1$, the bracket in K_B exceeds $(\sigma+2)(\sigma(\sigma+16)+16) - (\sigma+1)(\sigma+4)^2 = (3\sigma+4)^2$, whence $K_B > (3\sigma+4)^2(u_S - (2+\sigma)u_B)^2 \geq 0$ for $u_S > 0$, and $K_B = (\sigma+2)^2(3\sigma+4)^2 u_B^2 > 0$ at $u_S = 0$; this step uses only $2u_B u_S < 1$ and holds for all $\sigma \in (-1, \gamma]$. (i) If $a^{**} < 0$ then $u_B > \underline{h}(u_S) > u_S$, hence $(\sigma+2)u_B - u_S > 0$ and (A.15) is positive; if $u_S \geq \bar{h}_B(u_B)$, i.e. $(\sigma+2)u_B - u_S \leq 0$, then (A.15) is nonnegative for any $a^{**} \geq 0$. (ii) The root of the second factor of (A.16) is positive iff $u_S < \bar{h}_S(u_B)$, where $\sigma - 2u_S(2u_B + u_S) + 4 > 0 \Leftrightarrow u_S < \bar{h}_S(u_B)$; the case analysis (including that $a^{**} > 0$ and $u_S > \bar{h}_S(u_B)$ imply a gain, and that in the intermediate region the developer loses) follows as in the text. The ‘only if’ uses admissibility: the application stock is affine in the fee, so $2\Delta V_S(a) = (q_S(a) - q_S(0))(q_S(a) + q_S(0))$; where $u_S < \bar{h}_S(u_B)$ the stock falls in the fee, so a gain at $a^{**} > 0$ would require $q_S(a^{**}) < -q_S(0) < 0$, which $q_S(a^{**}) > 0$ excludes. The dividing curve of part (i) is in Online Appendix B.2: on each slice of constant σ the relevant cubic decreases and then increases, so it crosses zero exactly once, and the four endpoint signs that establish this hold as polynomial identities. $\square$

Proof of Proposition 5. (i) $\Pi^I(a^{**}) > \Pi^I(0)$ by strict concavity and $a^{**} \neq 0$; $\Delta\pi_2 > 0$ by Proposition 3; and $a^{**} < 0$ implies $u_B > \underline{h}(u_S) > u_S$, so that $\Delta V_B > 0$ and $\Delta V_S > 0$ by Proposition 4. The last of these takes one step beyond the frontier condition. The developer gains from a subsidy only if the coefficient $\sigma + 4 - 2u_S(2u_B + u_S)$ of the factorization (B.4) is positive, which it is wherever $a^{**} < 0$, since that region lies above the diagonal, and $u_S < u_B$ with $2u_B u_S < 1$ gives $\sigma + 4 - 2u_S(2u_B + u_S) > \gamma + 4 - (\gamma + 3) - 1 = 0$. Summing the four terms gives $\Delta W > 0$. (ii) $\Delta W(a) = \Delta V_B(a) + \Delta V_S(a) + \Delta\pi_2(a) + \Delta\Pi^I(a)$ is a quadratic polynomial in a vanishing at $a = 0$; its coefficients are in closed

form from (A.15)–(A.16), the rival’s markup, and Π^I . The marginal coefficient equals $\Delta W'(0) = -2(v+r)N_1/[(\sigma+4)^2(1-2u_B u_S)^2]$, with

$$N_1 = (\sigma+2)[(\sigma+6)u_B - 8u_B^2 u_S - 2u_S^3] + 2(\sigma^2 + 3\sigma + 4)u_B u_S^2 - 2u_S. \quad (\text{A.17})$$

Along the locus $a^{**} = 0$, N_1 is proportional, with strictly positive factor, to $(\gamma - \sigma)(\sigma + 1)(\sigma + 2)(\sigma + 4)P(\sigma, \gamma)$, where every coefficient of the polynomial P is positive (Online Appendix B.2): hence $N_1 > 0$, $\Delta W'(0) < 0$, and by continuity the merger reduces welfare just inside the taxing region, for every γ . Evaluating ΔW at a^{**} and eliminating u_B yields the quartic of the Online Appendix; its coefficient signs are polynomial identities, while the single-crossing property and the behavior in γ are grid verifications there. $\square$

A.3 Proof of Proposition 6

Throughout, π^{bench} denotes the benchmark profit (A.8) of a manufacturer and the tie-breaking conventions of Section 2.3 apply.

Lemma A.1 (The compensating-package profit). *In the configuration where M_1 is hosted at share β and M_2 is outside, let $\bar{\beta}(a)$ be defined by $\pi_1(\bar{\beta}(a), a) = \pi^{\text{bench}}$ and let $\Phi(a)$ be the platform’s profit at the stage-3 equilibrium under $(\bar{\beta}(a), a)$. Then, on the admissible set: $\bar{\beta}$ is affine with $\bar{\beta}(0) = 1$ and $\bar{\beta}'(0) > 0$; Φ is a quadratic polynomial with $\Phi(0) = 0$ and*

$$\Phi'' = -\frac{2[\gamma(\sigma+2) + 2(3\sigma+4)]}{8 + \sigma(8+\sigma)} < 0; \quad (\text{A.18})$$

sign $\Phi'(0)$ is given by (9); and the unconstrained maximizer is

$$\hat{a} = -\frac{\Phi'(0)}{\Phi''} = \frac{(v+r)(\sigma+2)[(8+\sigma(8+\sigma))u_S - (4+3\sigma)u_B]}{(1-2u_B u_S)(\sigma+4)[\gamma(\sigma+2) + 2(3\sigma+4)]}. \quad (\text{A.19})$$

Finally, writing $P(\beta, a) \equiv (1-\beta)rD_1 + aD_S$ for the platform’s profit in this configuration, P is strictly concave in β , with

$$\frac{\partial^2 P}{\partial \beta^2} = -\frac{r^2(\sigma+2)(8+\sigma(8+\sigma))}{(\sigma+4)(3\sigma+4)(1-2u_B u_S)} < 0, \quad \frac{\partial P}{\partial \beta} \Big|_{\beta=\bar{\beta}(a)} = -\frac{r(\sigma+2)[(v+r) - a(u_S - u_B)]}{(\sigma+4)(1-2u_B u_S)} : \quad (\text{A.20})$$

whenever $a(u_S - u_B) \leq v+r$ (for every $a \geq 0$ if $u_B \geq u_S$, hence everywhere in the region $u_B \geq h_\Phi(u_S)$, where $u_B/u_S \geq R_\Phi(\sigma) \geq 2$), the platform’s profit from any package (β, a) with $\beta \geq \bar{\beta}(a)$ is at most $\Phi(a)$: rents left to the host lower the platform’s take.

Proof. In the single-host pricing subgame, equilibrium markups are linear in $(\beta r, a)$ and the host’s equilibrium profit is proportional to the square of its own demand, itself linear in $(\beta r, a)$. The equation $\pi_1 = \pi^{\text{bench}}$ therefore pins D_1 at its benchmark value, an equation linear in $(\beta r, a)$: $\bar{\beta}$ is affine, and $\bar{\beta}(0) = 1$ because the benchmark is the $(\beta, a) = (1, 0)$ outcome. Substituting, $(1-\bar{\beta}(a))rD_1 + aD_S$ is a quadratic polynomial in a , with the a -independent second derivative (A.18), whose sign is immediate for $\sigma > -1$. The closed forms (9) and (A.19), and the sign of $\bar{\beta}'(0)$, are direct computations, verified symbolically in Online Appendix B.7. For the last part, D_1 and D_S are linear in (β, a) at the pricing

equilibrium, so P is a quadratic polynomial; (A.20) is direct computation, the second expression using that the calibration pins D_1 at its benchmark value. Strict concavity in β and a nonpositive derivative at $\bar{\beta}(a)$ bound P by $\Phi(a)$ on $\beta \geq \bar{\beta}(a)$; and $R_\Phi(\sigma) \geq 2$ follows from $8 + \sigma(8 + \sigma) - 2(4 + 3\sigma) = \sigma(2 + \sigma) \geq 0$. $\square$

Proof of Proposition 6(i), subsidy, capped and single-host families. Let $u_B \geq h_\Phi(u_S)$, so $\Phi'(0) \leq 0$ by (9), and consider any package posted by the platform. *Single-host taxes* ($a > 0$). Hosting one manufacturer requires leaving it at least its walk-off value, which equals π^{bench} when the other is outside: $\beta \geq \bar{\beta}(a)$. Since $u_B \geq h_\Phi(u_S)$ implies $u_B/u_S \geq R_\Phi(\sigma) \geq 2$, the last part of Lemma A.1 bounds the platform's profit from any such package by $\Phi(a)$. Now $\Phi(0) = 0$, $\Phi'(0) \leq 0$, and $\Phi'' < 0$: hence $\Phi(a) < 0$ for all $a > 0$, and the profit is negative. *Subsidies* ($a < 0$). The subsidy is paid unconditionally. With $a < 0$, walking dominates any hosted position with $\beta \leq 1$ weakly and with $\beta < 1$ strictly, so hosting requires $\beta \geq 1$, on which the platform's licensing margin is nonpositive; its profit is then at most $a q_S < 0$. This step uses neither $u_B \geq h_\Phi(u_S)$ nor the cap, so the subsidy family is excluded on the whole admissible set. *Both-host packages.* Consider finally a package pair (β^1, β^2, a) accepted by both manufacturers, relabeled so that $\beta^2 \leq \beta^1$. In the both-host pricing subgame the hosted manufacturer M_2 's equilibrium markup rises in its own share,

$$\frac{\partial \hat{m}_2}{\partial \beta^2} = \frac{r(8 + \sigma(8 + \sigma))}{(4 + \sigma)(4 + 3\sigma)} > 0, \quad (\text{A.21})$$

while at $\beta^2 = 1$ its position is outcome-equivalent to walking (the fee's incidence on the shared application stock is the same either way), so that its hosted profit at $\beta^2 = 1$ equals $\pi^{walk}(\beta^1, a)$ identically. Three cases follow. A share $\beta^2 < 1$ is refused: by (A.21) it leaves M_2 strictly below its walk-off value, the no-market-power logic of Section 3 again. A share $\beta^2 = 1$ makes the pair outcome-equivalent to the single-host package (β^1, a) , covered by the first step. A share $\beta^2 > 1$ adds an unconditional per-user transfer to the second manufacturer. Lemma B.1 of the Online Appendix places the interior optimum of this case, its second-order condition $\Delta > 0$ holding throughout $\mathcal{A}^*$ by Corollary B.3 there. On $u_B \geq h_\Phi(u_S)$ one has $u_B/u_S \geq R_\Phi(\sigma) \geq 2$, hence $u_S < u_B$, hence $a^\dagger < 0$ by (B.37): the interior optimum of the both-host program calls for a subsidy, not a tax, so it is not a package of this family. This places the interior optimum, not the family. Packages with $\beta^k > 1$ and $a > 0$ exist at which both-accept is a Nash equilibrium with positive platform profit, and what excludes them is the selection rule: joint refusal is an equilibrium too, and the one the manufacturers rank first. The set of packages the rule selects is not convex, so concavity does not bound the platform's profit on it, and we have no analytic argument within the bounds of the sweep of Online Appendix B.7, where the exclusion is a grid finding reported with its protocol (`rep33_bothhost_grid.py`); beyond those bounds concavity does bound the profit of every package, selected or not, below the single-host value, which is what justifies them. A marginal bound gives the conclusion at the margin: for the symmetric both-host surplus $J(a) \equiv \sum_k (\hat{p}_k + r)D_k + a q_S$ at $\beta^1 = \beta^2 = 1$, direct computation gives

$$\text{sign } J'(0) = \text{sign} [(\sigma + 4)u_S - 4u_B], \quad (\text{A.22})$$

a ratio frontier $R_J(\sigma) = (\sigma + 4)/4$ that lies strictly below $R_\Phi(\sigma)$, since $R_\Phi - R_J = (\sigma^2 + 16\sigma + 16)/(4(3\sigma + 4)) > 0$: where $u_B \geq h_\Phi(u_S)$, a marginal fee lowers even the fully consolidated surplus of the platform and both manufacturers, so no marginal package can finance out of fee proceeds the participation of those it taxes. Hence no single-host,

capped or subsidy package earns the platform a positive profit, no both-host package does on the grid, and the benchmark outcome stands. $\square$

Proof of Proposition 6(ii). Let $u_B < h_\Phi(u_S)$. By Lemma A.1, $\Phi'(0) > 0$: the package $(\bar{\beta}(a), a)$ that saturates the host's participation constraint earns the platform $\Phi(a) > 0$ for small $a > 0$; by strict concavity the maximizer $\hat{a} > 0$ of (A.19) is unique, with $\hat{\beta} = \bar{\beta}(\hat{a}) > 1$ since $\bar{\beta}' > 0$, and with the convention that an indifferent manufacturer accepts, the optimum is attained. By the last part of Lemma A.1, no single-host package leaving the host a rent improves on this family as long as $\hat{a}(u_S - u_B) \leq v + r$; on the reference grid the bound fails at $\hat{a}$ only at strongly developer-skewed points ($u_S \geq 1.18$ at the reference calibration), all of which lie inside the two-host range below, where the single-host family is dominated in any case (Online Appendix B.7). Wherever $u_B \geq u_S$: by (A.21) the platform cannot pull the second manufacturer's share below one, a share equal to one leaves the outcome unchanged, and a two-host package taxes at its interior optimum only where $u_S > u_B$, by (B.37); that no constrained taxing pair selected by the rule beats the single-host package there is, as in part (i), a grid finding (Online Appendix B.7). Where $u_S > u_B$ the comparison is between two closed forms, $\Phi(\hat{a})$ of (A.19) and Ψ of (B.36), subject to the pair being feasible, that is, to joint refusal failing to be an equilibrium and mutual acceptance being one. Writing $\pi_k(A, R)$ for manufacturer k 's profit when it accepts and the other refuses, and similarly for the other profiles, the two-host package is optimal if

$$\Psi > \Phi(\hat{a}), \quad \pi_k(A, R) > \pi^{bench}, \quad \pi_k(A, A) \geq \pi_k(R, A), \quad (\text{A.23})$$

all evaluated at $(\beta^\dagger, \beta^\dagger, a^\dagger)$: three explicit inequalities in closed forms, whose region is located on the reference grid in Online Appendix B.7, together with what the two-host package does there; whether another two-host package wins where this candidate is infeasible is a grid finding. In the single-host range, the closed form (A.19) parallels that of a^{**} , with the bracket of (9) in the role of the bracket of $\Pi^{I'}(0)$; the two fees track each other away from the band and separate inside it, where $\hat{a}$ would be negative. $\square$

A.4 The General Model

Let buyer demands $Q_B^k(p_k, p_\ell, n_S^k, n_S^\ell)$ be symmetric, with devices demand substitutes in prices, the direct price effect dominant, $\partial Q_B^k / \partial n_S^k > 0 > \partial Q_B^k / \partial n_S^\ell$, and the net application effect positive: a common increase in both stocks raises each device's demand, $\nu_k \equiv \partial Q_B^k / \partial n_S^k + \partial Q_B^k / \partial n_S^\ell > 0$ (in the linear model, $\nu_k = u_B > 0$). Let the developer have increasing convex cost C_S and per-buyer benefit u_S ; and assume the two-sided feedback is nonnegative and contracting, $0 \leq \frac{\partial Q_S}{\partial n_B} (\nu_1 + \nu_2) < 1$, together with the self-map condition of Assumption B.1, which is what delivers existence rather than uniqueness alone; and, for the buyer half of (G4), let buyer demand derive from a quasi-linear utility whose marginal value of an application on device k is ν_k per buyer. Then the stage-4 system has a unique solution defining demands D_k, D_S decreasing in own price and, by the net positivity of ν_k , in the fee. Assume further that the pricing subgame is well behaved (unique equilibrium, strategic complementarity with stability, $\partial \hat{p}_k / \partial a \leq 0$, pass-through of perceived benefits below one, and a reduced-form cross-price effect $\partial D_2 / \partial p_1 \geq 0$ once the application stock has adjusted). Under these assumptions:

- (G1) *Separation.* The free-riding argument of Proposition 1 applies verbatim: the platform earns zero and the outcome is $(\beta = 1, a = 0)$; only the monotonicity facts (i)–(ii) are used.

- (G2) *Fee decomposition.* The integrated platform's fee solves (7) as written, with the same interpretation; the sign of a^{**} is not determined at this level of generality.
- (G3) *Harm.* If $dp_k^I/da \leq 0$, then $d\pi_2/da < 0$: the merger harms the rival if and only if the fee rises.
- (G4) *Surplus decompositions.* Equations (A.13)–(A.14) hold with general derivatives, with $1/(1 - 2u_{BS})$ replaced by the general feedback multiplier.

Proofs are those given above, which nowhere use linearity except where flagged; the existence and uniqueness of the stage-4 solution are Lemma B.2 of Online Appendix B.10.

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Online Appendix (Not for Publication)

Creating the Bottleneck: Vertical Integration in Platform Markets

This Online Appendix collects the analysis and the computations supporting the results of the paper. Section B.1 characterizes the admissible parameter set; Section B.2 the geometry of the frontiers $\underline{h}$, $\bar{h}_B$ and $\bar{h}_S$ and the intermediate regions of Propositions 2 and 4; Section B.3 the complements range of Section 5.4 of the paper; Section B.4 the analysis with efficiency gains; Section B.5 a dominant platform; Section B.6 coordination motives and porting costs; Section B.7 the contract space once the cap on revenue sharing is lifted, including the choice between integration and separation; Sections B.8 and B.9 the extensions to N manufacturers and to an integrated rival; Section B.10 the welfare objects, the single-homing micro-foundation, and the lemmas of the general model; Section B.11 the numerical simulations. Throughout, “the paper” refers to the main text; propositions, corollaries and remarks stated here are numbered B.1, B.2 and so on, and those of the paper are cited by their own numbers. Unless otherwise stated, the simulations use $\gamma = 4$, $v = 2$, $r = 1/2$. Replication scripts for the computations and for the figures are available on the authors’ webpages.

B.1 The Admissible Parameter Set

We characterize the set of (u_B, u_S) on which the platform’s maximization problems are strictly concave.

Reference grid. Unless a passage says otherwise, every count reported in this Online Appendix is taken on one grid: step 0.02 in each direction over $[0.02, 1.40]^2$ (70×70), restricted to $\sigma \geq 0$ in the substitutes range and to $\sigma \in (-1, 0)$ in the complements range. On it, the set $\mathcal{A}$ of (A.6) has 1,380, 1,853, 2,252, 2,514, 2,654 and 2,737 points at $\gamma = 1/2$, 1, 2, 4, 8 and 16, the condition $D_2(a^{**}) > 0$ removing none at $\gamma \leq 2$ and 21, 51 and 67 points at $\gamma = 4$, 8 and 16. Subsets (single-host points, taxing points, band points) are named where they are used. Every count is taken at the six values of γ of the reference calibration; the computations confined to $\gamma = 4$ are illustrations and say so (the reference points and the by-channel table of Section B.5, the coarsened and deep sweeps of Section B.7, and the figures). Two passages use a different grid and say so: the map across γ below, on a 110×110 grid over $[0.02, 2]$; and the complements range, which is measured over $[0.02, 1.60]$. Consider the two-instrument problem of Section 5 of the paper (which nests the one-instrument problem of Section 3 as the restriction $\beta = 1$), $\max_{(\beta, a)} \pi^I(\beta, a)$ with $\pi^I(\beta, a) = (p_1^I + r)D_1 + (1 - \beta)rD_2 + aD_S$ evaluated at the pricing equilibrium. Computations show that:

- (i) $\partial^2 \pi^I / \partial \beta^2 = -\frac{4(\sigma + 1)(\sigma + 2)(\sigma(9\sigma + 32) + 32)r^2}{(\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)}$, which is strictly negative;
- (ii) $\partial^2 \pi^I / \partial a^2 = -2H / [(\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)]$, with H as defined in Appendix A.1 of the paper;
- (iii) the determinant of the Hessian equals $-4r^2(\sigma + 1)N_2 / ((\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)^2)$, where $N_2 = 9\sigma^3((u_B + u_S)^2 - 2) + 5\sigma^2(9u_B^2 + 22u_B u_S + 9u_S^2 - 20) + 16\sigma(5u_B^2 + 14u_B u_S + 5u_S^2 - 12) + 16((3u_B + u_S)(u_B + 3u_S) - 8)$.

The Hessian is thus negative definite if and only if the numerator in (ii) is positive and $N_2 < 0$. Comparing the two conditions after the change of variables described below shows that the difference of the two numerators is positive, so that $N_2 < 0$ is *necessary and sufficient* for the concavity of the two-instrument problem. Condition $N_2 < 0$ can be rewritten as

$$\frac{(u_B + u_S)^2 - 2}{(u_S - u_B)^2} < \frac{16 + \sigma(16 + 5\sigma)}{(2 + \sigma)(32 + \sigma(32 + 9\sigma))}. \quad (\text{B.1})$$

The rewriting (B.1) is valid off the balanced diagonal $u_S = u_B$; on the diagonal itself N_2 factors as $2(\sigma + 2)(2u_B^2 - 1)(9\sigma^2 + 32\sigma + 32)$, so $N_2 < 0$ there follows from the feedback condition $2u_B u_S < 1$ alone and the diagonal is admissible throughout. Since the right-hand side of (B.1) is positive, a simple sufficient condition is $(u_B + u_S)^2 < 2$; when $(u_B + u_S)^2 > 2$, (B.1) holds when network effects are sufficiently asymmetric. Note also that, by (ii), $N_2 < 0$ implies $H > 0$: the admissible set characterized here is strictly contained in the set on which the one-instrument problem of Section 3 is strictly concave. The two are therefore kept apart, as (A.6) and (A.7) state: Propositions 2 to 5 live on $\mathcal{A} = \{H > 0, D_2(a^{**}) > 0\}$, while the joint choice of β and a in Section 5 and the contract space of Section 4 live on the smaller $\mathcal{A}^* = \{N_2 < 0, D_2(a^{**}) > 0\}$ (in shorthand, the publication condition of (A.6) and (A.7) understood); for Section 4 the binding requirement is the second-order condition of the separated platform's two-host program, which is $N_2 < 0$ in disguise (Corollary B.3). Importing the stronger condition into the baseline of Section 3 as well would not be conservative but substantive: it empties the region in which the developer gains from the merger at every $\gamma \geq 4$, a region that is nonempty on $\mathcal{A}$. Relative to $\mathcal{A}$, the reference grid loses nothing at $\gamma = 1/2$ and 11% of its points at $\gamma = 16$; the unraveling region, where the substantive results of Section 4 live, loses 3 to 6%.

FRONTIERS OF THE ADMISSIBLE SET. Instead of working with (u_B, u_S) , it is easier to work with (σ, u_S) , using $u_B = (\gamma - \sigma)/(2(1 + \gamma)u_S)$, a $\mathcal{C}^1$ -diffeomorphism on the open quadrant $u_B, u_S > 0$, where $\sigma \in [0, \gamma]$ in the baseline; the endpoint $\sigma = \gamma$ corresponds to the axes and is treated as a limit, the polar cases $u_B = 0$ and $u_S = 0$ being analyzed separately in Sections B.4–B.6. Introducing $x = u_S^2$, condition $N_2 < 0$ becomes $g(x) < 0$ with

$$\begin{aligned} g(x) = & \frac{1}{x(1 + \sigma)(4 + \sigma)^2(4 + 3\sigma)^2} \Big(4x^2(1 + \gamma)^2(4 + 3\sigma)(12 + \sigma(11 + 3\sigma)) \\ & - 4x(1 + \gamma) \Big(128 + \gamma(4 + 3\sigma)(12 + \sigma(11 + 3\sigma)) + \sigma(272 + \sigma(212 + \sigma(73 + 9\sigma))) \Big) \\ & + (\gamma - \sigma)^2(4 + 3\sigma)(12 + \sigma(11 + 3\sigma)) \Big). \end{aligned}$$

The numerator of g is a degree-2 polynomial in x with a strictly positive discriminant, hence two real roots $\underline{x}(\sigma) < \bar{x}(\sigma)$; since the numerator is positive at $x = 0$, both roots are positive for $\sigma < \gamma$ (at the limit $\sigma = \gamma$ the constant term vanishes and $\underline{x}(\gamma) = 0$, the axis), and $g(x) < 0$ if and only if $\underline{x}(\sigma) < u_S^2 < \bar{x}(\sigma)$. Define $\underline{u}_S(\sigma) = \sqrt{\underline{x}(\sigma)}$, $\underline{u}_B(\sigma) = (\gamma - \sigma)/(2(1 + \gamma)\underline{u}_S(\sigma))$, and symmetrically $(\bar{u}_B(\sigma), \bar{u}_S(\sigma))$ from $\bar{x}(\sigma)$. The set $\{N_2 < 0\}$ is then the region between the two parametric curves $\underline{\mathcal{C}} = (\underline{u}_B(\sigma), \underline{u}_S(\sigma))$ and $\bar{\mathcal{C}} = (\bar{u}_B(\sigma), \bar{u}_S(\sigma))$, $\sigma \in [0, \gamma]$; the rival-demand and publication conditions of $\mathcal{A}^*$ remove a further sliver at strongly developer-skewed points (Figure OA-1). The slope of each curve is

$$\frac{\dot{u}_S(\sigma)}{\dot{u}_B(\sigma)} = - \frac{2(1 + \gamma)x(\sigma)}{(\gamma - \sigma) + 2x(\sigma)/x'(\sigma)},$$

and computations show $\underline{x}'(\sigma) < 0 < \bar{x}'(\sigma)$ together with $(\gamma - \sigma) + 2\underline{x}(\sigma)/\underline{x}'(\sigma) > 0$: both curves are strictly downward-sloping in the (u_B, u_S) -plane. At $\sigma = 0$ the roots are $\underline{x}(0), \bar{x}(0) = (8 + 3\gamma \mp 4\sqrt{4 + 3\gamma})/(6(1 + \gamma))$.

INTERIORITY AND PUBLICATION AT THE OPTIMUM. Of the interiority requirements, only one is imposed; the others are verified. On the reference grid (13,529 points with $\sigma \geq 0$ and $H > 0$ over the six values of γ), the rival's demand at a^{**} is the single condition that ever binds: it fails at 139 points, all at $u_S \geq 1.10$ with u_B small, and those points are precisely what the second condition of (A.6) removes. On the remaining 13,390 points of $\mathcal{A}$: the publication condition $u_S n_B^I \geq a^{**}$ holds everywhere (minimal slack 0.005, at the weak-effects corner of $\gamma = 16$), and demands, the application stock, and the rival's markup are strictly positive by construction. One object does change sign: the integrated firm's own device margin $p_1^I + r$ turns negative at 12% of the points, all deep in the developer-skewed range ($u_S \geq 0.78$); this is the loss-leader configuration in which the fee finances below-perceived-cost device pricing. No step of the analysis requires that margin to be positive (the participation and calibration arguments run through the *manufacturers'* markups, which stay positive), so these points are retained. Retaining them does not carry the headline shares of Section 3: on the subset of $\mathcal{A}$ where the margin is non-negative, the shares of points at which the rival, the buyers, the developer and total welfare gain from the merger move by at most six percentage points at every γ (at $\gamma = 4$, buyers gain at 47% of that subset instead of 53% of $\mathcal{A}$, welfare rises at 34% instead of 37%), the loss-leader points being developer-skewed ones where buyers gain (`rep1_frontiers.py`).

The composition of $\mathcal{A}$ moves with product-market competition, and monotonically across the values of γ computed, which is what licenses the statements by regime of γ made in the paper (grid 110×110 over $[0.02, 2]$, $\sigma \geq 0$):

γ	1/2	1	2	4	8	16
points in $\mathcal{A}$	1,907	2,552	3,074	3,431	3,658	3,813
rival and buyers both harmed	21%	25%	32%	42%	55%	69%
neither harmed	41%	40%	38%	35%	31%	24%
<i>exactly one</i> harmed	39%	35%	30%	23%	14%	6%
developer gains under a tax ($u_S \geq \bar{h}_S$)	5.9%	4.9%	2.2%	0.4%	0.1%	0.0%

Every column is monotone in γ , so the two statements the paper makes about regimes of competition, that the disconnect between harm to the rival and harm to consumers is a soft-competition phenomenon and that the configuration in which the developer gains under a tax is another, are read off a comparative static monotone across the values of γ computed rather than asserted at two calibrations. Script: `rep1_frontiers.py`.

INVARIANCE ACROSS THE PROBLEMS OF THE PAPER. The Hessian above does not depend on a or β (the objective is quadratic), and: replacing the per-user benefit r by $r_0 > r$ (efficiency gains, Section 5.1 of the paper) multiplies $\partial^2 \pi^I / \partial \beta^2$ and the determinant by $r_0^2 / r^2 > 0$; adding coordination motives α_B (Section 5.3) shifts only the linear terms of the objective. The concavity part of the admissible set characterized by (B.1) therefore serves for all the two-instrument problems in the paper; interiority is verified extension by extension where it is used.

EXTENSION TO $\sigma \in (-1, 0)$. Section 5.4 of the paper works on the complements range $\sigma \in (-1, 0)$, where the baseline restriction $\beta = 1$ applies and the relevant conditions are

(a) strict concavity of the fee problem, $\partial^2 \Pi^I / \partial a^2 < 0$, and (b) interiority of the optimum: positive demands, applications, and markups at a^{**} . We characterize the resulting set numerically on a grid of step 0.02 over $[0.02, 1.60]$ in each direction, wider than the reference grid because the complements range reaches further out: 1,144, 723, 399, 194, 94 and 42 of the 1,747, 1,218, 765, 438, 236 and 121 grid points with $\sigma \in (-1, 0)$ are admissible at $\gamma = 1/2, 1, 2, 4, 8$ and 16 (`rep2_statistics.py`). The admissible $\sigma < 0$ band hugs the $\sigma = 0$ locus and thins as $2u_B u_S \rightarrow 1$, where the two-sided feedback diverges; interiority (rather than concavity) is the binding requirement near the corners where network effects are strongly asymmetric. Figure OA-1 displays both parts of the admissible set; the characterization is a computational statement at the stated grid resolutions. Two features of this definition deserve to be stated plainly. First, it is not symmetric with the substitutes range: there the only interiority imposed is a positive rival demand at a^{**} , and the loss-leader points at which the integrated firm's own device margin turns negative are retained, no step of that analysis requiring its sign, whereas here full interiority is imposed. Second, the asymmetry is not innocuous. On a finer grid (step 0.01) at $\gamma = 4$, interiority removes 561 of the 1,342 concavity-admissible complements points; at the 493 of them where the remaining objects stay positive, all of which are taxing, the alignment of Proposition B.1 generally fails: buyers gain at 430, the developer at 307, and the rival at exactly the 75 points where $u_B/u_S \leq \psi(|\sigma|)$, as part (i) of that proposition predicts. Proposition B.1 is accordingly a statement about the interior admissible set: where the fee finances below-cost device pricing, the alignment of interests it describes is not to be expected. Script: `refine4.py`. One boundary, central to Proposition B.1(i), can be handled analytically: on the locus $u_B/u_S = \psi(s)$, $s = |\sigma|$, the one-instrument concavity numerator reduces to $(s-4)^2(s-1)(3s-4)(s(3\gamma+2)-2\gamma)/(s(\gamma+1))$, which is negative (so the locus violates admissibility) if and only if $s < s^*(\gamma) \equiv 2\gamma/(3\gamma+2)$. The exclusion extends to the whole developer-skewed side as a polynomial identity. On a σ -slice, $u_B u_S = w \equiv (\gamma+s)/(2(1+\gamma))$ is fixed and $\rho \equiv u_B/u_S$ parameterizes the slice; direct computation gives

$$m_1(a^{**}) H = \frac{(v+r)(3s-4)}{\rho(\gamma+1)} F(\rho), \quad F \text{ linear in } \rho,$$

where $m_1(a^{**})$ is the integrated device margin at the optimum, with $F(0) = 2(\gamma+s)(s^2-6s+6) > 0$ and $F(\psi) = (3\gamma+s)(4-s)(1-s) > 0$. Hence $F > 0$ on $(0, \psi]$ and $m_1(a^{**}) H < 0$ there: every point with $u_B/u_S \leq \psi(s)$ fails concavity ($H \leq 0$) or interiority ($m_1(a^{**}) < 0$), for every $s \in (0, 1)$ and every γ . The split of the excluded segment between the two failures is itself analytic: along a slice H is concave in ρ , and $H'(\psi) \propto A\psi^2 - C = 4w(1-s)(3s^2-16s+16) > 0$, so for $s < s^*$, where $H(\psi) < 0$ by the displayed factorization, the tangent bound at ψ gives $H < 0$ on the entire segment: concavity fails on the whole segment below s^* ; beyond it interiority fails at ψ and near it, while concavity still fails as $\rho \rightarrow 0$, where $H \rightarrow -\infty$. Scripts: `rep1_frontiers.py`, `verifJ.py`.

B.2 Geometry of the Frontiers

This section establishes the properties of the three frontiers used in Propositions 2, 4, 5 and B.5, and analyzes the intermediate regions of the welfare map.

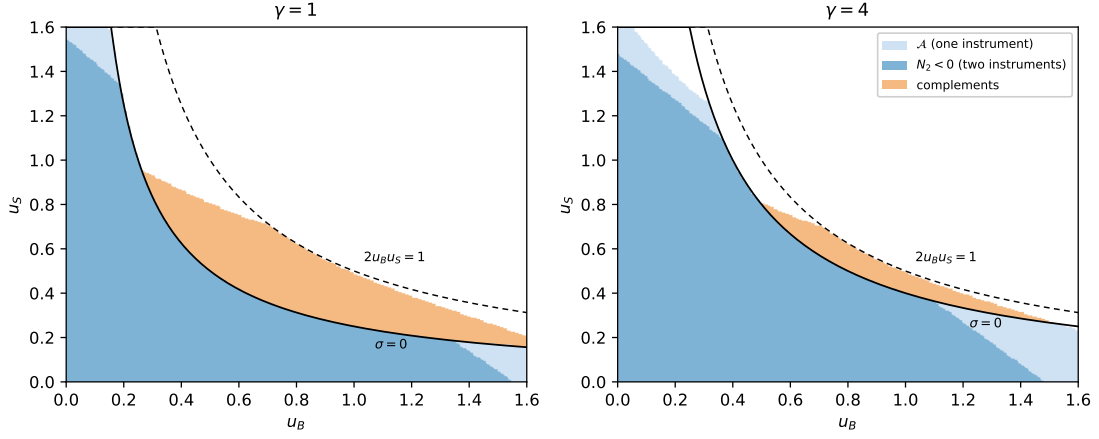


Figure OA-1: The admissible parameter sets for $\gamma = 1$ (left) and $\gamma = 4$ (right). Light blue: the baseline set $\mathcal{A}$ of (A.6), on which Propositions 2 to 5 are stated. Darker blue: the strictly smaller set $\{N_2 < 0\}$ of condition (B.1), required where the integrated firm chooses β and a jointly (Section 5). The light band between them, at the two ends of the substitutes range, is what separating the two sets recovers; it is where the configuration in which the developer gains despite a rising fee, $u_S \geq \bar{h}_S(u_B)$ in Proposition 4(ii), lives. Orange: complements range ($\sigma \in (-1, 0)$, concavity and full interiority at a^{**}). The solid curve is $\sigma = 0$; the dashed curve is $2u_B u_S = 1$.

THE CURVE $a^{**} = 0$ AND THE FRONTIER $\underline{h}$. From the fee first-order condition (Proposition 2 of the paper),

$$a^{**} = \frac{(3\sigma + 4)(v + r) \left[(3\sigma(\sigma(\sigma + 7) + 16) + 32)u_S - 2(\sigma + 2)(3\sigma + 4)u_B \right]}{H},$$

with $H > 0$ on the admissible set. The locus $a^{**} = 0$ can be expressed in (σ, u_S) -coordinates: substituting $u_B = (\gamma - \sigma)/(2(1 + \gamma)u_S)$, it becomes a second-degree polynomial equation in u_S with unique positive root

$$\hat{u}_S^{a^{**}=0}(\sigma) = \sqrt{\frac{(\sigma + 2)(3\sigma + 4)(\gamma - \sigma)}{(\gamma + 1)(3\sigma(\sigma(\sigma + 7) + 16) + 32)}}, \quad (\text{B.2})$$

and $\hat{u}_B^{a^{**}=0}(\sigma) = (\gamma - \sigma)/(2(\gamma + 1)\hat{u}_S^{a^{**}=0}(\sigma))$. The parametric curve $(\hat{u}_B^{a^{**}=0}(\sigma), \hat{u}_S^{a^{**}=0}(\sigma))$, $\sigma \in [0, \gamma]$, has the following properties. First, it always lies within the admissible set (the verification parallels the root comparisons of Section B.1). Second, $(\hat{u}_S^{a^{**}=0})^2 \leq (\hat{u}_B^{a^{**}=0})^2$ for all σ : the curve lies below the 45° line. Third, it runs from $(0, 0)$ (as $\sigma \rightarrow \gamma$) to $(\sqrt{\gamma/(\gamma + 1)}, \frac{1}{2}\sqrt{\gamma/(\gamma + 1)})$ (at $\sigma = 0$). Fourth, its slope equals I/J with $I = 2(\sigma + 2)(3\sigma + 4)(\gamma(3\sigma(\sigma(\sigma + 7) + 16) + 32) + 8) + \sigma(3\sigma(\sigma(11\sigma + 80) + 200) + 640) + 256 > 0$ and $J = (3\sigma(\sigma(\sigma + 7) + 16) + 32)((3\sigma(\sigma + 4) + 8)(3\sigma(\sigma(2\sigma + 9) + 16) + 32) - \gamma(3\sigma(\sigma(\sigma + 7) + 16) + 32))$; the denominator J vanishes when γ equals a threshold that is increasing in σ and equals 4 at $\sigma = 0$. Hence: if $\gamma < 4$ the curve is strictly increasing; if $\gamma > 4$ it is increasing then decreasing, with slope $(\gamma + 4)/(2(4 - \gamma))$ at $\sigma = 0$ (a vertical tangent, as the limit of that expression, in the knife-edge case $\gamma = 4$) and $2(\gamma + 2)(3\gamma + 4)/(3\gamma(\gamma + 7) + 16) + 32$ at $\sigma = \gamma$. For $\gamma > 4$

the parametric curve is therefore nonmonotone in u_B and some vertical lines $u_B = \text{const}$ meet it twice: the frontier is single-valued as a function of u_S , not of u_B . Single-valuedness along horizontal lines follows from the ratio form: $a^{**} > 0 \iff u_B/u_S < R_a(\sigma)$ with $R_a(\sigma) = (3\sigma(\sigma+7)+16)+32)/(2(\sigma+2)(3\sigma+4))$, whose derivative has the positive-coefficient numerator $9\sigma^4 + 60\sigma^3 + 138\sigma^2 + 144\sigma + 64$; since R_a is strictly increasing and σ is strictly decreasing in u_B at fixed u_S , the map $u_B \mapsto u_B/u_S - R_a(\sigma(u_B u_S))$ is strictly increasing and crosses zero at most once on each line $u_S = \text{const}$. It crosses on a line if and only if that line meets the parametric curve, and by the third property above the curve's highest point is $\frac{1}{2}\sqrt{\gamma/(\gamma+1)}$, reached at $\sigma = 0$: the crossing exists exactly for $u_S < \frac{1}{2}\sqrt{\gamma/(\gamma+1)}$, and above that height $a^{**} > 0$ at every admissible point. With the convention $\underline{h} = +\infty$ there, there exists a uniquely defined function $\underline{h} : u_S \mapsto \underline{h}(u_S)$ with

$$a^{**}(u_B, u_S) > 0 \iff u_B < \underline{h}(u_S),$$

and, since the curve lies below the 45° line, $\underline{h}(u_S) > u_S$.

THE WELFARE FACTORIZATIONS. For later reference we reproduce the two factorizations established in Appendix A.2 of the paper. With $\beta = 1$ and a fee moving from 0 to a ,

$$\Delta V_B(a) = a \left(a - \frac{2(\sigma+2)(3\sigma+4)^2(v+r)[(\sigma+2)u_B - u_S]}{K_B} \right) \frac{K_B}{(\sigma+4)^2(3\sigma+4)^2(1-2u_B u_S)^2}, \quad (\text{B.3})$$

$$\Delta V_S(a) = \frac{(\sigma - 2u_S(2u_B + u_S) + 4)^2}{2(\sigma+4)^2(1-2u_B u_S)^2} a \left(a - \frac{4(\sigma+2)u_S(v+r)}{\sigma - 2u_S(2u_B + u_S) + 4} \right), \quad (\text{B.4})$$

with $K_B > 0$ on $2u_B u_S < 1$.

THE FRONTIER $\bar{h}_B$. The buyer factorization (B.3) involves the sign of $(\sigma+2)u_B - u_S$. Using $\sigma = \gamma - 2u_B u_S(1+\gamma)$, the locus $(\sigma+2)u_B = u_S$ solves into

$$u_S = \bar{h}_B(u_B) \equiv \frac{(\gamma+2)u_B}{2(\gamma+1)u_B^2 + 1}.$$

$\bar{h}_B$ is first increasing then decreasing, reaching its maximum $u_S = (\gamma+2)/(2\sqrt{2(\gamma+1)})$ at $u_B = 1/\sqrt{2(\gamma+1)}$; these values satisfy $\sigma \geq 0$ iff $\gamma \geq 2$. In parametric form, $\bar{h}_B$ is $(u_B, u_S) = (\sqrt{(\gamma-\sigma)/(2(\gamma+1)(\sigma+2))}, \sqrt{(\sigma+2)(\gamma-\sigma)/(2(\gamma+1))})$, $\sigma \in [0, \gamma]$, which shows immediately that the curve lies strictly above the 45° line and runs from $(0,0)$ ($\sigma = \gamma$) to $(\sqrt{\gamma/(4(\gamma+1))}, \sqrt{\gamma/(\gamma+1)})$ ($\sigma = 0$); its portion in the admissible set, which loses its top at large γ (at $\gamma = 16$ the rival's demand condition fails at the maximum), suffices for what follows. Since $\underline{h}$ lies below and $\bar{h}_B$ above the 45° line, the region $\{a^{**} \geq 0\} \cap \{u_S \geq \bar{h}_B(u_B)\}$ of Proposition 4(i) is nonempty.

THE FRONTIER $\bar{h}_S$. The developer factorization involves $\sigma - 2u_S(2u_B + u_S) + 4$, which, using $\sigma = \gamma - 2u_B u_S(1+\gamma)$, is positive iff $4 + \gamma - 2u_S^2 - 2u_S u_B(\gamma+3) > 0$, i.e. iff $u_S < \bar{h}_S(u_B)$ with

$$\bar{h}_S(u_B) = \frac{1}{2} \left(\sqrt{2(\gamma+4) + (\gamma+3)^2 u_B^2} - (\gamma+3)u_B \right).$$

The following facts are immediate: (i) $\bar{h}_S$ is strictly decreasing and strictly convex; (ii) $\bar{h}_S(0) = \sqrt{2 + \gamma/2}$; (iii) $\bar{h}_S(u_B) = u_B$ iff $u_B = u_S = 1/\sqrt{2}$, which corresponds to $\sigma = -1$. Hence, for $\sigma > -1$, $\bar{h}_S$ lies strictly above the 45° line. In (σ, u_S) -coordinates the region

$u_S > \bar{h}_S(u_B)$ is $u_S^2 > \tilde{x}_a(\sigma) \equiv (\gamma(\sigma + 2) + 3\sigma + 4)/(2(\gamma + 1))$. This region intersects the admissible set $\mathcal{A}$ of (A.6), but thinly and less so as competition intensifies: on the wide grid (step 0.02 over $[0.02, 2]$ in each direction) it holds at 5.9, 4.8, 2.3, 0.4 and 0.1% of $\mathcal{A}$ at $\gamma = 1/2, 1, 2, 4$ and 8, and nowhere at $\gamma = 16$. Against the smaller two-instrument set it fails outright beyond $\gamma = 4$: $\tilde{x}_a(\sigma) \leq \bar{x}(\sigma)$ holds on the whole range of σ at $\gamma = 1/2$, on 79% of it at $\gamma = 1$, on 21% at $\gamma = 2$ and on none of it from $\gamma = 4$ on, the two coinciding exactly at $\gamma = 4, \sigma = 0$ ($\tilde{x}_a = \bar{x} = 1.2$). This is one reason the baseline propositions are stated on $\mathcal{A}$ and not on the two-instrument set (`rep2_statistics.py`).

INTERMEDIATE REGION: BUYERS. It remains to sign $\Delta V_B(a^{**})$ when $a^{**} \geq 0$ and $(\sigma + 2)u_B - u_S \geq 0$, i.e., in (σ, x) -coordinates with $x = u_S^2$, when $x \in [\underline{x}_a(\sigma), \bar{x}_a(\sigma)]$ where $\underline{x}_a(\sigma) \equiv (\sigma + 2)(3\sigma + 4)(\gamma - \sigma)/((\gamma + 1)(3\sigma(\sigma + 7) + 16) + 32)$ (from (B.2)) and $\bar{x}_a(\sigma) \equiv (\sigma + 2)(\gamma - \sigma)/(2(\gamma + 1))$; $\underline{x}_a < \bar{x}_a$ holds on $[0, \gamma]$ as a polynomial identity (`verifI.py`), so the interval is nonempty. On this interval, the sign of $\Delta V_B(a^{**})$ is the sign of

$$a^{**} - \frac{2(\sigma + 2)(3\sigma + 4)^2(v + r)[(\sigma + 2)u_B - u_S]}{K_B},$$

with $K_B > 0$ defined in Appendix A.2 of the paper. Substituting $u_B = (\gamma - \sigma)/(2(\gamma + 1))u_S$ and clearing the positive factors $v + r$, H and K_B , this expression has the sign of the cubic

$$\begin{aligned} h_1(x) = & (\sigma + 2)^3(3\sigma + 4)^3(\gamma - \sigma)^3 \\ & + 4(\gamma + 1)(\sigma + 2)(3\sigma + 4)x^2[\gamma^2(\sigma(\sigma(23\sigma + 183) + 507) + 584) + 240] \\ & + \gamma(\sigma(\sigma(\sigma(49 - 5\sigma) + 562) + 1618) + 1840) + 736 \\ & + \sigma(\sigma(\sigma(37 - 4\sigma) + 422) + 1184) + 1312 + 512] \\ & - 4(\gamma + 1)^2x^3[\gamma(3\sigma + 4)(\sigma(\sigma + 4)(\sigma(19\sigma + 65) + 88) + 128) \\ & - \sigma(\sigma(\sigma + 3)(\sigma + 8) + 16)(3\sigma(\sigma + 6) + 10) + 16)] \\ & - (\gamma + 1)(\sigma + 2)^2(3\sigma + 4)^2x(\gamma - \sigma)[3\gamma(3\sigma(\sigma + 7) + 16) + 32] \\ & + \sigma(\sigma(\sigma(3\sigma + 61) + 288) + 480) + 256]. \end{aligned}$$

The four endpoint signs $h_1(\underline{x}_a) < 0 < h_1(\bar{x}_a)$ and $h'_1(\underline{x}_a) < 0 < h'_1(\bar{x}_a)$ hold as polynomial identities on $0 \leq \sigma < \gamma$: each quantity is a rational function of (σ, γ) , and after the substitution $\gamma = \sigma + t$, $t > 0$, every coefficient of the expanded numerator has a single sign, as does the denominator (`verifI.py`). Since h'_1 is negative at $\underline{x}_a$ and positive at $\bar{x}_a$, and is a degree-2 polynomial with a single sign change there, h_1 first decreases then increases on the interval, and therefore admits a *unique* root $x_0^B(\sigma) \in (\underline{x}_a, \bar{x}_a)$. The parametric curve $((\gamma - \sigma)/(2(1 + \gamma)\sqrt{x_0^B(\sigma)}), \sqrt{x_0^B(\sigma)})$, $\sigma \in [0, \gamma]$, is injective, and continuously differentiable in σ on $[0, \gamma]$; it divides the intermediate region into two connected components. Since h_1 carries the sign of $\Delta V_B(a^{**})$ and is negative at $\underline{x}_a$ and positive at $\bar{x}_a$, buyers lose on the component adjacent to $\underline{x}_a$, that is, just inside the taxing frontier, and gain on the component adjacent to $\bar{x}_a$, at the developer-skewed end of the slice. This is the exact buyers' frontier inside the intermediate region. Figure 1 of the paper draws $\bar{h}_B$ rather than this curve, and the two run close together: on the reference grid the intermediate region covers 24% of the admissible set at $\gamma = 1/2$, rising to 80% at $\gamma = 16$, while buyers gain on fewer than 2% of its points, so the component on which they gain is a thin band along $\bar{h}_B$ (`rep32_buyer_region.py`).

INTERMEDIATE REGION: DEVELOPERS. When $a^{**} > 0$ and $u_S < \bar{h}_S(u_B)$ (i.e. $x \in [\underline{x}_a(\sigma), \tilde{x}_a(\sigma)]$), the developer loses. Indeed, $\Delta V_S(a^{**}) = 0$ reduces to $h_2(x) = 0$ with

$$\begin{aligned} h_2(x) = & (\sigma + 2)(3\sigma + 4)^2(\gamma - \sigma)(\gamma(\sigma + 2) - \sigma(2\sigma + 7) - 4) \\ & + 2(\gamma + 1)^2(\sigma(\sigma + 4)(\sigma(19\sigma + 65) + 88) + 128)x^2 \\ & - (\gamma + 1)(3\sigma + 4)x(\gamma(\sigma + 2)(3\sigma(3\sigma + 7) + 46) + 88) + \sigma(\sigma(31\sigma + 251) + 720) + 880 + 384, \end{aligned}$$

a strictly convex quadratic with positive discriminant; the endpoint signs $h_2(\underline{x}_a) < 0$ and $h_2(\tilde{x}_a) < 0$ hold as polynomial identities, by the same $\gamma = \sigma + t$ expansion as for h_1 (`verifI.py`), hence $h_2 < 0$ on the whole interval: $\Delta V_S(a^{**}) < 0$ there. Together with the two sufficient conditions established in Appendix A.2 of the paper, this completes the developer part of Proposition 4.

THE WELFARE FRONTIER. Total welfare aggregates the four factorizations already computed: $\Delta W(a) = \Delta V_B(a) + \Delta V_S(a) + \Delta \pi_2(a) + \Delta \Pi^I(a)$, a quadratic in a vanishing at 0. Its quadratic coefficients are, respectively, $K_B/[(\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)^2]$, $T^2/[2(\sigma + 4)^2(1 - 2u_B u_S)^2]$ with $T = \sigma + 4 - 2u_S(2u_B + u_S)$, $(2 + \sigma)(\hat{m}'_2)^2/[2(1 - 2u_B u_S)]$ with $\hat{m}'_2 = -[2u_B(4 + 3\sigma) + 2\sigma u_S]/[(4 + \sigma)(4 + 3\sigma)]$, and $-H/[(\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)]$; the marginal coefficient sums the corresponding linear terms and reduces to $\Delta W'(0) = -2(v + r)N_1/[(\sigma + 4)^2(1 - 2u_B u_S)^2]$ with N_1 as in (A.17) (component identity verified symbolically, `b5j.py`). Substituting $u_B = (\gamma - \sigma)/(2(1 + \gamma)u_S)$, the sign of $\Delta W(a^{**})$ on the taxing range is that of a quartic polynomial in $x = u_S^2$ with coefficients polynomial in (σ, γ, v, r) , generated and evaluated in the replication script. Two properties organize it. First, along the locus $a^{**} = 0$ the numerator of N_1 factors as

$$\frac{(\gamma - \sigma)(\sigma + 1)(\sigma + 2)(\sigma + 4)P(\sigma, \gamma)}{(3\sigma(\sigma(\sigma + 7) + 16) + 32)^2},$$

with $P(\sigma, \gamma) = \gamma(9\sigma^5 + 117\sigma^4 + 564\sigma^3 + 1284\sigma^2 + 1392\sigma + 576) + 27\sigma^5 + 345\sigma^4 + 1626\sigma^3 + 3564\sigma^2 + 3648\sigma + 1408$: every factor is strictly positive on $0 < \sigma < \gamma$, so the merger reduces welfare in a neighborhood of $\underline{h}$, as a polynomial identity. Second, the quartic's constant and cubic coefficients are strictly positive and its linear and quartic coefficients strictly negative, as polynomial identities under the substitution $\gamma = \sigma + t$, so it admits at most three positive roots; on dense σ -grids it changes sign at most once on each admissible taxing slice, always from negative to positive ($\gamma \in \{0.5, 1, 2, 3, 4, 5, 6, 8, 10, 16\}$: 1,200 slices, 793 crossings, all ascending, none multiple; the sign it delivers agrees with a direct evaluation of $\Delta W(a^{**})$ at every one of the 9,664 taxing points of $\mathcal{A}$ on the reference grid, at its six values of γ). The welfare-positive part of the taxing region hugs the strongly developer-skewed admissible boundary and recedes as γ grows: at the same 120 slices per γ it is present on all of them for $\gamma \leq 5$, on 59 at $\gamma = 6$, on 14 at $\gamma = 8$, and on none at $\gamma \in \{10, 16\}$. On the subsidizing side no verification is needed: Proposition 5(i) is analytic ($\Delta W > 0$ at all 3,726 subsidizing points of $\mathcal{A}$ on the reference grid, 411 to 754 per γ , as a check). Scripts: `b5j.py`, `rep11_welfare_sub.py`, `rep14_slices.py`.

CONTINUATION ACROSS $\sigma = 0$. Section 5.4 of the paper uses the fact that the frontier $\underline{h}$ extends to the complements range. The defining bracket of a^{**} , $(3\sigma(\sigma(\sigma + 7) + 16) + 32)u_S - 2(\sigma + 2)(3\sigma + 4)u_B$, has coefficients $3\sigma^3 + 21\sigma^2 + 48\sigma + 32$ and $2(\sigma + 2)(3\sigma + 4)$ that remain strictly positive on $\sigma \in (-1, 0]$ (at $\sigma = -1$ they equal 2 and 2), and $H > 0$ on the admissible set of Section B.1. The locus $a^{**} = 0$ is therefore defined by the same equation (B.2) with $\sigma \in (-1, 0]$, is a $\mathcal{C}^1$ curve in (u_B, u_S) (rational functions of (σ, u_S) with

nonvanishing denominators), and matches the baseline frontier at $\sigma = 0$: the function $\underline{h}$ continues smoothly across the boundary, as claimed in Section 5.4.

ALIGNMENT CASES FOR $\sigma \in (-1, 0)$. Proposition B.1 of the paper asserts that on the admissible complements range the sign of a^{**} is (for γ large) a sufficient statistic for every constituency. The mechanism is geometric: the factorizations (B.3)–(B.4) below extend verbatim to $\sigma \in (-1, 0)$, so misalignment requires a^{**} to exceed a constituency’s threshold: for the developer this is exactly $u_S \geq \bar{h}_S(u_B)$; for buyers it is certain beyond $\bar{h}_B$ and possible below it, at the dividing curve of Section B.2. On the grid the first never happens: $u_S \geq \bar{h}_S(u_B)$ requires u_S close to the feedback boundary $2u_B u_S = 1$, which violates interiority. The second happens only for small γ . On that grid:

γ	1/2	1	2	4	8	16
admissible points	1,144	723	399	194	94	42
of which taxing ($a^{**} > 0$)	565	349	193	90	43	20
taxing points at which the developer gains	0	0	0	0	0	0
taxing points at which buyers gain	209	86	14	0	0	0
taxing points at which welfare rises	41	2	0	0	0	0

The rival and the developer align exactly with the sign of a^{**} at every admissible point and every γ , and so do buyers for $\gamma \geq 4$; for $\gamma \leq 2$ buyers gain at some taxing points (37, 25 and 7% of them at $\gamma = 1/2, 1$ and 2), which are the points where $u_S \geq \bar{h}_B(u_B)$ remains admissible, the low- γ exception flagged in Proposition B.1(iii); at every subsidizing point all three gain. Total welfare follows the same statistic. On the subsidizing side this is analytic; on the taxing side, where the merged entity’s gain is set against the others’ losses, the losses dominate at every admissible point for $\gamma \geq 2$ and at all but 41 and 2 points at $\gamma = 1/2$ and 1, every one of them among the points at which buyers gain: the same low- γ exception seen once more, not a separate one. Two clarifications on status. ‘ γ large’ is operational, not a closed-form threshold: what is needed is that the taxing part of the admissible band stay below the buyers’ dividing curve, which the grid confirms for $\gamma \geq 4$ and which fails partly for $\gamma \leq 2$; and the frontier non-intersections are grid checks at the stated resolutions, not a continuous-domain certificate. The low- γ buyer exceptions also show that complementarity alone does not align buyers with the rival: the strong-competition condition of Proposition B.1(iii) is doing real work. Script: `rep2_statistics.py`.

B.3 Complements

Section 5.4 of the paper relaxes Assumption 1 and lets $\sigma \in (-1, 0)$, the lower bound being where the two-sided feedback diverges ($2u_B u_S \rightarrow 1$). For $\sigma < 0$ device prices become strategic substitutes, with best-response slopes $\sigma/(2(2 + \sigma)) \in (-\frac{1}{2}, 0)$, and the pricing subgame retains a unique equilibrium. The admissible set is not the $\mathcal{A}$ of the substitutes range. There the only interiority imposed is a positive rival demand at a^{**} , and the loss-leader points at which the integrated firm’s own device margin turns negative are retained; here we impose full interiority at a^{**} , which at $\gamma = 4$ removes 561 of the 1,342 concavity-admissible points of the finer grid (step 0.01) of Section B.1. The restriction is not cosmetic: where the fee finances below-cost device pricing the alignment described below generally fails. Every statement here is on that interior admissible set.

The architecture of Section 3 of the paper survives intact. The separation benchmark extends, a fee still hurting both manufacturers and the free-riding argument being untouched, and so does the absence of market power over the rival: the participation constraint still reduces to $\beta^2 \geq 1$ on the whole range $\sigma \in (-1, 0)$.¹⁸ The fee a^{**} is still given by (A.9), and the frontier $u_B = \underline{h}(u_S)$ separating taxation from subsidization continues smoothly across the $\sigma = 0$ boundary. What changes is the welfare map, and it changes by becoming simpler: for γ large the four regions of Figure 1 of the paper collapse into two, separated by that same frontier, which turns back toward the 45° line and meets it exactly at $\sigma = -1$, where $u_B = u_S = 1/\sqrt{2}$ for every γ .

Proposition B.1 (Complements: interests align). *Let $\sigma \in (-1, 0)$.*

- (i) *The merger harms the rival if and only if the fee rises ($a^{**} > 0$).*
- (ii) *The developer gains from the merger if and only if the rival does ($a^{**} < 0$): analytically when $a^{**} < 0$, and on the grid of Online Appendix B.3 when $a^{**} > 0$.*
- (iii) *Buyers gain whenever $a^{**} < 0$; when $a^{**} > 0$ they lose at every admissible point of that grid for $\gamma \geq 4$, and gain at some for $\gamma \leq 2$.*

Proof of Proposition B.1. (i) Write $\sigma = -s$, $s \in (0, 1)$. The bracket of (8) is proportional to $-u_B + \frac{s}{2} \left[\frac{2u_B + u_S}{4-s} + \frac{u_S}{4-3s} \right]$, which is negative iff $u_B(4-2s)(4-3s) > 2s(2-s)u_S$, i.e. iff $u_B/u_S > \psi(s)$ as stated. On the admissible set of Online Appendix B.1 the condition holds everywhere: every point with $u_B/u_S \leq \psi(s)$ fails either the concavity requirement or interiority of the optimum (Online Appendix B.1). (ii)–(iii) The factorizations (A.15)–(A.16) and the positivity of K_B extend verbatim to $\sigma \in (-1, 0)$, and so do the sufficiency arguments of Proposition 4; a negative fee then benefits every constituency. On the taxing side alignment requires the taxing region to satisfy $u_S < \bar{h}_S(u_B)$, which is the developer’s exact frontier, and to lie below the buyers’ dividing curve of Online Appendix B.2, of which $u_S < \bar{h}_B(u_B)$ is only a sufficient condition; both are grid findings, the second holding for $\gamma \geq 4$ and failing for $\gamma \leq 2$ on the part of the taxing region where $u_S \geq \bar{h}_B(u_B)$ remains admissible. Details in Online Appendix B.2. $\square$

Part (i) and the subsidizing halves of parts (ii) and (iii) are identities: on $a^{**} < 0$ the two factorizations of Proposition 4 sign both surpluses on $\sigma \in (-1, 0)$ with no side condition. The taxing halves are verified on the complements grid of Online Appendix B.1, at its six values of γ : the developer’s alignment holds at every admissible point, and buyers’ for $\gamma \geq 4$.

For γ large, the sign of the fee is then a sufficient statistic for the welfare of every constituency: integration either benefits the rival, the buyers and the developer together, or harms them together.¹⁹ Total welfare follows the same statistic. On the subsidizing

¹⁸The derivative of the rival’s profit in β^2 has the sign of $2 + 3\sigma/(4 + \sigma) + \sigma/(4 + 3\sigma)$, which is strictly positive on $(-1, 0)$ and vanishes only in the degenerate limit $\sigma = -1$. One familiar force does flip. Under substitutes a firm that keeps a licensing margin on its rival’s sales competes less hard against it, which is the softening described in Section 5 of the paper; with complements the same margin makes it price its own device *more* aggressively. This matters for the efficiency-gains analysis, not for the baseline studied here.

¹⁹‘Large’ is quantified in Online Appendix B.2: what is needed is that the taxing part of the admissible band stay below the buyers’ dividing curve, of which $\bar{h}_B$ of Proposition 4 is a sufficient condition. For $\gamma \geq 4$ this holds at every admissible grid point; for $\gamma \leq 2$ buyers gain at 37, 25 and 7% of the admissible taxing points at $\gamma = 1/2$, 1 and 2, the exception acknowledged in part (iii).

side it does so analytically: when $a^{**} < 0$, parts (i)–(iii) and the strict profitability of integration make all four components of ΔW positive, exactly as in Proposition 5(i). On the taxing side the merged entity’s gain is set against the others’ losses and the sign is not determined analytically, but the losses dominate at every taxing grid point for $\gamma \in \{2, 4, 8, 16\}$, and at all but 41 of 565 and 2 of 349 at $\gamma = 1/2$ and 1, every exception lying among the low- γ points where buyers gain (Online Appendix B.2).

B.4 Efficiency Gains

This section supports Section 5.1 of the paper. With efficiency gains, the integrated system yields the per-user benefit $r_0 > r$ while the outside option still yields r ; let $\Delta = r_0 - r$. The integrated platform solves

$$\max_{(\beta, a)} \pi^I(r_0, \beta r_0, a) \quad \text{s.t.} \quad \pi_2^I(\beta r_0, r_0, a) \geq \pi_2^{out}(r, r_0, a), \quad 0 \leq \beta \leq 1, \quad (\text{B.5})$$

where π_2^I is the rival’s profit when hosted at share β and π_2^{out} its profit under self-supply at benefit r , the integrated platform keeping benefit r_0 and the fee a applying in both configurations.

THE PARTICIPATION THRESHOLD $\underline{\beta}(r_0)$. In the linear model the rival’s profit is proportional to the square of its equilibrium markup, $\pi_2 = [(2 + \sigma)/(2(1 - 2u_B u_S))] \hat{m}_2^2$, in every configuration. In the hosted configuration the markup is $\hat{m}_2^I = \hat{p}_2 + \beta r_0$ with $\hat{p}_2$ the equilibrium price against the accommodating integrated firm; in the outside configuration it is $\hat{m}_2^{out} = \hat{p}_2 + r$ against the non-accommodating one. Both markups are linear in the fee with the *same* slope $-(2u_B(4 + 3\sigma) + 2\sigma u_S)/((4 + \sigma)(4 + 3\sigma))$: the fee shifts the two configurations’ markups equally and cancels from the comparison. Equating the (positive) markups then yields a condition linear in β , whose solution is

$$\underline{\beta}(r_0) = 1 - \left(1 - \frac{r}{r_0}\right) \frac{8 + \sigma(8 + \sigma)}{8(1 + \sigma)}, \quad (\text{B.6})$$

so that $\underline{\beta}(r_0) < 1$ iff $r_0 > r$, decreasing in r_0 , independent of a . This is the floor (11) of the paper, with premium $\Delta = r_0 - r$ measured against r_0 and $\theta = \Lambda/(8(1 + \sigma))$; the same symbol $\underline{\beta}$ is used for it in every premium environment below, the premium and its scale factor changing with the source. (Symbolic verification: script `rep4_efficiency.py`.)

THE PRICING POLICY: BINDING CONSTRAINTS. Let (β^*, a^*) denote the unconstrained optimum of (B.5) (first-order conditions in both instruments), $\bar{a}$ the optimal fee when $\beta = \underline{\beta}$ is imposed, and $\bar{a}(r_0)$ the optimal fee when $\beta = 1$ is imposed. The concavity conditions are those of Section B.1. Three curves organize the solution in the (u_B, u_S) -plane.

Curve $\beta^ = 1$.* Imposing $\beta = 1$ reduces (B.5) to the fee-only problem of Section 3 with r replaced by r_0 ; the locus $\beta^* = 1$ is where the β first-order condition of the full two-instrument program vanishes at that restricted solution, a second-degree equation in u_S with positive root

$$\hat{u}_S^{\beta^*=1}(\sigma) = \sqrt{X_{cap}(\sigma)}, \quad \hat{u}_B^{\beta^*=1}(\sigma) = \frac{\gamma - \sigma}{2(\gamma + 1)\hat{u}_S^{\beta^*=1}(\sigma)}, \quad (\text{B.7})$$

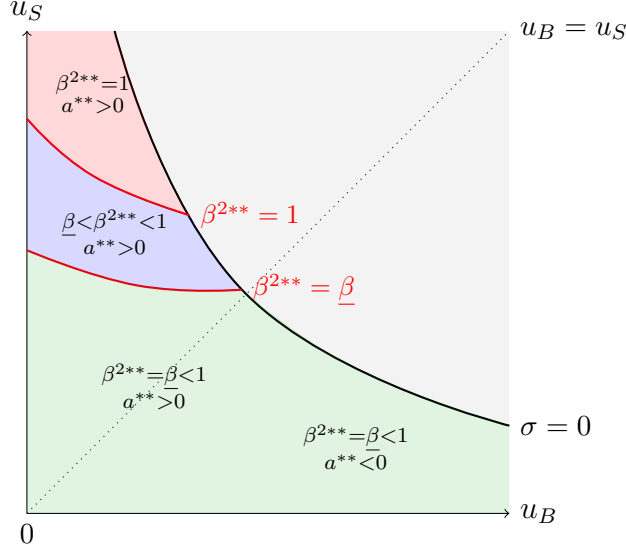


Figure OA-2: The integrated platform's pricing policy with efficiency gains ($r_0 > r$), in the (u_B, u_S) plane. In the red region the cap $\beta^2 \leq 1$ binds; in the blue region no constraint binds; in the green region the rival's participation constraint binds.

with X_{cap} given by (B.19). The curve is the same in both readings of the model: by Proposition B.3(iii) it is independent of v , r and the size of the gain. One shows: the curve lies strictly above the 45° line; it is strictly downward-sloping (its slope is $-C/D$ with $C, D > 0$ given by polynomial expressions in (σ, γ)); it starts and ends inside the admissible set, since $(\hat{u}_S^{\beta^*=1}(0))^2 = (8 + 3\gamma)/(6(1 + \gamma)) \in [\underline{x}(0), \bar{x}(0)]$ and $(\hat{u}_S^{\beta^*=1}(\gamma))^2 = 1 + (\gamma + 4)/(\gamma(3\gamma + 11) + 12) < \bar{x}(\gamma)$. Above this curve the cap $\beta \leq 1$ binds.

Curve $\beta^ = \underline{\beta}$.* No continuity argument is needed here, and none is used: by Proposition B.3(i), which is this program under the relabeling $(r_0, r) \mapsto (r, r - \delta)$, the reduced derivative φ is affine and strictly decreasing, so the participation constraint binds if and only if $\varphi(\underline{\beta}r_0) \leq 0$: one explicit inequality, valid at every gain. Its zero set, in (σ, x) -coordinates with $x = u_S^2$, defines $\hat{u}_S^{\beta^*=\underline{\beta}}(\sigma)$ and the associated parametric curve, which coincides with (B.7) in the limit $r_0 \rightarrow r$, where $\underline{\beta} \rightarrow 1$. Its slope at $r_0 = r$ equals $-A/B < 0$ with $A, B > 0$ polynomial. The two curves cannot cross when $r_0 > r$ (a crossing would force $\underline{\beta} = 1$), and comparing them at $\sigma = \gamma$ shows that $\beta^* = \underline{\beta}$ lies below $\beta^* = 1$. Below the curve $\beta^* = \underline{\beta}$ the participation constraint binds.

Conclusion. In the (u_B, u_S) -plane: above the curve (B.7), the solution is $\beta^{**} = 1$ with $a^{**} = \bar{a}(r_0) > 0$; between the two curves, the solution is interior, $(\beta^{**}, a^{**}) = (\beta^*, a^*)$; below the curve $\beta^* = \underline{\beta}$, the solution is $\beta^{**} = \underline{\beta}$ with $a^{**} = \bar{a}$; Figure OA-2 draws the three regions. The map is drawn under the maintained restriction

$\underline{\beta}(r_0) \geq 0$, which by (B.6) is $\Delta \leq 8r(1 + \sigma)/\sigma^2$ (at the polar boundary $u_B = 0$, $\Delta \leq 8r(1 + \gamma)/\gamma^2$); for larger gains the lower bound $\beta \geq 0$ binds and the participation constraint goes slack, which (B.18) covers as the lower end of the admissible range rather than as a separate regime.

Curve $\bar{a} = 0$. Tedious computations show that $\bar{a} \geq 0$ reduces, in (σ, x) -coordinates, to an inequality linear and decreasing in x ; the resulting frontier $\tilde{x}(\sigma)$ has an explicit (unwieldy) closed form, linear in (v, r, Δ) , reported in the replication script. The associated parametric curve lies below the 45° line, and for Δ small it has the shape of the

$a^{**} = 0$ locus of Section B.2, upward-sloping for $\gamma \leq 4$ and increasing then decreasing for $\gamma > 4$ (both properties hold at $\Delta = 0$, where the curve is that locus, and extend by continuity). Above it the fee is a tax, below it a subsidy. A similar argument yields a downward-sloping curve $\beta^* r_0 = r$ separating the region where the rival's *perceived* benefit rises with integration from the region where it falls.

POLAR CASE $u_S = 0$ (BUYER-SKEWED EFFECTS). When $u_S = 0$ the platform extracts the full premium: $\beta^{**} = \underline{\beta}$ and $a^{**} = \bar{a}$, with

$$\bar{a} = - \frac{u_B \left(16(v + r_0)(2 + \gamma)(4 + 3\gamma) + (16 + 3\gamma(4 + \gamma))(8 + \gamma(8 + \gamma))\Delta \right)}{8(4 + 3\gamma)((4 + \gamma)^2 - 2u_B^2(2 + \gamma))} < 0,$$

a subsidy to developers financed by taxing the buyer side; concavity requires $u_B \leq \bar{u}_B = \sqrt{2(2 + \gamma)(32 + \gamma(32 + 9\gamma)) / ((4 + 3\gamma)(12 + \gamma(11 + 3\gamma)))}$. *Rival*. $\Delta\pi_2 = \pi_2^I(\beta r_0, r_0, \bar{a}) - \pi_2^S$ has the sign of a function $f(\beta, \bar{a})$ whose numerator is increasing in u_B , negative at $u_B = 0$ (where the merger is pure extraction) and positive at $u_B = \bar{u}_B$: there is a unique cutoff $\hat{u}_B \in (0, \bar{u}_B)$ such that the rival loses iff $u_B < \hat{u}_B$. For small Δ the cutoff is small: the merged firm subsidizes the developer there, and the rival gains from the larger application stock once buyer-side effects are strong, even though its retained benefit $\beta r_0 = r - \Delta\sigma^2/[8(1 + \sigma)]$ lies slightly below r . *Buyers*. $dV_B^I(\beta r_0, \bar{a})/du_B > 0$ (the direct effect and the induced fee effect are both positive, and $d\beta/d\bar{u}_B = 0$), and $\Delta V_B > 0$ at $u_B = \bar{u}_B$: buyers gain except possibly in a neighborhood of $u_B = 0$, where the sign is that of a quadratic in Δ (positive for Δ small). *Developer*. With $u_S = 0$, $V_S = a^2/2$, and since separation entails $a = 0$, $\Delta V_S = \bar{a}^2/2 \geq 0$: the developer always gains.

POLAR CASE $u_B = 0$ (DEVELOPER-SKEWED EFFECTS). When $u_B = 0$, concavity requires $u_S < \bar{u}_S$ with $\bar{u}_S^2 = 2(2 + \gamma)(32 + \gamma(32 + 9\gamma)) / ((4 + 3\gamma)(12 + \gamma(11 + 3\gamma)))$, and the constraint pattern follows the three regimes above, with explicit thresholds $\hat{u}_S^{\beta^*=\underline{\beta}}(\gamma) < \hat{u}_S^{\beta^*=1}(\gamma)$:

$$(\beta^{**}, a^{**}) = \begin{cases} (\underline{\beta}, \bar{a}) & \text{if } u_S \leq \hat{u}_S^{\beta^*=\underline{\beta}}(\gamma), \\ (\beta^*, a^*) & \text{if } \hat{u}_S^{\beta^*=\underline{\beta}}(\gamma) < u_S \leq \hat{u}_S^{\beta^*=1}(\gamma), \\ (1, \bar{a}(r_0)) & \text{if } u_S > \hat{u}_S^{\beta^*=1}(\gamma), \end{cases}$$

with, at $u_B = 0$,

$$\begin{aligned} \underline{\beta} &= \frac{8r(1 + \gamma) - \gamma^2\Delta}{8(1 + \gamma)(r + \Delta)}, \\ \bar{a} &= \frac{u_S[8(r + v)(4 + 3\gamma)(32 + 3\gamma(16 + \gamma(7 + \gamma))) + (512 + \gamma(4 + \gamma)(256 + \gamma(4 + \gamma)(20 - 9\gamma)))\Delta]}{8(4 + \gamma)^2(4 + 3\gamma)^2 - 16u_S^2(32 + \gamma(64 + \gamma(40 + 7\gamma)))}, \\ \beta^* &= \frac{v(4 + 3\gamma)(-16 - 3\gamma(4 + \gamma) + u_S^2(12 + \gamma(11 + 3\gamma))) + (r + \Delta)(64 + \gamma(96 + \gamma(52 + 9\gamma)))}{(2(2 + \gamma)(32 + \gamma(32 + 9\gamma)) - u_S^2(4 + 3\gamma)(12 + \gamma(11 + 3\gamma)))(r + \Delta)}, \\ a^* &= \frac{u_S(4 + 3\gamma)(12 + \gamma(11 + 3\gamma))(r + v + \Delta)}{2(2 + \gamma)(32 + \gamma(32 + 9\gamma)) - u_S^2(4 + 3\gamma)(12 + \gamma(11 + 3\gamma))}, \\ \bar{a}(r_0) &= \frac{u_S(4 + 3\gamma)(32 + 3\gamma(16 + \gamma(7 + \gamma)))(r + v + \Delta)}{(4 + \gamma)^2(4 + 3\gamma)^2 - 2u_S^2(32 + \gamma(64 + \gamma(40 + 7\gamma)))}. \end{aligned}$$

Rival. $\pi_2^I = 2(2 + \gamma)(v(4 + 3\gamma) - au_S\gamma + r_0(4(1 + \gamma)\beta - \gamma))^2 / ((4 + \gamma)^2(4 + 3\gamma)^2)$ and $\pi_2^S = 2(r + v)^2(2 + \gamma)/(4 + \gamma)^2$. Tracking $\pi_2^I(\beta^{**}r_0, r_0, a^{**})$ across the three regimes

(decreasing in u_S in the first, increasing in the second, decreasing in the third, and $\Delta\pi_2 < 0$ at $u_S = 0$) shows that the set where the rival gains is an interval $[\underline{u}_S^{M_2}, \bar{u}_S^{M_2}]$ around $\hat{u}_S^{\beta^*=1}(\gamma)$, nonempty iff $\Delta\pi_2 > 0$ at $u_S = \hat{u}_S^{\beta^*=1}(\gamma)$, that is, iff the efficiency gain is large enough relative to the fee increase. *Buyers.* $\Delta V_B(\beta^{**}r_0, a^{**})$ is increasing in u_S , positive at $u_S = \bar{u}_S$, and its sign at $u_S = 0$ is that of an explicit quadratic in Δ (positive for Δ small): there is a cutoff $\tilde{u}_S^B$ such that buyers gain iff $u_S \geq \tilde{u}_S^B$. *Developer.* ΔV_S has the sign of $2u_S(v(2+\gamma) + r_0(1+\beta(1+\gamma))) - a(4+\gamma - 2u_S^2) - 2u_S(2+\gamma)(v+r)$; it is negative in the first regime and increasing in u_S in the third, with a sign at $u_S = \bar{u}_S$ governed by Δ . These two facts do not by themselves pin the sign pattern in the middle regime, so we record the scan: at $\gamma = 4$, $\Delta \in \{0.3, 0.6\}$, the expression stays negative on the whole domain (the would-be cutoff is degenerate at $\bar{u}_S$: the developer never gains in the polar case); at calibrations where it does turn positive, the crossing was unique at every one checked. The cutoff description is thus a computational statement at the stated calibrations.

SIMULATIONS. Figure OA-9 in Section B.11 maps $\Delta\pi_2$ and ΔV_B at the constrained optimum for $\Delta \in \{0.3, 0.6\}$: rival harm is unchanged as Δ grows while buyer harm recedes ($\Delta\pi_2 < 0$ at 75% of the admissible grid at both $\Delta = 0.3$ and $\Delta = 0.6$; $\Delta V_B < 0$ at 44% and 35%), and rival harm is undone in the buyer-skewed, subsidizing range; these are the findings that organize Section 5.1 of the paper.

B.5 A Dominant Platform

This section supports Section 5.2 of the paper. The outside option yields $r - \delta$, $\delta \in (0, r)$, rather than r : the platform's advantage exists *before* any merger, so the benchmark against which integration must be judged is no longer the free-riding outcome of Proposition 1. We characterize the two sides of the comparison, for every δ , and then compare them. The benchmark is the best package the separated platform can post, and that package is generally asymmetric: the platform buys one manufacturer's adoption with part of the premium and holds the other at its outside value.

Throughout, $\Lambda \equiv 8 + \sigma(8 + \sigma)$, $\kappa \equiv \Lambda/(2u_B(3\sigma + 4))$, $\bar{\delta} \equiv u_S(v + r)/(u_B + u_S)$, and, with $M \equiv 1 - 2u_B u_S$, every statement is made on the set $\mathcal{A}^*$ of Appendix A.1 of the paper, on which $N_2 < 0$ and both two-instrument problems are strictly concave.

The section is organized as follows. Section B.5.1 derives, from the adoption game, the two constraints the platform faces and with them its feasible set. Section B.5.2 characterizes the benchmark policy and gives the closed form of each regime. Section B.5.3 reads that policy against the case $\delta = 0$, in which the platform earns nothing, and locates the band in which only integration taxes the developer. Section B.5.4 gives the integrated policy, Section B.5.5 the effect of the premium on separation welfare, and Section B.5.6 the comparison of the two. Section B.5.7 collects the verification, the quantification and the simulations. Propositions B.2 to B.4 and their corollaries are analytic; every quantitative statement names its grid and its script.

B.5.1 The Adoption Game and the Two Constraints

Throughout, π_δ^{bench} denotes a manufacturer's profit in the symmetric duopoly at perceived benefit $r - \delta$ with the fee dead (the joint-rejection outcome), and configurations share the application stock as in Section 2. The platform posts manufacturer-specific shares,

and we label the manufacturers so that $\beta^1 \geq \beta^2$. The benchmark policy solves

$$\max_{\beta^2 \leq \beta^1 \leq 1, a} (1-\beta^1)r D_1 + (1-\beta^2)r D_2 + a q_S \quad \text{s.t.} \quad \beta^2 r \geq r - \delta, \quad a \leq a_{kill}(\beta^1, \delta), \quad (\text{B.8})$$

where

$$a_{kill}(\beta, \delta) = \frac{(\beta r - (r - \delta))(8 + \sigma(8 + \sigma))}{2u_B(4 + 3\sigma)} \quad (\text{B.9})$$

is the fee at which a lone adopter at share β is just indifferent, so that the second constraint is what keeps joint refusal from being an equilibrium, and it bears on the better-treated manufacturer alone. For $a < 0$ the subsidy is paid under joint rejection as well, adoption is then governed by the floor alone, and the floor implies $a_{kill} \geq 0 > a$, so the feasible set is as displayed. The next paragraph derives both constraints and shows that they exhaust the adoption game's requirements.

The benchmark program. Joint rejection is an equilibrium of the adoption game iff no manufacturer gains from unilateral adoption. A lone adopter at (β^k, a) earns the host-one profit against a rival at $r - \delta$; equating it to π_δ^{bench} and solving for the fee gives the affine threshold (B.9), positive whenever $\beta^k r > r - \delta$: the premium itself finances the killing of joint rejection, with no need of the $\beta > 1$ rebates of Proposition 6. It suffices that *one* manufacturer's adoption alone beat refusal, and the platform buys one: for $a \leq a_{kill}(\beta^1, \delta)$ joint rejection is not an equilibrium. Both-accept is then an equilibrium iff $\beta^k r \geq r - \delta$ for each k (by the markup monotonicity (A.21), a hosted manufacturer's profit crosses its walk-off value exactly at $\beta^k r = r - \delta$, the fee being active in both positions), and the second manufacturer, held exactly at $r - \delta$, accepts once its rival is on the system, by the acceptance convention of Section 2.3; a configuration in which it refuses is not an equilibrium, since it would accept, and both-accept is the selected outcome. On this route the platform solves (B.8). The other route, a package at which joint refusal survives but the manufacturers' joint profit under adoption is at least $2\pi_\delta^{bench}$, so that the selection rule retains adoption, never does better: for symmetric packages this is the compensated locus of Section B.5.3, which lies inside the feasible set, and for asymmetric packages it is a grid finding, the enumerated adoption game returning the value of (B.8) at every point of the reference grid coarsened to step 0.1, at the six values of γ and three premia (Section B.5.7). Restricting the platform to a common share, $\beta^1 = \beta^2$, defines the *symmetric restriction* of the program, which hands the premium back to both manufacturers to buy one adoption; the benchmark improves on it strictly at most points where it taxes, by a median of 3 to 12% of the platform's profit and up to 47% (Section B.5.7).

B.5.2 The Separation Benchmark

Write $x_k \equiv \beta^k r - (r - \delta)$ for the part of the premium the platform hands back to manufacturer k , with $x_1 \geq x_2$. The benchmark policy solves (B.8), which in these variables is the maximization of a quadratic Π_δ on the constraint set

$$\mathcal{P}_\delta \equiv \{(x_1, x_2, a) : 0 \leq x_2 \leq x_1 \leq \delta, \quad a \leq \kappa x_1\}, \quad (\text{B.10})$$

which is convex, and on which the fee is bounded above but not below.

Proposition B.2 (The separation benchmark under dominance). *On $\mathcal{A}^*$, and for every $\delta \in (0, r)$:*

(i) Π_δ is strictly concave on $\mathcal{P}_\delta$. The Kuhn–Tucker conditions are therefore necessary and sufficient, and the benchmark is the unique point of $\mathcal{P}_\delta$ at which the multipliers of the four constraints are non-negative.

(ii) It is one of the following policies, the first four in closed form:

	shares	fee	
floor	$x_1 = x_2 = 0$	$a = \min\{a_{sub}, 0\}$	no fee, or subsidizing
taxing edge	$x_1 = x_1^* \in (0, \delta), x_2 = 0$	$a = \kappa x_1^*$	taxing, M_2 at its floor
cap corner	$x_1 = \delta, x_2 = 0$	$a = \kappa \delta$	at the cap, M_2 at its floor
seam	$x_1 = x_2 = x^\circ$	$a = a^\circ < \kappa x^\circ$	no constraint binding
residual faces	$0 < x_2 \leq x_1$		both handed back

with a_{sub} , x_1^* and (x°, a°) given by (B.16), (B.14) and (B.12), and the region on which each is optimal given by the signs of the associated multipliers. The residual faces are those on which the second manufacturer is handed back part of the premium as well, $0 < x_2 \leq x_1$: the cap with the fee constraint slack (both manufacturers at $\beta = 1$, the monetizing regime), the taxing edge or the cap corner with $x_2 > 0$. They are solved by the same enumeration and occupy, on the reference grid, 1 to 7% of the admissible set at $\delta = 0.1$ and 4 to 17% at $\delta = 0.3$, the share rising with γ .

(iii) As $\delta \rightarrow 0$ the interior policies vanish and the benchmark converges to two branches, separated by the ratio $R_\Phi(\sigma) = \Lambda/(4 + 3\sigma)$ of Proposition 6:

$$\begin{aligned} \beta^1 = 1, \quad \beta^2 r = r - \delta, \quad a = \frac{\delta \Lambda}{2u_B(3\sigma + 4)} & \quad \text{if } u_B/u_S < R_\Phi(\sigma), \\ \beta^1 r = \beta^2 r = r - \delta, \quad a = 0 & \quad \text{if } u_B/u_S > R_\Phi(\sigma). \end{aligned}$$

On each branch the policy is exactly affine in δ , up to the premium at which the regime changes.

Proof. (i) The Hessian of Π_δ in (x_1, x_2, a) is that of the two-host program of Appendix A.3: its first diagonal entry is $-(\sigma + 2)\Lambda/((\sigma + 4)(3\sigma + 4)M) < 0$, its leading 2×2 minor is $4(\sigma + 1)(\sigma + 2)^2/((\sigma + 4)(3\sigma + 4)M^2) > 0$, and its determinant is $-4(\sigma + 1)(\sigma + 2)^2\Delta_\delta/((\sigma + 4)^2(3\sigma + 4)M^3)$, negative if and only if $\Delta_\delta > 0$, which holds on $\mathcal{A}^*$ by the identity (B.13) (`rep35_dominance_asym.py`, test 1; the script is cited as `rep35` below). A strictly concave programme over a convex set has a unique maximizer, at which the Kuhn–Tucker conditions hold; conversely they are sufficient. (ii) The Lagrangian carries four multipliers, for $a \leq \kappa x_1$, $x_1 \leq \delta$, $x_2 \geq 0$ and $x_2 \leq x_1$, and the sixteen configurations of binding constraints reduce to the rows stated: $x_2 = 0$ together with $x_2 = x_1$ is the floor, on which the fee constraint reads $a \leq 0$, so the best fee is a_{sub} if $a_{sub} \leq 0$ and 0 otherwise; $x_2 = 0$ with the fee constraint is the taxing edge; $x_2 = 0$ with the cap and the fee constraint is the cap corner; no constraint binding is the seam, symmetric by the symmetry of Π_δ in (x_1, x_2) ; the cap with $x_2 = 0$ and the fee constraint slack is never optimal on the reference grid, for the reason given below; and the remaining configurations, all with $x_2 > 0$, are the residual faces. The closed forms and the multiplier signs are derived below. (iii) As $\delta \rightarrow 0$ the interval $[0, \delta]$ shrinks to a point, so no interior x_k survives and the fee constraint binds together with the floor or with the cap. By (B.14), $x_1^* > 0$ as $\delta \rightarrow 0$ if and only if $u_S\Lambda - u_B(3\sigma + 4) > 0$, that is $u_B/u_S < R_\Phi(\sigma)$; x_1^* is then truncated to δ , which is $\beta^1 = 1$, the fee constraint binds at $a = \kappa\delta = a_{kill}(1, \delta)$, and

$x_2 = 0$ since the derivative of Π_δ in x_2 at that corner, whose sign is given in (B.15), is negative for δ small. Otherwise $x_1^* \leq 0$, the floor binds, and $a_{sub} > 0$ for δ small enough (since $a_{sub} \rightarrow (\sigma + 2)u_S(v + r)/(\sigma + 4 - 4u_B u_S) > 0$), so the fee is truncated to zero. Affinity in δ is immediate on both branches, the policy being $(\beta^1, \beta^2, a) = (1, 1 - \delta/r, \kappa\delta)$ and $(1 - \delta/r, 1 - \delta/r, 0)$ respectively. $\square$

Two comments. First, the dividing line of (iii) is the frontier h_Φ of Proposition 6: as $\delta \rightarrow 0$ the dominant platform taxes the developer exactly where the single-host package of Section 4 was profitable, and for the same reason. That package bought one manufacturer's adoption with a rebate above the cap and taxed the developer; here the premium pays for the adoption, since a share of r against an outside option of $r - \delta$ is a rebate of δ , and the fee $\kappa\delta$ it buys is what the cap allows. The symmetric restriction, which pays for the adoption twice, taxes only on $u_B/u_S < R_\delta(\sigma) \equiv \Lambda/(2(3\sigma + 4)) = R_\Phi/2$, a ratio that reappears below as the factor by which the fee that kills joint refusal exceeds the fee that merely compensates; the chain of ratio frontiers of the paper reads $R_p < R_\delta < R_J < R_\Phi < R_a$ for $\sigma > 0$, with $R_J - R_\delta = \sigma^2/(4(3\sigma + 4))$, and at $\sigma = 0$ the first three equal 1 and the last two equal 2. Second, the dividing line has a direct reading. The fee depresses device demand at rate u_B and yields at rate u_S ; the platform therefore monetizes the developer where the developer side is the more responsive one, and where buyers are, a fee would cost more in device demand than the share concession it requires, so the platform lives on the premium alone.

The seam. With no constraint binding the optimum is the unconstrained stationary point of Π_δ , symmetric by the symmetry of Π_δ in (x_1, x_2) , so that $x_1 = x_2 = x^\circ$; the two first-order conditions in (x, a) are linear and solve jointly in closed form. Writing

$$\Delta_\delta \equiv 2(\sigma + 4) - 8u_B u_S - (\sigma + 2)(u_B + u_S)^2, \quad (\text{B.11})$$

the interior stationary point is

$$a^\circ = \frac{(v + r)(\sigma + 2)(u_S - u_B)}{\Delta_\delta}, \quad x^\circ = \frac{(u_B + u_S)a^\circ - (v + r) + 2\delta}{2}, \quad (\text{B.12})$$

the second expression being the x first-order condition displayed below solved for x . The Hessian has $\partial^2 \Pi / \partial x^2 = -4(\sigma + 2)/[(\sigma + 4)(1 - 2u_B u_S)] < 0$ and determinant $4(\sigma + 2)\Delta_\delta/[(\sigma + 4)^2(1 - 2u_B u_S)^2]$, so that $\Delta_\delta > 0$ is the second-order condition. It holds throughout: writing N_2 for the concavity polynomial of Section B.1,

$$(18\sigma^3 + 100\sigma^2 + 192\sigma + 128)\Delta_\delta = -2(\sigma + 4)N_2 + 2(13\sigma^3 + 64\sigma^2 + 112\sigma + 64)(u_B - u_S)^2 \quad (\text{B.13})$$

identically, both bracketed polynomials having positive coefficients, so that $N_2 < 0$ implies $\Delta_\delta > 0$: the benchmark programme is strictly concave on the whole admissible set, and no point has to be discarded as a saddle. The interior point is feasible if and only if

$$(v + r) - 2\delta < (u_B + u_S)a^\circ < v + r \quad \text{and} \quad a^\circ < \kappa x^\circ,$$

the first pair being $0 < x^\circ < \delta$ written out. Its upper half is the negation of the condition $a(u_B + u_S) \geq v + r$ under which both manufacturers are worth keeping at the cap, which locates the seam exactly: the cap has stopped binding and the fee constraint has not started. Two features are worth recording. The sign rule is the one of Lemma B.1, since $\Delta_\delta > 0$ makes $\text{sign}(a^\circ) = \text{sign}(u_S - u_B)$; when $2\delta < v + r$, as at every calibration we

use, the lower bound above forces $a^\circ > 0$: the seam is developer-skewed and it taxes, but strictly below a_{kill} , so it is not divide-and-conquer.

Where the fee constraint binds with the second manufacturer at its floor, that is along $x_2 = 0$, $a = \kappa x_1$, the objective is a strictly concave quadratic in x_1 alone, its second derivative being $-\Lambda(\Lambda - 2u_B u_S(5\sigma + 6))/(4u_B^2(3\sigma + 4)^2 M)$, negative on the feedback set since $2u_B u_S(5\sigma + 6) < 5\sigma + 6 < \Lambda$; it is maximized at

$$x_1^* = \frac{2u_B(\sigma + 2)(3\sigma + 4)}{(\sigma + 4)\Lambda(\Lambda - 2u_B u_S(5\sigma + 6))} \left((v+r)[u_S\Lambda - u_B(3\sigma + 4)] - \delta[u_S\Lambda + \sigma(\sigma + 2)u_B] \right) \quad (\text{B.14})$$

truncated to $[0, \delta]$ (**rep35**, test 2). The benchmark fee is then $a_{sep} = a_{kill}(\beta^{1*}, \delta) = x_1^* \Lambda / (2u_B(3\sigma + 4))$. On this edge the favored manufacturer earns exactly π_δ^{bench} , being indifferent between adopting alone and joint refusal, and the other earns strictly less, held at its floor with the fee active (**rep35**, test 6). The symmetric restriction's taxing branch, which hands x back to both manufacturers along $a = \kappa x$, has a maximizer of the same form with $2u_B(3\sigma + 4)$ in place of $u_B(3\sigma + 4)$ in the first bracket, hence the frontier $R_\delta = R_\Phi/2$; it is dominated wherever it taxes.

The cap corner is $x_1 = \delta$, $x_2 = 0$, $a = \kappa\delta$: the favored manufacturer keeps the whole per-user benefit and the fee is the largest one compatible with its adoption alone. At the cap with the fee constraint slack the share term on device 1 vanishes and the objective reduces to $\delta D_2 + a q_S$, whose maximizer in the fee is $a_{sub}(\delta/2)$, with a_{sub} as in (B.16) (**rep35**, test 3); this configuration is never the optimum on the reference grid, because where the fee constraint goes slack at the cap the platform prefers to hand back part of the premium to the second manufacturer as well. The derivative of Π_δ in x_2 along the edge $x_2 = 0$, $a = \kappa x_1$ has the sign of

$$x_1 [u_B(3\sigma^2 + 12\sigma + 8) + u_S\Lambda] - 2u_B(3\sigma + 4)(v + r - 2\delta) \quad (\text{B.15})$$

(**rep35**, test 4), negative for x_1 small and positive once x_1 is large relative to u_B : lowering the second manufacturer's price enlarges the application stock the fee is levied on, and where the fee is large enough this outweighs the premium handed back. The residual faces of Proposition B.2(ii) are the trace of this, and they sit in the developer-skewed corner where u_B is small and u_S large: among them the case in which both manufacturers sit at $\beta = 1$ with the fee constraint slack, the monetizing regime, holds at 0.7 to 3.8% of the admissible set at $\delta = 0.1$ and 2.8 to 8.2% at $\delta = 0.3$, the share rising with γ , and with the fee constraint binding at a further 0.2 to 0.9% and 0 to 0.8%; the taxing edge and the cap corner with $0 < x_2 < x_1$ hold 0.2 to 2.0% and 1.4 to 8.1% (**rep35**, block B).

The same route runs with $a < 0$, and there it is the platform's best policy in the buyer-skewed corner. With a subsidy the fee is not dead in the joint-rejection outcome: by (2) the developer collects $-a > 0$ on the platform's store whatever the adoption profile, so the application stock is the same function of prices in every configuration, and the comparison a manufacturer makes reduces to its own perceived benefit. Monotonicity fact (i) then makes adoption strictly dominant at any $\beta r > r - \delta$, so joint rejection is not an equilibrium and a_{kill} , positive, never binds. The platform therefore posts the smallest feasible share, $\beta r = r - \delta$ (the supremum over $\beta r = r - \delta + \varepsilon$, closed by the acceptance convention), and solves $\max_a \Pi_{sub}(a) = \delta(D_1 + D_2) + a q_S$ at the symmetric equilibrium with perceived benefit $r - \delta$. Direct computation gives

$$\Pi'_{sub}(0) = \frac{2(\sigma + 2)[u_S(v + r) - \delta(u_B + u_S)]}{(\sigma + 4)(1 - 2u_B u_S)}, \quad \Pi''_{sub} = -\frac{2(\sigma + 4 - 4u_B u_S)}{(\sigma + 4)(1 - 2u_B u_S)},$$

the second strictly negative on the admissible set (where $4u_B u_S < 2 < \sigma + 4$), whence

$$a_{sub} = \frac{(\sigma + 2)[u_S(v + r) - \delta(u_B + u_S)]}{\sigma + 4 - 4u_B u_S}, \quad a_{sep}|_{x=0} = \min\{a_{sub}, 0\}, \quad (\text{B.16})$$

a_{sub} being negative, so that the benchmark subsidizes the developer, if and only if $\delta > \bar{\delta}$; as on the compensating path of Appendix A.3, $\Pi''_{sub} = 2 dq_S/da$. The truncation matters: it is the case “no fee” of the floor row. The argument just given runs on $a < 0$ only: with $a > 0$ the fee *is* dead under joint rejection, a_{kill} is back, and at $x_1 = 0$ it caps the fee at $\kappa \cdot 0 = 0$. So where $\delta \leq \bar{\delta}$ and the taxing edge does not pay, the floor carries no fee at all: the platform posts $\beta^k r = r - \delta$ to both, sets $a = 0$, and lives on the premium alone, collecting δ per user on both devices and using neither of its two instruments. The benchmark policy is therefore the one of Proposition B.2, and its rows are states of the fee constraint and of the second manufacturer’s share: the fee constraint binds strictly on the taxing edge and at the cap corner, where the fee is valuable and one manufacturer is held at its floor; it binds at zero on the floor, where the platform posts no fee at all and lives on the premium; it goes slack in the buyer-skewed corner where $\delta > \bar{\delta}$, and the platform subsidizes; on the seam neither it nor the cap binds; and on the residual faces the second manufacturer is handed back part of the premium too. The maximum is verified against the enumerated adoption game, symmetric and asymmetric packages, at every point of the coarsened reference grid, and the comparison $a_{sep} \leq a^{**}(\delta)$ is quantified, in Section B.5.7. Every identity and closed form of this section is analytic; all of them are verified symbolically and on the reference grid by `rep_core.py`, `rep35_dominance_asym.py`, `rep10_interior.py` and `verifL.py`, and the symmetric restriction by `rep21_dominance_analytic.py` and `rep12_xi.py`.

B.5.3 What Dominance Buys under Separation

What dominance changes is easiest to see against the case $\delta = 0$ of Section 3, where the platform earns nothing at all.

TWO USES OF THE SAME PREMIUM. At $\delta = 0$ the developer fee is dead, and not because the platform lacks the instrument: a lone refuser free-rides on the application stock its rival’s adoption pays for, so joint refusal survives every fee the platform might post, and killing it would take handing back more than the whole per-user benefit, $\beta > 1$, which the cap forbids (Proposition 6). The premium removes exactly that obstacle, and it does so twice over. It is first income in its own right: posting $\beta^k r = r - \delta$ collects δ per user on *both* devices without any instrument being used, which is the floor of Proposition B.2. And it makes joint refusal cheap to kill, since $a_{kill}(\beta^1, \delta)$ is positive as soon as $\beta^1 r > r - \delta$: the premium finances what Section 4 needed rebates above the cap for, and it needs to finance it once, since one manufacturer’s adoption alone breaks joint refusal and the other, held at its outside value, follows. Dominance, in other words, relaxes the very constraint the cap imposed, and the divide-and-conquer package it allows is the single-host deal of Proposition 6 with the premium in the role of the rebate.

The platform’s problem is then an arbitrage between two uses of one premium: keep it as margin on both devices, or hand back x_1 of it on one device to buy the right to charge κx_1 to the developer. The fee is worth more the larger u_S , and costs more the larger u_B , since it depresses device demand at that rate; the regimes of Proposition B.2 are the corners of that arbitrage, and R_Φ is the ratio at which the two uses balance.

THE COMPENSATED LOCUS. Holding each manufacturer at its benchmark profit while charging the developer pins a symmetric package to the locus $\beta r = r - \delta + u_B a$, along which a manufacturer's markup, and hence its profit, is held at its $(r - \delta)$ -benchmark level. This locus lies inside the symmetric face of $\mathcal{P}_\delta$ with the fee constraint slack for $\sigma > 0$: on it $x = u_B a$, so

$$\frac{a_{kill}}{a} = \frac{\Lambda}{2(3\sigma + 4)} = R_\delta(\sigma) > 1, \quad (\text{B.17})$$

the fee lies below what would kill joint refusal, strictly for $\sigma > 0$. The locus is therefore one path among those Proposition B.2 optimizes over, not a separate route, and the ratio gives R_δ its meaning: it is the factor by which the fee that just kills joint refusal exceeds the fee that merely compensates. The same locus reappears in Section B.6 as one of the two constraints of the benchmark under coordination motives, the other being the equilibrium condition of joint adoption, since coordination benefits accrue only when both manufacturers adopt.

WHERE ONLY INTEGRATION TAXES. The benchmark taxes the developer on a set the diagonal does not describe. Writing $\Lambda \equiv 8 + \sigma(8 + \sigma)$, (B.14) gives $a_{sep} > 0$ if and only if

$$(v + r) [u_S \Lambda - u_B(3\sigma + 4)] > \delta [u_S \Lambda + \sigma(\sigma + 2) u_B],$$

which as $\delta \rightarrow 0$ reads $u_B/u_S < R_\Phi(\sigma)$ with $R_\Phi(\sigma) = \Lambda/(3\sigma + 4)$ as in Proposition B.2(iii); on the reference grid the sign of the bracket agrees with the sign of a_{sep} at every point, at every γ and $\delta \in \{0.01, 0.1, 0.3\}$ (`rep35`, block D). Here as elsewhere the diagonal is only the boundary case: $R_\Phi(0) = 2$, and R_Φ is strictly increasing. The integrated firm taxes on $u_B/u_S < R_a(\sigma)$, and $R_a - R_\Phi = \sigma^2(\sigma + 1)/(2(\sigma + 2)(3\sigma + 4)) \geq 0$ as a polynomial identity (Appendix A.3), so the band in which the developer fee turns positive *only* with integration is, as $\delta \rightarrow 0$, $R_\Phi(\sigma) < u_B/u_S < R_a(\sigma)$: the band of Proposition 6(i), on which the single-host package of Section 4 does not pay and the dominant platform, which prices the same package with the premium, does not tax either. It is thin: of the 545 admissible points of $\{u_S \leq u_B < \underline{h}(u_S)\}$ at $\gamma = 4$ the benchmark taxes at 521, and at 91 to 99% of that set across the six values of γ (`rep26_dominance_numbers.py`); the symmetric restriction, whose frontier is R_δ , taxed at 32% of it at $\gamma = 4$.

B.5.4 The Integrated Policy

Under integration the platform owns M_1 , keeps the benefit r on its own device, and licenses M_2 at the perceived benefit $b_2 = \beta^2 r$, subject to the rival's participation constraint and to $0 \leq \beta^2 \leq 1$. Write $a(b_2)$ for the fee that maximizes the integrated profit at a given b_2 , and $\varphi(b_2) \equiv \partial \Pi^I / \partial b_2$ evaluated along $a = a(b_2)$.

Proposition B.3 (The integrated policy under dominance). *On $\mathcal{A}^*$, and for every $\delta \in (0, r)$:*

- (i) Π^I is a strictly concave quadratic in (b_2, a) on $\mathcal{A}^*$. Hence $a(b_2)$ is affine in b_2 , and φ is affine with $\varphi' = \det H / \Pi_{aa}^I < 0$, so it has a unique root b_2^* .
- (ii) The participation floor in benefit units is $r \underline{\beta}(\delta)$; write $\underline{b}_2(\delta) \equiv \max\{r \underline{\beta}(\delta), 0\}$ for

the lower end of the admissible range of b_2 . The optimum is $a^{**} = a(b_2^{**})$ with

$$b_2^{**} = \begin{cases} \underline{b}_2(\delta) & \text{if } b_2^* \leq \underline{b}_2(\delta), \\ b_2^* & \text{if } \underline{b}_2(\delta) < b_2^* < r, \\ r & \text{if } b_2^* \geq r. \end{cases} \quad (\text{B.18})$$

No restriction on the size of δ is needed. That the rival's demand stays positive at the optimum, and that the developer publishes on the platform's store there, are grid findings: both hold at every point of the reference grid, at every γ , for every $\delta \in \{0.05, 0.1, 0.2, 0.3, 0.4, 0.45\}$, publication with a slack of at least 0.008 (`rep22_integration_delta.py`, test 8).

(iii) b_2^* does not depend on δ . The cap region $\{b_2^* \geq r\}$, equivalently $\{\varphi(r) \geq 0\}$, is therefore a fixed curve of the (u_B, u_S) -plane, independent of v , r and δ ; in (σ, X) -coordinates with $X = u_S^2$ it is $X \geq X_{cap}(\sigma)$ with

$$X_{cap}(\sigma) = \frac{3\gamma\sigma^2 + 11\gamma\sigma + 12\gamma + 3\sigma^3 + 19\sigma^2 + 44\sigma + 32}{2(\gamma + 1)(3\sigma^2 + 11\sigma + 12)}. \quad (\text{B.19})$$

The premium moves only the participation frontier $\{b_2^* = r\beta(\delta)\}$.

Proof. (i) Π^I is a quadratic in (b_2, a) with $\Pi_{aa}^I = -2H/[(\sigma + 4)^2(3\sigma + 4)^2M] < 0$ and $\det H = -4(\sigma + 1)N_2/[(\sigma + 4)^2(3\sigma + 4)^2M^2] > 0$ on $\mathcal{A}^*$. Solving $\partial\Pi^I/\partial a = 0$ gives $a(b_2)$, affine because the problem is quadratic; substituting, $\varphi' = \det H/\Pi_{aa}^I < 0$. (ii) The feasible set for b_2 is the interval $[\underline{b}_2(\delta), r]$ and the reduced objective $\Pi^I(b_2, a(b_2))$ is strictly concave with derivative φ ; a strictly concave function of one variable on an interval attains its maximum at its unconstrained maximizer when that point is interior, and at the nearer endpoint otherwise, which is (B.18). Where $r\beta(\delta) < 0$ the participation constraint is slack and the lower end is 0: the corner $\beta^{2**} = 0$ is the first line of (B.18), not an exception to it. (iii) δ enters the integrated problem only through the rival's outside option, which appears in $\underline{\beta}(\delta)$ and nowhere in Π^I ; hence b_2^* is free of δ . Since φ is affine and decreasing, $b_2^* \geq r$ if and only if $\varphi(r) \geq 0$; writing $u_B = w/u_S$ with $w = u_B u_S = (\gamma - \sigma)/(2(\gamma + 1))$, the numerator of $\varphi(r)$ is quadratic in $X = u_S^2$ with a single real root, which is (B.19). $\square$

Corollary B.1 (Three cells). *The cap region of Proposition B.3 is strictly contained in the taxing region of Proposition B.2(iii). As $\delta \rightarrow 0$ the admissible set therefore splits into exactly three cells: no point has the benchmark at the floor and the integrated firm at the cap.*

Proof. By Proposition B.2(iii) the floor region is $u_B/u_S \geq R_\Phi$, that is $X \leq w/R_\Phi$; by Proposition B.3(iii) the cap region is $X \geq X_{cap}$. The two are disjoint if and only if $X_{cap}(\sigma) > w/R_\Phi(\sigma)$ on $0 \leq \sigma \leq \gamma$, an inequality in (σ, γ) alone. Writing $\gamma = \sigma + t$ with $t \geq 0$, the difference equals $(\sigma + 1)(6\sigma^4 + 3\sigma^3t + 72\sigma^3 + 23\sigma^2t + 272\sigma^2 + 56\sigma t + 448\sigma + 48t + 256)$ divided by $2(\sigma + t + 1)(\sigma^2 + 8\sigma + 8)(3\sigma^2 + 11\sigma + 12)$; every term of both is positive, so the difference is positive, strictly so since the numerator has the constant term 256 (`rep28_containment.py`, which also records the same inequality for the symmetric restriction's frontier R_δ). $\square$

Two remarks. The threshold itself is the one of Appendix (B.6) relabeled: $(r_0, r) \mapsto (r, r - \delta)$ gives $\beta(\delta) = 1 - (\delta/r)\Lambda/(8(1 + \sigma))$, truncated to $[0, 1]$. And the relabeling runs the other way too: the efficiency-gains program of Section B.4 is (B.5) with $(r, r - \delta)$ in place of (r_0, r) , so Proposition B.3 covers that case as well, and Section B.4 uses it rather than the reverse. The two-instrument policy therefore needs solving once, not twice.

Every identity and closed form of this section is analytic. The verifications, symbolic and on the reference grid, are carried out in the scripts `rep22_integration_delta.py` and `rep28_containment.py`.

B.5.5 The Cost of Dominance under Separation

At $\delta = 0$ both branches of Proposition B.2(iii) reduce to $(\beta^1 r, \beta^2 r, a) = (r, r, 0)$, which is Proposition 1, and by Proposition B.3 the integrated policy reduces to $b_2 = r$ with the fee-only optimum, which is Proposition 2. The comparison at $\delta = 0$ is therefore Propositions 3–5 of the paper, and what the premium adds is their derivative. On each branch the policy is affine in δ , so every equilibrium object is an exact quadratic in δ and the derivatives below are exact, not truncations: they are the exact local marginal effects at $\delta = 0$, and since every object is quadratic in δ the derivatives are affine on each branch.

Proposition B.4 (Dominance is costly under separation). *On $\mathcal{A}^*$, along either branch of Proposition B.2(iii), the separation welfare is strictly decreasing in the premium at $\delta = 0$:*

$$\text{floor branch: } \left. \frac{dW_{sep}}{d\delta} \right|_0 = - \frac{4(v+r)(\sigma+2)(\sigma u_B u_S + \sigma u_S^2 + 2u_S^2 + 1)}{(\sigma+4)^2 M^2} < 0, \quad (\text{B.20})$$

$$\text{taxing branch: } \left. \frac{dW_{sep}}{d\delta} \right|_0 = - \frac{(v+r)(\sigma+2)\mathcal{B}}{(\sigma+4)^2(3\sigma+4)M^2} < 0, \quad (\text{B.21})$$

with $\mathcal{B} = \sigma^3 + 14\sigma^2 + 62\sigma + 56 + 2u_S^2(\sigma^3 + 13\sigma^2 + 34\sigma + 24) - 2u_B u_S(\sigma^2 + 28\sigma + 32)$.

Proof. Along each branch substitute the policy of Proposition B.2(iii) into the stage-3 equilibrium, form $W_{sep} = V_B + V_S + \pi_1 + \pi_2 + \Pi$, and differentiate at $\delta = 0$; the two displays follow (rep35, test 5). Welfare is the sum of the two user surpluses and of every firm's profit, so that $W_{sep} = V_B + V_S + \pi_1 + \pi_2 + \Pi$ under separation and $W^I = V_B + V_S + \pi_2 + \Pi^I$ under integration, with the objects of Appendix A.1. In (B.20) every term of the bracket is non-negative and the last is 1, so the bracket is positive. In (B.21), $2u_B u_S < 1$ gives $\mathcal{B} > \sigma^3 + 13\sigma^2 + 34\sigma + 24 + 2u_S^2(\dots) > 0$; $3\sigma + 4 > 0$ completes the sign. $\square$

Corollary B.2 (Where the integrated outcome ignores the premium). *In the cap region of Proposition B.3, $b_2^{**} = r$ and $a^{**} = a(r)$, neither of which involves δ ; the integrated outcome, and hence W^I , is exactly independent of the premium. There*

$$\left. \frac{d(W^I - W_{sep})}{d\delta} \right|_0 = - \left. \frac{dW_{sep}}{d\delta} \right|_0 > 0 :$$

the merger's welfare increment rises with the premium at exactly the rate at which the counterfactual deteriorates.

WHAT MOVES, AND WHO PAYS. On the floor branch, where the platform uses neither instrument, the pieces are as follows, all evaluated at $\delta = 0$, that is at $\beta = 1$ and $a = 0$:

	value at $\delta = 0$	sign
device demand $dD_1/d\delta$	$-\frac{\sigma + 2}{(\sigma + 4)M}$	< 0
buyers $dV_B/d\delta$	$-\frac{2(v + r)(\sigma + 2)^2}{(\sigma + 4)^2 M^2}$	< 0
developer $dV_S/d\delta$	$-\frac{4u_S^2(v + r)(\sigma + 2)^2}{(\sigma + 4)^2 M^2}$	< 0
manufacturers $d(2\pi)/d\delta$	$-\frac{8(v + r)(\sigma + 2)}{(\sigma + 4)^2 M}$	< 0
platform $d(2\delta D_1)/d\delta$	$+\frac{2(v + r)(\sigma + 2)}{(\sigma + 4)M}$	> 0

The last four lines sum to (B.20) identically.

The platform's gain exceeds the manufacturers' loss whenever $\sigma > 0$, their difference being $2\sigma(v + r)(\sigma + 2)/[(\sigma + 4)^2 M] \geq 0$: at first order the *industry* is not made worse off by dominance, and is made better off as soon as devices are substitutes ($\sigma > 0$); the difference vanishes at $\sigma = 0$. The entire welfare loss, and more, falls on the two user sides, and it is of first order because the downstream margin $p_1 + \beta r = 2(v + r)/(\sigma + 4)$ is already strictly positive at $\delta = 0$. The premium is an upstream margin stacked on a downstream one: the manufacturers price against a perceived benefit $r - \delta$ while the platform keeps δ , output falls at rate $(\sigma + 2)/[(\sigma + 4)M]$, and the loss it causes is linear in δ rather than quadratic. Integration removes that stacked margin on the device it owns; where the integrated policy is itself insensitive to the premium, Corollary B.2 makes the two rates equal and opposite, and Section B.5.6 reports how the merger's welfare increment moves elsewhere.

Both displays of Proposition B.4, the five lines of the table and the identity that closes them are analytic. The floor branch and the table are verified symbolically in `rep27_slopes_closed.py`, the taxing branch in `rep35_dominance_asym.py`.

B.5.6 The Comparison

Propositions B.2 and B.3 give both sides of the comparison in closed form, and Proposition B.4 settles the direction in which the counterfactual moves. The rest is computation, and we report it in full, including what does not come out uniform. Write

$$G(\delta) \equiv \Pi^I(\delta) - [\Pi_{sep}(\delta) + \pi_1(\delta)] \quad (\text{B.22})$$

for the private gain from the merger: the merged entity is the platform *together with* M_1 , so what integration must beat is the sum of the two separation profits. Under the asymmetric benchmark the two manufacturers are not in the same position before the merger, and a convention is needed: we take the merger partner to be the manufacturer the platform favors, M_1 in the labeling of (B.8), and the rival to be the one held at its floor. The convention is conservative for the private gain, since the favored manufacturer's separation profit is the larger of the two, and it changes nothing for buyers, the developer and total welfare. Unless stated otherwise the reference grid is the step-0.02 grid over $[0.02, 1.40]^2$ screened on $\mathcal{A}^*$, which holds 1843, 2186, 2373 and 2445 points at $\gamma = 1, 2, 4, 8$; the cell tables below use the finer step-0.01 grid.

THE THREE CELLS. By Proposition B.2(iii) the interior separation regimes vanish as $\delta \rightarrow 0$, and by Corollary B.1 the surviving partition has three cells:

	n	(taxing, cap)	(taxing, participation)	(floor, participation)
$\gamma = 1/2$	5371	7.6%	54.0%	38.4%
$\gamma = 1$	7225	10.0%	53.8%	36.2%
$\gamma = 2$	8602	12.1%	55.0%	33.0%
$\gamma = 4$	9344	13.7%	58.0%	28.4%
$\gamma = 8$	9631	14.6%	63.5%	21.9%
$\gamma = 16$	9687	15.1%	71.3%	13.6%

The cell (floor, cap) is empty at every one of these 47,860 points, as Corollary B.1 requires.

THE SIGNS OF THE SLOPES. Each increment is an exact quadratic in δ inside a cell, so its derivative at $\delta = 0$ is exact. The table reports the share of each cell at which that derivative is *positive* (`rep25_sign_uniformity.py`):

γ	cell	G	ΔW	$\Delta \pi_2$	ΔV_B	ΔV_S	Δa
1/2	(taxing, cap)	0.0	100	100	100	100	0.0
	(taxing, participation)	0.1	100	100	100	100	0.0
	(floor, participation)	100	100	69.3	100	100	0.0
1	(taxing, cap)	0.0	100	100	100	100	0.0
	(taxing, participation)	0.6	100	100	100	100	0.0
	(floor, participation)	100	100	67.8	100	100	0.0
2	(taxing, cap)	0.0	100	100	100	100	0.0
	(taxing, participation)	2.0	100	100	100	100	0.0
	(floor, participation)	100	100	66.3	100	100	0.0
4	(taxing, cap)	0.0	100	100	100	100	0.0
	(taxing, participation)	5.4	100	100	100	100	0.0
	(floor, participation)	100	98.9	64.6	92.9	100	0.0
8	(taxing, cap)	0.0	100	100	100	100	0.0
	(taxing, participation)	12.3	100	100	100	100	0.0
	(floor, participation)	100	98.0	62.3	82.1	100	0.0
16	(taxing, cap)	0.0	100	100	100	100	0.0
	(taxing, participation)	21.6	100	100	100	100	0.0
	(floor, participation)	100	98.9	57.0	74.6	100	0.0

Three readings are uniform across every cell and every γ computed. The merger’s fee increment always *shrinks* with the premium: $d(\Delta a)/d\delta < 0$ at every point. The developer’s position relative to the counterfactual always *improves*: $d(\Delta V_S)/d\delta > 0$ at every point. And in the two taxing cells every constituency’s increment other than the merged firm’s moves in the same direction as welfare’s. A fourth is uniform up to a small exception: $d(\Delta W)/d\delta > 0$ everywhere except at 1.1 to 2.0% of the floor cell at $\gamma \geq 4$.

The private gain is the one object whose slope changes sign across cells. It is negative throughout the (taxing, cap) cell, negative at 78 to 100% of (taxing, participation), and positive throughout (floor, participation): dominance erodes the merger’s private value where the platform can already tax the developer without it, and adds to that value where the platform cannot. The regularity is strong and its direction is the economically expected one, but it is not a theorem: in the floor cell the numerator of $dG/d\delta$ is a degree-seven polynomial in σ with coefficients of mixed sign, and it does not factor

(rep27_slopes_closed.py). We state it as what it is, a computational regularity on the grids reported.

OVER THE WHOLE RANGE OF THE PREMIUM. Beyond the neighbourhood of zero the comparison is made with the exact optima of Propositions B.2 and B.3 on the reference grid, at the six values of γ ; the table shows $\gamma = 4$ (2,373 points), and the one below it the welfare column at every γ :

δ	median G	min G	$G \leq 0$	median ΔW	$\Delta W > 0$	median Δa
0.00	0.4365	0.0000	0%	-0.5949	33.2%	+0.4734
0.05	0.2350	0.0012	0%	-0.0168	45.6%	+0.1077
0.10	0.1843	0.0050	0%	+0.0516	54.6%	+0.0047
0.20	0.1909	0.0197	0%	+0.4121	75.0%	-0.0129
0.30	0.2355	0.0566	0%	+0.9168	86.1%	-0.0221
0.45	0.2175	0.0891	0%	+1.5893	100%	-0.0148

$\Delta W > 0$, share of the admissible set	$\delta = 0$	0.05	0.10	0.20	0.30	0.45
$\gamma = 1/2$ (1,380 points)	56.0%	85.0%	98.1%	100%	100%	100%
$\gamma = 1$ (1,843)	52.0%	75.4%	89.0%	100%	100%	100%
$\gamma = 2$ (2,186)	44.7%	64.0%	79.4%	96.0%	100%	100%
$\gamma = 4$ (2,373)	33.2%	45.6%	54.6%	75.0%	86.1%	100%
$\gamma = 8$ (2,445)	20.0%	29.8%	34.9%	44.2%	55.7%	98.9%
$\gamma = 16$ (2,446)	10.5%	17.4%	22.1%	33.9%	43.3%	75.3%

Two facts stand out. The private gain never vanishes: $G > 0$ at every admissible point, every γ and every premium computed, up to $\delta = 0.45$ with $r = 1/2$, although it comes close to zero at the first premia, its minimum over the admissible set being below 2×10^{-3} at $\delta = 0.05$ for $\gamma \leq 8$. Integration is always privately worthwhile, however dominant the platform already is. Its median is nonetheless U-shaped at every γ , falling by 51 to 70% before recovering, with a trough near $\delta \approx 0.05$ to 0.20: at small premia the two taxing cells, which hold 62 to 86% of the plane and whose slopes are negative, set the direction; past the trough the floor cell and the level effect take over. For $\gamma \geq 4$ the median falls once more beyond the premium at which the rival's floor reaches zero, where the corner $\beta^2 = 0$ takes over. Meanwhile the welfare effect moves one way only. The share of the admissible set on which the merger raises welfare rises with the premium at every γ , from 10 to 56% when the platform has no pre-existing advantage, depending on γ , to 75 to 100% at $\delta = 0.45$, and to 100% at every $\gamma \leq 2$ from $\delta = 0.3$ on. The private signal and the social signal do not move together: the private gain is U-shaped in the premium while the share of welfare-improving mergers rises with it (rep24_merger_incentive.py).

B.5.7 Verification, Quantification and Simulations

We record here the verification of the benchmark and the quantification. The benchmark is computed at every point of the reference grid by the enumeration of the active sets of $\mathcal{P}_\delta$, exact for a concave quadratic, at the six values of γ and for $\delta \in \{0.05, 0.1, 0.2, 0.3\}$; on the reference grid coarsened to step 0.1 (46 to 88 points per γ) and for $\delta \in \{0.1, 0.2, 0.3\}$, the enumerated adoption game, with its four configurations, its Nash equilibria and the selection rule, swept over symmetric and asymmetric packages (β^k in 41 values from $1 - \delta/r$ to 1, a in 81 values from -1 to 3), never returns more than the program does, at

any of the 446 point-premium pairs (`rep35_dominance_asym.py`, block C). The regimes of Proposition B.2 occupy, at $\delta = 0.1$ and across γ , 17 to 39% of the admissible set for the floor (2.7 to 4.9% of it subsidizing), 28 to 40% for the taxing edge, 20 to 39% for the cap corner, up to 0.2% for the seam and 1 to 7% for the residual faces; at $\delta = 0.3$ the taxing edge holds 46 to 54%, the cap corner 0 to 11%, the floor 24 to 41% (11 to 20% of it subsidizing), the seam about 1% and the residual faces 4 to 17% (`rep26_dominance_numbers.py`). The symmetric restriction is strictly beaten at 55 to 82% of the admissible points, depending on γ and δ , never at a point where it sits at the floor without a fee; the median gain is 3 to 12% of the platform’s profit and the largest 15 to 48%, and the benchmark fee exceeds the symmetric restriction’s fee at 39 to 73% of the points where the latter taxes (`rep35`, block B). All counts below are on the reference grid and use the exact optima of Propositions B.2 and B.3. Whether the benchmark taxes beyond the integrated fee depends on product-market competition and on the premium: $a_{sep} > a^{**}(\delta)$ at 17 to 18% of the taxing points at $\gamma = 1/2$, at 18 to 21% at $\gamma = 1$, 22 to 30% at $\gamma = 2$, 19 to 49% at $\gamma = 4$, 32 to 66% at $\gamma = 8$ and 48 to 72% at $\gamma = 16$, for δ from 0.1 to 0.3, which is the merger’s fee increment turning negative at strong rivalry and large premia (`rep26_dominance_numbers.py`). The benchmark fee is not everywhere non-negative: it subsidizes wherever $\delta > \bar{\delta}(u_B, u_S)$, which happens at 67 to 68, 169 to 180 and 271 to 297 points for $\delta = 0.1, 0.2$ and 0.3 depending on γ (2.7 to 4.9%, 6.9 to 13.0% and 11.1 to 19.9% of the admissible set): a strongly buyer-skewed corner that widens with the premium, and in which the dominant platform tilts the price structure as the integrated firm does (`verifL.py`).

The merger increment against this benchmark splits by channel ($\gamma = 4, v = 2, r = 1/2$; each entry is the integrated value minus the separation- δ value, at the respective optima), for premia up to $\delta = 0.3$.²⁰

(u_B, u_S)		$\delta = 0$	0.1	0.2	0.3
(0.1, 0.1)	fee increment $a^{**} - a_{sep}$	0.137	−0.002	−0.007	−0.012
	rival $\Delta\pi_2$	−0.016	−0.064	−0.123	−0.174
	welfare ΔW	−0.074	−0.019	−0.040	−0.065
(0.5, 0.18)	fee increment	0.072	0.005	−0.028	−0.078
	rival $\Delta\pi_2$	−0.048	−0.069	−0.115	−0.143
	welfare ΔW	−0.231	−0.029	+0.067	+0.205
(0.15, 1.0)	fee increment	2.307	1.314	0.490	0.556
	rival $\Delta\pi_2$	−1.071	−0.645	−0.414	−0.379
	welfare ΔW	−2.516	+1.134	+4.605	+5.256

The taxing-side fee increment falls with the first units of dominance to about zero, and turns negative at balanced effects, where the benchmark’s taxing edge already collects what the integrated firm would; at developer-skewed points it stays positive and turns back up once the benchmark’s own taxing branch runs into the cap; the subsidy-side increment grows (the integrated fee turns negative at (0.5, 0.18) for $\delta \geq 0.2$ while $a_{sep} = 0$,

²⁰The rival’s participation threshold stays non-negative at every admissible point as long as $\delta < 8r(1 + \gamma)/(8 + \gamma(8 + \gamma))$, which is 0.357 at $\gamma = 4$; the bound is $8r(1 + \sigma)/\Lambda$, decreasing in σ , and $\sigma \leq \gamma$. Beyond that premium, at points where network effects are weak (σ close to γ), $\beta(\delta) < 0$: the rival would remain on the platform’s system even at a zero share, its participation constraint stops binding, and the integrated firm sits at the corner $\beta^{2**} = 0$, which (B.18) covers as the lower end of the admissible range. The table below stops at $\delta = 0.3$; Section B.5.6 carries the comparison to $\delta = 0.45$.

the premium there falling short of $\bar{\delta} = 0.66$); the rival's loss deepens at balanced effects and shrinks at developer-skewed ones; and the share of points at which the merger reduces welfare falls monotonically in the premium, at every γ computed, to zero at $\gamma \leq 2$ (`rep26_dominance_numbers.py`):

	$\delta = 0$	0.1	0.2	0.3
$\gamma = 1/2$	44.0%	1.9%	0.0%	0.0%
$\gamma = 1$	48.0%	11.0%	0.0%	0.0%
$\gamma = 2$	55.3%	20.6%	4.0%	0.0%
$\gamma = 4$	66.8%	45.4%	25.0%	13.9%
$\gamma = 8$	80.0%	65.1%	55.8%	44.3%
$\gamma = 16$	89.5%	77.9%	66.1%	56.7%

The corresponding maps close this section, in Figure [OA-3](#).

HOW FAR THE SIGN OF PROPOSITION [B.4](#) CARRIES. That proposition signs $dW_{sep}/d\delta$ at $\delta = 0$, where the benchmark never subsidizes, a_{sub} at $\delta = 0$ being positive. Away from zero the sign can reverse, and the instrument the benchmark uses distinguishes most of the points at which it does. Evaluating W_{sep} at the optimum of Proposition [B.2](#) along $\delta \in [0, 0.48]$ in steps of 0.01, separation welfare falls throughout at 78 to 84% of the admissible set at every γ and rises somewhere at the remaining 16 to 22% of points (383 at $\gamma = 4$). At all but 21, 46, 45, 9, 0 and 206 of them, for $\gamma = 1/2$ to 16, the benchmark is subsidizing the developer at the premium where the rise occurs, which by ([B.16](#)) requires $\delta > \bar{\delta}$: this is the decomposition of Section [B.5.5](#) read at larger premia, the premium being costly when the platform spends it on taxing the developer or keeps it as margin, since it then only stacks a margin on a margin, and paying for a correction to the underprovision of applications where it is large enough to fund a subsidy instead. The exceptions are of two kinds. At $\gamma \leq 4$ the rise occurs where the benchmark enters a residual face and hands part of the premium back to the second manufacturer, whose price falls; at $\gamma = 16$ it occurs along the taxing edge at strongly buyer-skewed points, where the fee, small there, falls with the premium faster than the margin stacks. The merger comparison is made against the benchmark itself, so it is unaffected either way (`rep29_Wsep_monotone.py`).

The seam of Section [B.5.2](#), where no constraint binds, is the unconstrained optimum of Π_δ , symmetric by symmetry, and it is the same point for the benchmark and for its symmetric restriction. On the reference grid the benchmark sits there at 7, 7, 7, 9, 10 and 12 points at $\delta = 0.2$ and 15, 17, 18, 18, 23 and 28 at $\delta = 0.3$, for $\gamma = 1/2$ to 16: 0.3 to 1.1% of the admissible set. At every one of them the closed form ([B.12](#)) returns the numerical optimum to machine precision, and that optimum strictly exceeds the best of the candidates at which a constraint binds, by between 10^{-7} and 3×10^{-3} in relative value. Scripts: `rep_core.py`, `rep5_dominance.py`, `rep10_interior.py`.

SIMULATIONS. The map of the welfare effect completes the picture. Conventions are those of Section [B.11](#): the admissible set of Section [B.1](#), blue marking a rise and red a fall relative to the separation benchmark. Figure [OA-3](#) maps the welfare effect of the merger against the separation- δ benchmark, for $\delta \in \{0, 0.15, 0.3\}$: the harm region (red) shrinks monotonically toward weak, balanced network effects as the platform's pre-existing advantage grows; this is the narrowing described in Section [5.2](#) of the paper.

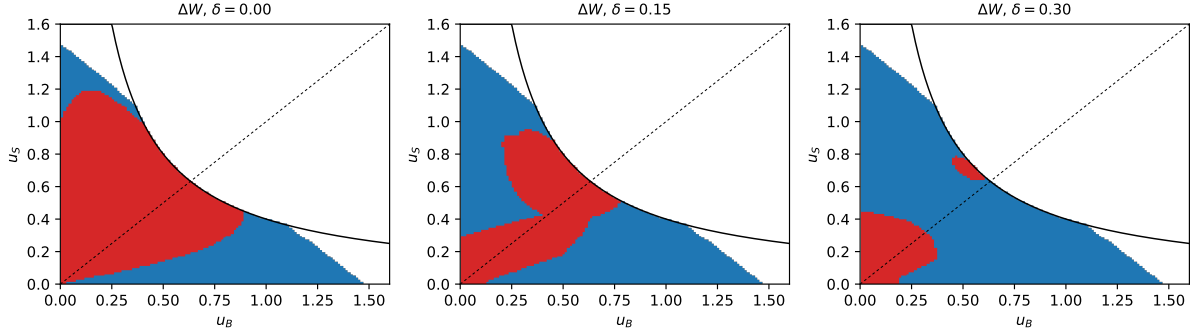


Figure OA-3: Welfare effect of integration against the dominant-platform benchmark of Proposition B.2, for $\delta \in \{0, 0.15, 0.3\}$ (blue: rise; red: fall), at $\gamma = 4$. The dotted line is the diagonal $u_S = u_B$. Drawn by `rep5_figure.py`.

B.6 Coordination Motives and Porting Costs

This section supports Section 5.3 of the paper. Coordination motives, a direct network effect among buyers, are modeled as an extra gross benefit $\alpha_B \geq 0$ accruing to every buyer when both manufacturers run the same operating system, which, the outside options being distinct systems, happens only on the platform's system. Under integration with M_2 hosted, demand intercepts are $v + \alpha_B$; if M_2 walks, they are v on both devices.

THE PARTICIPATION THRESHOLD. The rival's participation constraint compares its hosted markup (with intercept $v + \alpha_B$ and share βr) with its walk-off markup (intercept v , benefit r , the fee still active); both are linear in a with the same slope, so the fee cancels as in Section B.4, and the constraint reduces to $\beta \geq \underline{\beta}$ with

$$\underline{\beta} = 1 - \frac{\alpha_B}{r} \cdot \frac{4 + 3\sigma}{4(1 + \sigma)}, \quad (\text{B.23})$$

the floor (11) of the paper at $\Delta = \alpha_B$ and $\theta = (4 + 3\sigma)/(4(1 + \sigma))$ (symbolic verification: `rep4_efficiency.py`). The premium α_B plays exactly the role of the efficiency gain, with one difference: it enters the buyers' utility directly rather than the licensing revenue, so there is no gain to pass through.

THE PRE-MERGER BENCHMARK UNDER COORDINATION. For $\alpha_B > 0$ the separation benchmark is itself distorted: both manufacturers on the platform's system enjoy α_B under separation as well, and the independent platform can price the premium. What it can price is limited by the adoption game, through two constraints. The first is that joint adoption be an equilibrium of the adoption game. A manufacturer that walks alone loses the premium, which requires both manufacturers on the same system, but keeps the whole per-user benefit r and the shared stock of applications, the fee remaining active on the platform's store. Its markup and the hosted markup respond to the fee identically, so the fee cancels as in (B.23), and the constraint is a floor on the share alone,

$$\beta \geq \beta_{NE} \equiv 1 - \frac{\alpha_B}{r} \cdot \frac{2(4 + 3\sigma)}{\Lambda}, \quad (\text{B.24})$$

with $\Lambda = 8 + \sigma(8 + \sigma)$ (symbolic verification: `rep34_coordination.py`, cited as `rep34` below). The second is the selection rule. Joint refusal is an equilibrium at every package

with shares at or below one: a lone adopter gains no premium, so its comparison with refusal is that of Proposition 1, a subsidy being collected under refusal as well. The rule therefore retains adoption only where the manufacturers' joint profit under adoption is at least their joint profit under refusal, that is where the hosted markup $\hat{m}(\beta, a) = 2(v + \alpha_B + \beta r - u_B a)/(4 + \sigma)$ is at least the refusal markup $2(v + r - u_B \min\{a, 0\})/(4 + \sigma)$:

$$\beta r \geq r - \alpha_B + u_B \max\{a, 0\}, \quad (\text{B.25})$$

the compensated locus of Section B.5 for a positive fee, a flat floor for a subsidy. The two floors are ordered, $\beta_{NE} r - (r - \alpha_B) = \alpha_B \sigma(2 + \sigma)/\Lambda \geq 0$ with equality at $\sigma = 0$ only, so (B.24) binds for every subsidy and for fees up to the crossing $u_B a_\times = \alpha_B \sigma(2 + \sigma)/\Lambda$, and (B.25) beyond it. At the compensated package with no fee, in particular, a walker's markup exceeds the hosted markup by $\alpha_B \sigma(2 + \sigma)/((4 + \sigma)(4 + 3\sigma))$: that package is not adoptable, and a benchmark built on (B.25) alone would overstate what the independent platform extracts (rep34, tests 1 to 5).

The benchmark program. Under joint adoption at share β and fee a , each manufacturer's demand is $D = (2 + \sigma)(v + \alpha_B + \beta r - u_B a)/((4 + \sigma)(1 - 2u_B u_S))$, the application stock is $q_S = [2u_S(2 + \sigma)(v + \alpha_B + \beta r) - a(\sigma + 4 - 4u_B u_S)]/((4 + \sigma)(1 - 2u_B u_S))$, and the platform earns $\Pi(\beta, a) = 2(1 - \beta)rD + a q_S$, a quadratic, concave in the fee, to be maximized on the convex polygon that (B.24), (B.25) and the cap $\beta \leq 1$ delimit. Its two lower edges have closed forms. Along the floor $\beta = \beta_{NE}$ the share is fixed and the fee alone moves; writing $\delta_{NE} \equiv (1 - \beta_{NE})r = 2(4 + 3\sigma)\alpha_B/\Lambda$ for the coordination rent per buyer, the profit is concave in a with maximizer

$$a_{NE} = \frac{(2 + \sigma)[u_S(v + r + \alpha_B) - \delta_{NE}(u_B + u_S)]}{\sigma + 4 - 4u_B u_S}, \quad (\text{B.26})$$

the fee of the subsidy branch of Proposition B.2 at premium δ_{NE} and intercept $v + \alpha_B$. Along the compensated locus $\beta r = r - \alpha_B + u_B a$ the hosted markup is held at its refusal level, so each manufacturer's demand is constant at its benchmark level $\bar{D}$ and the application stock falls one for one with the fee, $q_S(a) = q_0 - a$: the profit is the strictly concave quadratic

$$\Pi_{comp}(a) = 2(\alpha_B - u_B a)\bar{D} + a(q_0 - a), \quad (\text{B.27})$$

with unconstrained maximizer

$$a_{comp} = \frac{(u_S - u_B)(v + r)(2 + \sigma)}{(1 - 2u_B u_S)(4 + \sigma)}, \quad (\text{B.28})$$

the cap $\beta \leq 1$ truncating the locus at $a = \alpha_B/u_B$ (rep34, test 6). The optimum is the best of the candidates the polygon offers, the interior stationary point where the Hessian is negative definite, the stationary points of the three edges and the two vertices; the maximization is exact for a quadratic. The program is checked against the enumerated adoption game, four configurations, Nash equilibria and the selection rule, at 18 point-premium pairs across the six values of γ , where the best enumerated package never exceeds the analytic optimum, and against a coarse sweep of asymmetric packages (β^1, β^2, a) at three points, where nothing beats the symmetric optimum (rep34, block B).

The benchmark policy. On the reference grid the optimum falls in six regimes, according to which constraints bind, counted here at $\alpha_B \in \{0.2, 0.4\}$ (rep34, block B):

γ	1/2	1	2	4	8	16
admissible points ($\mathcal{A}^*$)	1,380	1,843	2,186	2,373	2,445	2,446
$\alpha_B = 0.2$						
floor β_{NE} , subsidy	153	135	114	83	50	22
floor β_{NE} , tax below $a_\times$	157	249	355	449	537	589
kink: both floors, fee $a_\times$	484	711	867	962	991	1,006
compensated locus	503	472	380	272	179	107
no constraint binding	5	5	4	4	3	2
cap $\beta = 1$	78	271	466	603	685	720
$\alpha_B = 0.4$						
floor β_{NE} , subsidy	295	311	263	196	125	65
floor β_{NE} , tax below $a_\times$	255	394	554	709	848	939
kink: both floors, fee $a_\times$	329	497	653	743	770	784
compensated locus	445	515	440	326	211	128
no constraint binding	18	21	17	12	11	7
cap $\beta = 1$	38	105	259	387	480	523

The kink is the most frequent regime at $\alpha_B = 0.2$. There both constraints bind: the share is β_{NE} and the fee is $a_\times = \alpha_B \sigma(2 + \sigma)/(\Lambda u_B)$, the largest fee at which the floor (B.24) still satisfies the selection rule (zero at the one or two grid points with $\sigma = 0$); the platform would tax more along the floor, but beyond $a_\times$ every further unit of fee has to be handed back to the manufacturers as a higher share along the locus, and the locus is not worth following. Two regimes deserve comment. The independent platform *subsidizes* the developer at the strongly buyer-skewed points ($u_B/u_S \geq 13$ at $\alpha_B = 0.2$ and ≥ 6.4 at 0.4, all at $u_S \leq 0.18$): along the floor β_{NE} the derivative of the profit at $a = 0$ is negative if and only if $u_B/u_S > (v + r + \alpha_B)\Lambda/(2(4 + 3\sigma)\alpha_B) - 1$ (rep34, test 7). The free-riding argument of Proposition 1 does not protect the developer here: a manufacturer that walked to enjoy the subsidized commons at full benefit would lose the premium, and the coordination rent, earned on both devices, is what the subsidy enlarges. And at the developer-skewed points the platform leaves the whole per-user benefit and taxes: on the cap its profit is the fee revenue alone, and it prefers the cap once $a(u_B + u_S) > v + \alpha_B - r(1 - 2\beta)$, where lower device prices bring the buyers whose applications it taxes. The benchmark policy thus has the floor, interior and cap regimes of the integrated policy of Section B.4, with the compensated locus inserted between the floor and the cap. The structure is analytic; the counts are grid findings.

Two consequences carry over to the comparison with the merger. First, the independent platform collects the premium only in part, $\delta_{NE} = 2(4 + 3\sigma)\alpha_B/\Lambda$ per buyer, the whole of α_B at $\sigma = 0$ and less as σ grows, and the integrated firm collects more from the rival, since its floor (B.23) lies below (B.24):

$$\beta_{NE} - \underline{\beta} = \frac{\alpha_B}{r} \cdot \frac{(4 + 3\sigma)\sigma^2}{4(1 + \sigma)\Lambda} \geq 0, \quad (\text{B.29})$$

with equality at $\sigma = 0$ only (rep34, test 8). The reason is the fee: facing a rival that walks, the integrated firm internalizes the fee it collects on its own buyers and prices lower than a hosted manufacturer would, so the rival's outside option is worse under integration than under separation. This is an increment of the premium channel that the merger creates, and at the origin of the plane, where the fee is worth nothing to either firm, it is all the merger does. Second, the fee is on both sides of the comparison. Where

the benchmark taxes at the floor β_{NE} the integrated firm taxes less than the independent platform at most points (69 to 98% of them, according to γ and α_B), and wherever the independent platform subsidizes, the integrated firm subsidizes more; at the kink the two fees are ordered either way; on the locus and at the cap the integrated firm taxes more at every point but a handful, and at the cap it leaves the rival the whole share at half of the points or more, so that the merger there is the fee channel alone (**rep34**, block C). The maps of Section B.11 are computed against this benchmark, as those of Section B.5 are under dominance; the shares against the undistorted benchmark are reported there for comparison with the efficiency case. Integration remains privately profitable at every admissible point, at the six values of γ and both premia, the merged entity's profit exceeding the sum of the manufacturer's and the platform's under separation (**rep34**, block E).

THE PRICING POLICY. The Hessian of the unconstrained problem does not depend on α_B : the admissible set is that of Section B.1. The unconstrained optimum (β^*, a^*) is given by the Section-3 formulas with $v + r$ replaced by $v + r + \alpha_B$; in particular the loci $\beta^* = 1$ and $a^* = 0$ are *unchanged*. Two new curves matter:

Curve $\beta^ = \underline{\beta}$.* In (σ, x) -coordinates, $\beta^* \geq \underline{\beta}$ reduces to a strictly convex quadratic inequality in x whose left-hand side is negative at $x = 0$: it holds iff $x \geq \bar{x}_{cm}(\sigma)$, where the root $\bar{x}_{cm}(\sigma)$ has an explicit closed form (reported in the replication script; writing $\alpha_B = \alpha(v + r)$ makes it a function of (σ, γ, α) alone). The associated parametric curve satisfies: $\hat{u}_B(\gamma) = 0$; $\hat{u}_B(0) < \hat{u}_S(0)$ iff $\alpha \leq 1$; and its slope converges, as $\alpha \rightarrow 0$, to the (negative) slope of the $\beta^* = 1$ curve: for α small the curve is strictly downward-sloping. Below it the participation constraint binds.

Curve $\tilde{a} = 0$. With $\beta = \underline{\beta}$ imposed, the optimal fee $\tilde{a}$ is positive above a frontier that is linear in x : $\tilde{a} \geq 0$ iff $x \geq \tilde{x}_{cm}(\sigma)$ with

$$\tilde{x}_{cm}(\sigma) = \frac{(\gamma - \sigma)(4 + 3\sigma)(8(r + v)(2 + \sigma) + \alpha_B(4 + \sigma)(8 + 3\sigma))}{2(1 + \gamma)(4(r + v)(32 + 3\sigma(16 + \sigma(7 + \sigma))) + \alpha_B(4 + \sigma)(16 + \sigma(20 + 3\sigma)))}$$

(**rep34**, test 9; as $\alpha_B \rightarrow 0$ the frontier reduces to the locus $a^{**} = 0$ of Section 3, $\underline{x}_a(\sigma)$ in Appendix B.2). As $\alpha_B \rightarrow 0$ the associated parametric curve has positive slope and lies below the 45° line (the difference $\hat{u}_B^2 - \hat{u}_S^2$ is proportional to $\gamma - \sigma \geq 0$); by continuity both properties persist for α_B small.

The resulting policy map has four regions (Figure OA-4): $\beta^{**} = 1$ with a positive fee (developer-skewed effects); interior solution; $\beta^{**} = \underline{\beta}$ with a positive fee; and $\beta^{**} = \underline{\beta}$ with a subsidy (buyer-skewed effects). Contrary to the efficiency-gains case, the region where the rival is left a share strictly above $\underline{\beta}$ carries no pass-through of any merger-created gain: it merely reflects the platform's own interest in boosting buyer participation.

POLAR CASES. The two axes are read against the corrected benchmark, numerically at $\gamma = 4$, along the grid rows $u_S = 0.02$ and $u_B = 0.02$, 70 points each (**rep34**, block E). Near $u_S = 0$ a positive fee forfeits publication ($u_S n_B^I \geq a$ fails), so both the independent platform and the integrated firm subsidize or charge next to nothing, and the integrated firm subsidizes more, recouping on its own device margin: the developer gains at every point. The other constituencies gain only away from the origin, where the subsidized commons is worth something to them: the rival loses at $u_B < 0.40$ (0.54 at $\alpha_B = 0.4$), buyers at $u_B < 0.20$ (0.26), and welfare falls at $u_B < 0.10$ (0.14). What remains near the origin is the premium channel alone: the rival passes from β_{NE} to the lower $\underline{\beta}$, the

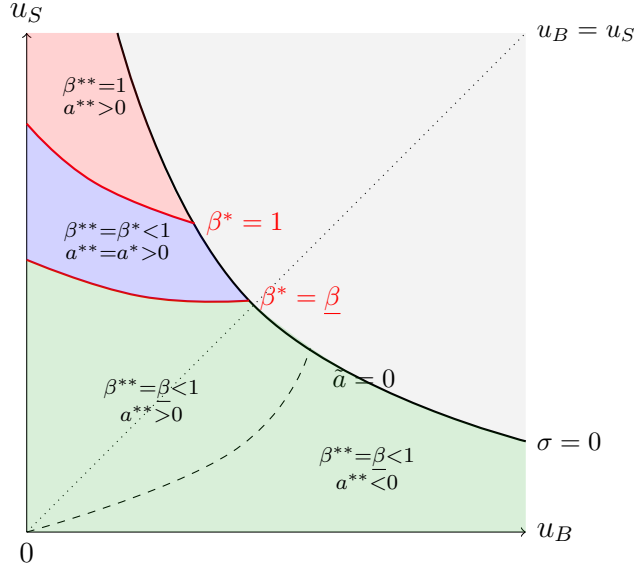


Figure OA-4: The integrated platform's pricing policy with coordination motives, in the (u_B, u_S) -plane (schematic, α_B small).

integrated firm earns a margin on the rival's sales and competes less hard, both prices rise, and the rival, the buyers and welfare lose. Near $u_B = 0$ the fee is the main instrument on both sides, and the three regimes $(\underline{\beta}, \tilde{a}) \rightarrow (\beta^*, a^*) \rightarrow (1, \tilde{a}(1))$ follow one another as u_S increases, for the integrated firm as for the independent platform. The rival loses at every point of the row, to the lower floor where the floor binds and to the higher fee where the cap binds on both sides. The developer gains at every point: the independent platform taxes it already, and where the integrated firm taxes more its lower device price brings the developer more buyers. Buyers lose at $u_S < 0.24$ (0.34) and gain beyond, and welfare falls at $u_S < 0.22$ (0.32). The two polar cases thus meet at the origin, where the merger is extraction from the rival and nothing else, and part company away from it: extraction without pass-through does its worst where effects are developer-skewed, while in the buyer-skewed polar case the subsidized commons carries every constituency once u_B is large enough. How the two forces net out on the interior of the admissible set is quantified in Section B.11.

PORTING COSTS. Let the developer bear a unit cost $c > 0$ to serve each operating system beyond the first. If M_2 is hosted, one system carries all applications and the developer's marginal condition is $q_S = u_S(n_B^1 + n_B^2) - a$. If M_2 walks, its fork is a second system: the marginal application must also cover the porting cost, so $q_S = u_S(n_B^1 + n_B^2) - a - c$, and the application stock, shared by *both* devices, shifts down by c at the margin. The walk-off configuration is therefore isomorphic to a fall of both demand intercepts by $u_B c$, and the participation constraint reduces, by the same computation as (B.23), to $\beta \geq \underline{\beta}$ evaluated at the premium $\Delta = c u_B$:

$$\underline{\beta} = 1 - \frac{c u_B}{r} \cdot \frac{4 + 3\sigma}{4(1 + \sigma)},$$

the porting-cost threshold quoted in Section 5.3 of the paper (symbolic verification: `rep4_efficiency.py`). The analysis of the pricing policy then parallels the coordination case with α_B replaced by $c u_B$.

Porting costs also erode the separation benchmark itself, and they do so *immediately*, not beyond a threshold. Under separation with two outside systems, the developer’s marginal condition is $q_S = u_S(n_B^1 + n_B^2) - c$ (one system beyond the first, no fee): the benchmark stock of applications is lower than in the costless-porting baseline. Hosting *both* manufacturers eliminates the duplicated system and saves the porting cost, an intercept-equivalent premium of u_{BC} per buyer, which the independent platform can price along the compensated locus of Section B.5, with $\delta = u_{BC}$ in the role of the premium: the locus is $\beta r = r - u_{BC} + u_B a$, and the premium converts into taxing capacity at most $\delta/u_B = c$, the saved cost itself. Two qualifications apply. The construction presumes that applications are still ported to the second system under separation, which requires $u_S n_B^2 \geq c$; where that fails, availability is asymmetric across systems and neither the shared-stock formula nor the intercept equivalence behind the threshold describes the equilibrium. And, as with coordination motives and unlike dominance, the benchmark is bounded by the adoption game on both sides: a unilateral adopter saves no porting cost, since its rival’s system is still there to be served, so joint refusal survives every capped package and the selection rule imposes (B.25) with u_{BC} in the role of α_B ; and a unilateral walker, whose fork is a second system, bears the porting cost while keeping the whole per-user benefit, so joint adoption is an equilibrium only above the floor (B.24) with u_{BC} in the role of α_B . The benchmark is thus the program of this section with (v, α_B) replaced by $(v - u_{BC}, u_{BC})$, both floors included (rep34, test 10), and the independent platform collects $2(4 + 3\sigma) u_{BC}/\Lambda$ per buyer, the saved cost in full at $\sigma = 0$ only. The separation benchmark is no longer free-riding proof once $c > 0$: within those limits, for every $c > 0$ and $u_B > 0$ a share $\beta < 1$ is consistent with adoption under separation, and the benchmark distortion grows with c toward the dominant-platform environment of Section 5.2 of the paper.

B.7 Contracts without the Cap

This section supports Section 4 of the paper and Appendix A.3, and states the two propositions that Section 4 reports in words and proves what is analytic in them. The platform posts manufacturer-specific packages (β^1, β^2, a) with the cap $\beta^k \leq 1$ removed; manufacturers play the adoption game; the selection rule of Section 2.3 picks, among its Nash equilibria, the one maximizing joint manufacturer profit. Throughout, π^{bench} is the benchmark profit of a manufacturer, $\Pi^I(a)$ the integrated objective of Section 3 evaluated at the pricing equilibrium, and the linear model is maintained, on the admissible set $\mathcal{A}^*$ of (A.7); every share reported below is a share of that set. All closed forms below are generated and verified symbolically in `rep_core.py` and `rep3_symbolic.py`; the game-consistent searches reported here are in `rep3_contracts.py`; the comparison of the two arrangements is in `rep18_contract_vs_merger.py` and `rep18b_welfare_factor.py`.

CONFIGURATIONS AND TWO IDENTITIES. Three pricing configurations matter: *host-one* (M_1 hosted with perceived benefit $\beta^1 r$, M_2 outside with benefit r , fee a active), *both-host* (both hosted at (β^1, β^2) , fee active), and the *benchmark* (both outside, fee dead). Because applications are a public good across operating systems and the fee operates on the total stock, a manufacturer hosted at share one is in exactly the position of a walker: writing $\pi_2^{both}(\beta^1, \beta^2, a)$ for the hosted rival’s profit and $\pi^{walk}(\beta^1, a)$ for the outsider’s profit against a lone host,

$$\pi_2^{both}(\beta^1, 1, a) \equiv \pi^{walk}(\beta^1, a), \quad (\text{B.30})$$

identically in (β^1, a) , and the host's profit and the platform's profit coincide across the two configurations at $\beta^2 = 1$ as well: a package pair with $\beta^2 = 1$ is *outcome-equivalent* to the single-host package (β^1, a) . Moreover the hosted rival's equilibrium markup satisfies

$$\frac{\partial \hat{m}_2}{\partial \beta^2} = \frac{r(8 + \sigma(8 + \sigma))}{(4 + \sigma)(4 + 3\sigma)} > 0,$$

independently of (β^1, a) : since π_2^{both} is proportional to $\hat{m}_2^2$, any $\beta^2 < 1$ leaves the rival strictly below its walk-off value (B.30) and is refused. This is the no-market-power result of Section 3 in contract form: nothing can be extracted from the second manufacturer.

THE HOST'S PROBLEM. Let $\bar{\beta}(a)$ solve $\pi_{host}(\beta, a) = \pi^{bench}$ (the host's participation constraint, saturated), and let

$$\Phi(a) = S(\bar{\beta}(a), a) - \pi^{bench}, \quad S(\beta, a) = [(p_1 + r)D_1 + a q_S] \Big|_{\text{host-one eq.}},$$

the platform's profit along that constraint. Because the host's equilibrium profit is proportional to the square of its demand, itself linear in $(\beta r, a)$, the equation $\pi_{host} = \pi^{bench}$ is linear in βr given a : $\bar{\beta}$ is *affine*, with $\bar{\beta}(0) = 1$ and $\bar{\beta}'(0) > 0$ at every admissible point (the fee depresses the shared stock, so the host must be compensated). Three facts, established symbolically (`rep3_symbolic.py`):

- (i) $\Phi(0) = 0$, and Φ is a quadratic polynomial in a with the a -independent second derivative

$$\Phi'' = - \frac{2[\gamma(\sigma + 2) + 2(3\sigma + 4)]}{8 + \sigma(8 + \sigma)} < 0 \quad \text{for all } \sigma > -1 :$$

strict concavity holds on the whole admissible set, as a polynomial identity rather than a grid check.

- (ii) The slope at the origin is, in closed form,

$$\Phi'(0) = \frac{2(v + r)(\sigma + 2) [(8 + \sigma(8 + \sigma))u_S - (4 + 3\sigma)u_B]}{(1 - 2u_B u_S)(\sigma + 4)(8 + \sigma(8 + \sigma))},$$

so that $\text{sign } \Phi'(0) = \text{sign}[(8 + \sigma(8 + \sigma))u_S - (4 + 3\sigma)u_B]$, the frontier h_Φ of the paper, and the unconstrained maximizer is

$$\hat{a} = - \frac{\Phi'(0)}{\Phi''} = \frac{(v + r)(\sigma + 2) [(8 + \sigma(8 + \sigma))u_S - (4 + 3\sigma)u_B]}{(1 - 2u_B u_S)(\sigma + 4) [\gamma(\sigma + 2) + 2(3\sigma + 4)]}. \quad (\text{B.31})$$

- (iii) The *naive envelope identity is false*: $\Phi'(0) \neq \Pi'(0)$. Although the compensating and integrated equilibria coincide at $a = 0$, both derivatives carry a first-order term in the *rival's* price response dp_2/da (the pair's surplus is not at a p_2 -optimum), and that response differs across the two regimes, because the host's reaction to the fee runs through the compensation $\bar{\beta}'(0)$ while the integrated firm's runs through the fee base aD_S . The wedge $\Phi'(0) - \Pi'(0)$ takes both signs on the admissible set (nonnegative at 680, 885, 1,008, 1,024, 952 and 845 of the 1,380, 1,843, 2,186, 2,373, 2,445 and 2,446 points of $\mathcal{A}^*$ at $\gamma = 1/2$ to 16 on the reference grid; `rep3_contracts.py`); its zero is not at $\underline{h}$, and the resulting frontier ordering is the subject of the next block.

- (iv) Off the participation constraint, rents to the host lower the platform's take: $P(\beta, a) = (1 - \beta)rD_1 + aD_S$ is strictly concave in β , and its derivative at $\bar{\beta}(a)$ is proportional to $-[(v + r) - a(u_S - u_B)]$, with positive factor $r(\sigma + 2)/[(\sigma + 4)(1 - 2u_B u_S)]$ (symbolic; `b5i.py`). Hence $P \leq \Phi(a)$ on $\beta \geq \bar{\beta}(a)$ whenever $a(u_S - u_B) \leq v + r$: always in the region $u_B \geq h_\Phi$, where $u_B \geq 2u_S$; and at $\hat{a}$ throughout the single-host range: on the admissible grid the bound fails at $\hat{a}$ only for $u_S \geq 1.18$, points at which the two-host family below dominates the single-host value in any case (`b5i.py`).

Together: in the region $u_B \geq h_\Phi(u_S)$, $\Phi(a) < 0$ for all $a > 0$ (Proposition 6(i)); in the region $u_B < h_\Phi(u_S)$, Φ attains the unique interior maximum (B.31) (Proposition 6(ii)). Because the wedge takes both signs, $\hat{a}$ can fall short of or exceed a^* , and $\Phi(\hat{a})$ can even slightly exceed the integrated gain (101.5% of it at $(u_B, u_S) = (0.1, 0.3)$), while it falls to zero as $u_B \uparrow h_\Phi$; Table B.1 quantifies.

One byproduct of the computations deserves note: the pair surplus $S(\beta, a)$ is *not* bounded by the integrated value $\Pi^I(a)$. At $a = 0$, for $\sigma > 0$ and β close enough to one, shares $\beta < 1$ soften the host, the rival responds softly, and $S(\beta, 0) > \pi^{bench} = \Pi^I(0)$: delegation to a soft host is a commitment device in the sense of the strategic-delegation literature. Those packages are irrelevant here only because a host offered $\beta < 1$ walks; the platform can never collect the softness rent, and the binding calibration forces $\bar{\beta}(a) > 1$, aggression rather than softness, as soon as $a > 0$. The first-order cost or benefit of that aggression, relative to integrated pricing, is exactly the wedge of item (iii): deep in the unraveling region the compensating share doubles as a profitable commitment (whence the ratios above 100% in Table B.1), while near $\underline{h}$ the compensation-induced aggression outruns the integrated firm's own and the wedge turns negative; hence the band $h_\Phi \leq u_B < \underline{h}$ of the paper.

THE COMPENSATING PATH: WELFARE. Along the compensating family $(\bar{\beta}(a), a)$ the equilibrium has an exact structure: the host's demand is pinned at its benchmark level (this is the calibration), so the host's profit is constant at π^{bench} , the compensation is affine with $\bar{\beta}'(a) = 2u_B(3\sigma + 4)/[r(\sigma^2 + 8\sigma + 8)]$, and $\Phi'' = 2dq_S/da$: the compensating problem is concave exactly where the fee shrinks the application stock along the path. Every other equilibrium object is affine in a , and the four surplus changes are quadratics vanishing at $a = 0$ whose marginal coefficients are, in closed form,

$$\frac{dm_2}{da} = -\frac{4u_B(\sigma + 1)}{\sigma^2 + 8\sigma + 8}, \quad \Delta V'_B(0) = -\frac{2u_B(v + r)(\sigma + 1)(\sigma + 2)^2}{(\sigma + 4)(1 - 2u_B u_S)^2(\sigma^2 + 8\sigma + 8)},$$

$$\Delta V'_S(0) = q_S^{bench}(dq_S/da), \text{ and}$$

$$\Delta W'(0) = -\frac{2u_B(v + r)(\sigma + 2)[(\sigma + 1)(\sigma + 2)(1 + 2u_S^2) + (7\sigma + 8)(1 - 2u_B u_S)]}{(\sigma + 4)(1 - 2u_B u_S)^2(\sigma^2 + 8\sigma + 8)},$$

whose bracket is a sum of two positive terms. All four are strictly negative on the admissible set: at first order, the outsider, the developer, the buyers and total welfare all fall with no frontier condition, while the platform gains where $u_B < h_\Phi(u_S)$, which is the region of the statement (the integrated counterpart of the last statement requires $N_1 > 0$). At $\hat{a}$ itself the four signs are a numerical statement: all negative at every admissible single-host grid point (1,169 points at $\gamma = 1$ and 1,689 at $\gamma = 4$, no exception). Script: `verifK.py`.

BOTH-HOST SURPLUS, h_J , AND THE RATIO-FORM COMPARISON. For symmetric both-host packages, define the consolidated surplus $J(a, \beta) = \sum_k (\hat{p}_k + r) D_k + a q_S$ at the symmetric delegated equilibrium. Direct computation gives

$$\frac{\partial J}{\partial \beta} \Big|_{\beta=1} = \frac{2r(\sigma+2)[\sigma(u_B + u_S) + 4u_S]}{(1 - 2u_B u_S)(\sigma+4)^2} (a - a_J), \quad a_J \equiv \frac{(v+r)\sigma}{\sigma(u_B + u_S) + 4u_S}, \quad (\text{B.32})$$

so that pushing both hosted manufacturers into aggression destroys consolidated surplus at all fees below the threshold a_J (at the reference points of Table B.1, $a_J = 2.65, 2.53, 0.91$ and 2.74 respectively), and

$$J'(0) \Big|_{\beta=1} = \frac{2(v+r)(\sigma+2)}{(1 - 2u_B u_S)(\sigma+4)^2} \left[(4+\gamma)u_S - (4+2(1+\gamma)u_S^2)u_B \right], \quad (\text{B.33})$$

whose bracket defines the frontier h_J of Appendix A.3. All three frontiers of the contract analysis admit a common *ratio form*: at any point, using $2(1+\gamma)u_B u_S = \gamma - \sigma$,

$$\begin{aligned} J'(0) > 0 &\iff \frac{u_B}{u_S} < R_J(\sigma) \equiv \frac{4+\sigma}{4}, & \Phi'(0) > 0 &\iff \frac{u_B}{u_S} < R_\Phi(\sigma) \equiv \frac{8+\sigma(8+\sigma)}{4+3\sigma}, \\ a^{**} > 0 &\iff \frac{u_B}{u_S} < R_a(\sigma) \equiv \frac{3\sigma(\sigma+7)+16+32}{2(\sigma+2)(3\sigma+4)}, \end{aligned}$$

and the chain $R_J < R_\Phi < R_a$ holds as a pair of polynomial identities:

$$R_\Phi - R_J = \frac{\sigma^2 + 16\sigma + 16}{4(3\sigma+4)} > 0, \quad R_a - R_\Phi = \frac{\sigma^2(\sigma+1)}{2(\sigma+2)(3\sigma+4)} \geq 0, \quad (\text{B.34})$$

the second vanishing only at $\sigma = 0$ and increasing in it, its derivative being $\sigma(3\sigma^3 + 20\sigma^2 + 34\sigma + 16)$ over $2(\sigma+2)^2(3\sigma+4)^2$. Since each R is increasing in σ and σ is decreasing in u_B at fixed u_S , each ratio condition crosses a fixed- u_S line exactly once, so the chain translates pointwise into $h_J < h_\Phi \leq \underline{h}$, the last with equality only at $\sigma = 0$: below h_J , even the fully consolidated structure of the platform and both manufacturers gains from a marginal tax; between h_J and h_Φ , the compensating single-host deal and integration both do and the consolidated structure does not; between h_Φ and $\underline{h}$, the *band* of Proposition 6(i), integration alone does. Two cautions. First, (B.32)–(B.33) are marginal statements: J is *not* globally concave in the fee on the admissible set (its second derivative changes sign at scattered points, mostly with $\sigma < 0$; `rep3_contracts.py`), so the exclusion of nonlocal pairs rests on the search below, not on J . Second, the consolidated bound speaks to symmetric pairs at $\beta = 1$; asymmetric two-host packages are covered by the search. A participation-based bound cannot replace the search either: the consolidated surplus net of the two unilateral walk-off payoffs is strictly concave in (β^1, β^2, a) and admits a closed-form maximum, but that maximum exceeds $\max\{\Phi(\hat{a}), 0\}$ at every admissible point tested (`verifK.py`), because what disciplines both-host pairs is the equilibrium structure of the adoption subgame and the selection rule, not participation alone.

THE GAME-CONSISTENT SEARCH. To check the constructions above against the posted-contract game on finite grids, we sweep packages (β^1, β^2, a) , compute the pure Nash equilibria of the 2×2 adoption game at each package (using the closed-form configuration values, with interiority enforced and the ‘indifferent accepts’ convention), apply the joint-profit selection rule, and record the platform’s value at the selected equilibrium; the grid

includes slivers $\beta^1 \in \{\bar{\beta}(a) + \varepsilon\}$ along the compensating path so that the participation-binding packages are hit exactly (`rep3_contracts.py`). At the four reference points the search reproduces the single-host optimum to grid accuracy, *with the second share exactly at one*:

(u_B, u_S)	(0.1, 0.1)	(0.3, 0.1)	(0.5, 0.18)	(0.15, 1.0)
$\Phi(\hat{a})$	0.01818	0.00051	0.00233	4.60545
game maximum	0.01817	0.00051	0.00233	4.60539
at (β^1, β^2, a)	(1.016, 1.000, 0.136)	(1.008, 1.000, 0.023)	(1.028, 1.000, 0.045)	(1.394, 1.000, 1.954)

At the single-host package $(\bar{\beta}(\hat{a}), \hat{a})$ the developer publishes on the platform's store, $u_S D_1 \geq \hat{a}$, at every taxing point of $\mathcal{A}$ at the six values of γ , with a slack of at least 0.006 (`rep3_contracts.py`, block H). On the reference grid coarsened to step 0.1, the taxing region at $\gamma = 4$ contains 67 admissible points; at 52 of them the game maximum reproduces the truncated value $\Phi(\max\{\hat{a}, 0\})$ (at band points both are zero), and at the other 15, all with $u_S \geq 0.9$, a two-host package does strictly better (below). At the 27 subsidizing points of that grid the game maximum is 0 (`rep2_statistics.py`). Dedicated sweeps at band points $((0.34, 0.10), (0.543, 0.17), (0.719, 0.25)$ and $(0.885, 0.40)$ at $\gamma = 4$) also return zero, and they show the rule doing on the band what it does everywhere else: pairs exist whose mutual acceptance is a Nash equilibrium with positive platform profit (e.g. $\beta^1 = \beta^2 \approx 1.07$ financed by a moderate fee), but joint refusal is then an equilibrium as well and is what the manufacturers rank first, so the rule retains it and the platform earns nothing, which is Proposition 6(i) on the band. Lemma B.1 places their interior optimum: with $u_S < u_B$ there, $a^\dagger < 0$, so it is not a taxing pair at all. The exclusion of the family itself rests on the selection rule and is a grid finding. `rep33_bothhost_grid.py` does it on the whole of $\mathcal{A}^* \cap \{u_B \geq u_S\}$ at every γ of the reference calibration (700 to 1,240 points per γ , of which 343 to 729 in the region of Proposition 6(i)): at each point it sweeps $a \in [0.01, 4]$ (40 values), $\beta^1 \in [1, \bar{\beta}(a) + 2]$ (20 values) and $\beta^2 \in [1, \beta^1]$ (8 values), solves the adoption game with the conventions of Section 2.3, and compares the best package at which mutual acceptance is selected with the single-host value $\max\{\Phi(\hat{a}), 0\}$; the former never exceeds the latter (10.6 million packages in all), and in the region of Proposition 6(i) no selected package earns anything. The bounds of that sweep are not arbitrary. The platform's profit at a both-host package, $\sum_k (1 - \beta^k) r D_k + a q_S$ at the pricing equilibrium, is a quadratic in (β^1, β^2, a) , concave on $\mathcal{A}^*$ (Corollary B.3), and every package with shares at or above one that the sweep does not cover lies in one of two convex regions, $\{a \geq 4\}$ and $\{a \leq 4, \beta^1 \geq \bar{\beta}(a) + 2\}$ up to relabeling; the exact maximum of the quadratic on each, computed at every grid point, falls short of the single-host value by at least 2.0 (`rep33_bothhost_grid.py`). Beyond the bounds, then, no package hosting both manufacturers earns as much as the single-host package, whether the rule selects it or not, and packages with a share below one reduce to capped or single-host ones, which Appendix A.3 treats analytically; what rests on the grid is the interior of the swept region. These are computational statements on finite grids, and the ranges differ by experiment: the reference-point sweeps run $\beta^k \leq 1.6$ with the fee up to three times $\hat{a}$, the band sweeps $\beta^k \leq 2$ and $a \leq 1.5$, and the deep sweep below $\beta \leq 20$, $a \leq 25$. Each is stated where it is used; their status is recorded in the closing note of Appendix A.3.

Lemma B.1 (The both-host program). *On the configuration where both manufacturers are hosted, the platform's profit $\Pi(\beta^1, \beta^2, a) = \sum_k (1 - \beta^k) r D_k + a q_S$ at the stage-3*

equilibrium is a quadratic polynomial. Its critical point is unique, and it is symmetric: $\beta^{1\dagger} = \beta^{2\dagger}$ holds identically in the primitives. Writing

$$\Delta \equiv 2(\gamma + 4) - (\sigma + 2)(u_B^2 + u_S^2) - 2(2\gamma + \sigma + 8)u_B u_S, \quad (\text{B.35})$$

the critical point and its value are

$$a^\dagger = \frac{(v + r)(\sigma + 2)(u_S - u_B)}{\Delta}, \quad \Psi \equiv \Pi(\beta^\dagger, \beta^\dagger, a^\dagger) = \frac{(v + r)^2(\sigma + 2)}{\Delta}. \quad (\text{B.36})$$

By symmetry the Hessian is block-diagonal in the directions $(a, \beta^1 + \beta^2, \beta^1 - \beta^2)$; the last carries the eigenvalue $-2r^2(\sigma + 1)(\sigma + 2)/[(3\sigma + 4)M] < 0$, and on the symmetric block the (a, a) entry is negative and the determinant equals $4r^2(\sigma + 2)\Delta$ over positive factors, so $\Delta > 0$ is exactly the second-order condition, which we impose as we impose its analogue in Section B.5. It follows that

$$\text{sign } a^\dagger = \text{sign}(u_S - u_B): \quad (\text{B.37})$$

a two-host package taxes the developer at its interior optimum only where network effects are strictly developer-skewed.

Proof. The pricing equilibrium is linear in $(\beta^1 r, \beta^2 r, a)$, hence so are D_1 , D_2 and q_S , and Π is quadratic. Its three first-order conditions are linear with a unique solution, and $\beta^{1\dagger} - \beta^{2\dagger}$ reduces to zero identically. The closed forms (B.35)–(B.36) and the Hessian identity are direct computation (`rep9_dnr.py`). Since $v + r > 0$ and $\sigma + 2 > 0$, (B.37) follows from $\Delta > 0$. $\square$

Corollary B.3 (The second-order condition holds on the admissible set). $\Delta > 0$ everywhere on $\mathcal{A}^*$ of (A.7), so Lemma B.1 applies without further qualification wherever Section 4 of the paper is stated.

Proof. Substituting $\gamma = (\sigma + 2u_B u_S)/(1 - 2u_B u_S)$, the Δ of (B.35) is the same polynomial as the concavity object Δ_δ of Section B.5.4,

$$\Delta = 2(\sigma + 4) - 8u_B u_S - (\sigma + 2)(u_B + u_S)^2,$$

identically in (u_B, u_S, σ) . The identity established there,

$$(18\sigma^3 + 100\sigma^2 + 192\sigma + 128)\Delta = -2(\sigma + 4)N_2 + 2(13\sigma^3 + 64\sigma^2 + 112\sigma + 64)(u_B - u_S)^2,$$

has positive coefficients in both bracketed polynomials, so $N_2 < 0$ implies $\Delta > 0$. Symbolic verification: `rep9_dnr.py` and `rep21_dominance_analytic.py`. $\square$

The two objects coincide for a reason. Both ask when the platform's profit is concave in the pair (share, fee) it posts to a manufacturer whose participation constraint is the binding one, and the constraint has the same shape in the two environments; the two-host program of this section and the benchmark program of Section B.5 therefore have the same Hessian up to positive factors, and the divide-and-conquer package of the latter is the single-host package of this section with the premium in the role of the rebate. That $\Delta > 0$ is not implied by $H > 0$ alone is easy to see at $\sigma = 0$, where $H > 0$ reads $u_B^2 + 4u_B u_S + u_S^2 < 4$ while $\Delta > 0$ reads $u_B^2 + 6u_B u_S + u_S^2 < 4$: the two differ by $2u_B u_S$, which is $\gamma/(1 + \gamma)$ there. This is why Section 4 is stated on $\mathcal{A}^*$ and not on $\mathcal{A}$.

Remark B.1 (No envelope shortcut). *At $a = 0$ the compensating and the integrated equilibria coincide, which invites the envelope conclusion $\Phi'(0) = \Pi^I'(0)$, and with it $h_\Phi = \underline{h}$. The conclusion is false. Both derivatives contain a term in dp_2/da : the pair's surplus is not at a p_2 -optimum, so the rival's response enters at first order, and it differs across the two regimes because the host's price response to the fee, which runs through the compensation $\bar{\beta}'(0)$, differs from the integrated firm's. The wedge $\Phi'(0) - \Pi^I'(0)$ takes both signs on the admissible set; what is unambiguous is the frontier ordering (B.34), whence the band of Proposition 6(i). The band is the same identity read at $K < 0$: there the compensating share that keeps the host at its benchmark would commit it to fiercer pricing than the merged firm chooses, the rival's answering cut falls on the deal, and near $\underline{h}$, where the fee's own gain is thin, the deal loses money at fee levels at which integration still gains. It closes at $\sigma = 0$, where there is no rivalry and no answering cut. The wedge is quantified below.*

TWO-HOST PACKAGES. Deep in the developer-skewed range the optimum changes nature. The platform posts *two* generous shares ($\beta^1 = \beta^2 > 1$) financed by a high fee. Unilateral acceptance of either package is strictly profitable ($\pi_{host}(\beta^k, a) > \pi^{bench}$), so joint refusal is no longer a Nash equilibrium of the adoption game, and mutual acceptance is one: the selection rule has nothing to select. On the reference grid the pair beats the single-host package at 29, 66, 112, 162, 213 and 205 of the 847, 1,170, 1,457, 1,692, 1,899 and 2,103 admissible points of the unraveling region at $\gamma = 1/2$ to 16, all with $u_S \geq 1.30, 1.22, 1.14, 1.04, 0.96$ and 0.90 respectively (`rep9_dnr.py`). What it does there is recorded in `rep19_remark_table.py`: both manufacturers earn more than their benchmark at 76 to 97% of those points and so do buyers, both devices being cheaper; the developer is below its benchmark at 75 to 100%, total welfare at 35 to 72%. The family is a coalition of the three firms financed by the application commons rather than a deeper distortion of the device market, and the divide-and-rule reading of Segal [1999], in which both manufacturers are held below their benchmark, applies only at the 3 to 24% of those points where the pair's rebates fall short of that level; the point (0.215, 1.106) used as an illustration in an earlier draft was one of them. The package sells devices at a negative price on the strength of the rebate, the margin $p_k + \beta^k r$ remaining positive, at none of those points at $\gamma = 1/2$, a third at $\gamma = 1$, and 74, 95, 100 and 100% at $\gamma = 2$ to 16, which is where the linear model stops describing anything. The switch between the two families occurs, at $\gamma = 4$, near $u_S \approx 1$: on a u_S -grid of step 0.01, between 1.07 and 1.08 at $u_B = 0.1$, between 1.02 and 1.03 at $u_B = 0.15$, and between 0.95 and 0.96 at $u_B = 0.25$ (`rep2_statistics.py`).

Deeper still, the optimum survives but runs away in magnitude. A dedicated sweep of symmetric pairs ($\beta \leq 20$, $a \leq 25$, regime-aware payoffs, the adoption game solved and the selection rule applied at each package; `b5i.py`) finds an *interior* maximizer at every admissible grid point in the developer-skewed range, the optimal pair growing from about (2, 2.7) at $u_S \approx 1.1$ to $(\beta, a) \approx (7, 6)$ near the admissibility frontier. Beyond the admissible set the drift continues: at (0.316, 1.194), a point that lies outside $\mathcal{A}^*$ because $N_2 > 0$ there, the maximizer sits near $(\beta, a) \approx (9.3, 7.2)$ with value ≈ 31.6 ; the profit function does eventually turn, but at rebates and fees several times the market's scale, where the globally linear demand system, and hence the model, has stopped being meaningful. We read this runaway as one more face of the fragility of unrestricted two-part contracting in these environments. We state the comparison of Propositions B.5 and B.6 on the single-host family. Wherever the two-host optimum exceeds the integration

gain, it sells both devices at a negative price; we impose no price floor, and read this as the linear model's own signal that it has been pushed beyond its range.

Remark B.2 (Discriminatory vs. uniform posting). *The single-host construction uses manufacturer-specific shares. Under uniform posting ($\beta^1 = \beta^2 = \beta$), the host-one outcome is generally not implementable: at the package $(\bar{\beta}(\hat{a}), \hat{a})$ the second manufacturer strictly prefers joining to walking at every point of the unraveling region, and analytically so: $\bar{\beta}(\hat{a}) > 1$ whenever $\hat{a} > 0$, and $\partial \hat{m}_2 / \partial \beta^2 > 0$, so a manufacturer offered more than the full per-user benefit does better than the walker it would otherwise be. The best uniform both-accept package then earns less than $\Phi(\hat{a})$ at the four reference points (0.006 against 0.018 at (0.1, 0.1); 0 at (0.3, 0.1) and (0.5, 0.18); 4.545 against 4.605 at (0.15, 1.0); `rep2_statistics.py`), and wherever the symmetric two-host package, itself uniform, does not win (Remark B.3). Discrimination is thus needed for the replication of integration (the single-host optimum), not for the existence of rents: some uniform packages earn positive (if smaller) profit, and the two-host packages deep in the developer-skewed range are themselves symmetric.*

THE CHOICE BETWEEN INTEGRATION AND SEPARATION. On the unraveling region of Proposition 6(ii) the platform has two ways of taxing the developer, and two questions follow: which one it prefers, and which one costs the market more. Both gains are in closed form, the separation gain $\Phi(\hat{a})$ and the integration gain $\Pi^I(a^{**}) - \Pi^I(0)$ of Section 3 of the paper, and their comparison is a single polynomial inequality.

Proposition B.5 (Integration or separation). *Let $u_B < h_\Phi(u_S)$. The platform prefers integration to separation at its best single-host package if and only if (B.39) holds. As network effects vanish at a fixed ratio u_B/u_S , the condition becomes*

$$u_B/u_S > R_p(\sigma) \equiv \frac{R_\Phi(\sigma)}{2 + \sigma} = \frac{\Lambda}{(2 + \sigma)(4 + 3\sigma)}. \quad (\text{B.38})$$

Proof of Proposition B.5. Both gains are the vertices of concave quadratics with zero constant term. By Lemma A.1, $\Phi(\hat{a}) = \Phi'(0)^2 / (-2\Phi'')$ with $\Phi'(0) = 2(v + r)(\sigma + 2)[\Lambda u_S - (4 + 3\sigma)u_B] / [(1 - 2u_B u_S)(\sigma + 4)\Lambda]$ and Φ'' as in (A.18); by (A.9) and the second derivative displayed before it, $\Pi^I(a^{**}) - \Pi^I(0) = \Pi^{I'}(0)^2 / (-2\Pi^{I''})$ with $\Pi^{I'}(0) = 2(v + r)[\Theta u_S - 2(\sigma + 2)(3\sigma + 4)u_B] / [(\sigma + 4)^2(3\sigma + 4)(1 - 2u_B u_S)]$, where $\Theta \equiv 3\sigma(\sigma + 7) + 16 = 2(\sigma + 2)(3\sigma + 4)R_a(\sigma)$, and $\Pi^{I''} = -2H / [(\sigma + 4)^2(3\sigma + 4)^2(1 - 2u_B u_S)]$. Hence integration is preferred to separation if and only if

$$(1 - 2u_B u_S) \Lambda [\gamma(\sigma + 2) + 2(3\sigma + 4)] [\Theta u_S - 2(\sigma + 2)(3\sigma + 4)u_B]^2 \geq (\sigma + 2)^2 H [\Lambda u_S - (4 + 3\sigma)u_B]^2, \quad (\text{B.39})$$

the separation gain minus the integration gain being the right side of (B.39) minus the left, divided by $(1 - 2u_B u_S)^2(\sigma + 4)^2\Lambda[\gamma(\sigma + 2) + 2(3\sigma + 4)]H/(v + r)^2 > 0$.

For the limit, let $u_B, u_S \rightarrow 0$ at fixed $\rho = u_B/u_S$: then $\sigma \rightarrow \gamma$, $1 - 2u_B u_S \rightarrow 1$, $H \rightarrow (\sigma + 4)^2(3\sigma + 4)^2$, and $\gamma(\sigma + 2) + 2(3\sigma + 4) \rightarrow \sigma(\sigma + 2) + 2(3\sigma + 4) = \Lambda$. Dividing by u_S^2 , the right side of (B.39) minus the left becomes $\sigma^2 [(2 + \sigma)(4 + 3\sigma)\rho - \Lambda] L(\rho)$, where L is affine in ρ with $L(0) = -(3\sigma + 8)\Lambda(2\sigma^2 + 9\sigma + 8) < 0$ and $L(R_\Phi) = -\sigma^2(\sigma + 1)\Lambda \leq 0$, using $\Theta - 2\Lambda = (\sigma + 1)(3\sigma + 4)(\sigma + 4)$. So $L < 0$ on $[0, R_\Phi]$, and integration is preferred if and only if $(2 + \sigma)(4 + 3\sigma)\rho > \Lambda$, which is (B.38).

The identities behind (10): substituting $\bar{\beta}(a) = 1 + 2u_B a(3\sigma + 4)/(r\Lambda)$ and $\beta^2 = 1$ into (A.2), and $b_2 = r$ into (A.3)–(A.4), the single-host package and integration at a common fee a differ by

$$(p_1^s - p_1^I, p_2^s - p_2^I, D_S^s - D_S^I) = \frac{aK}{(4 + \sigma)\Lambda} \left(\frac{4(2 + \sigma)}{4 + 3\sigma}, \frac{2\sigma}{4 + 3\sigma}, -\frac{2u_S}{1 - 2u_B u_S} \right), \quad (\text{B.40})$$

s standing for the separation package. On $K = 0$ the stage-3 outcomes coincide at every fee. The host is held at its benchmark, $(p_1^s + \bar{\beta}r)D_1^s = \pi^{\text{bench}}$, so its compensation $(\bar{\beta} - 1)rD_1^s$ equals $\pi^{\text{bench}} - (p_1^s + r)D_1^s = -[(p_1^I + r)D_1^I - \Pi^I(0)]$ there, and $\Phi(a) = aD_S^s - (\bar{\beta} - 1)rD_1^s = \Pi^I(a) - \Pi^I(0)$ identically in a ; the two vertices coincide, $\hat{a} = a^{**}$, and so does every surplus. $\square$

In words: integration is preferred when network effects are strong, balanced, or tilted toward buyers; separation when they are weak in level and tilted toward the developer, the buyers' network benefit falling short of $R_p(\sigma)$ times the developer's, 81% at $\sigma = 1$ and 42% at $\sigma = 16$. The stronger the rivalry, the more the market must lean toward the developer for separation to win. On the reference grid the platform prefers separation on 4.9% of the admissible set at $\gamma = 1/2$, then 6.6, 8.3, 9.6, 10.2 and 11.3% at $\gamma = 1, 2, 4, 8, 16$, that is on 8 to 13% of the region where both instruments are active. Its preference is a thin one, a median 0.1 to 0.7% of the integration gain and never above 3.2%: on that region the two arrangements are near-copies of one another.

Proposition B.6 (What the choice costs). *Let $u_B < h_\Phi(u_S)$, and let separation be at its best single-host package. Total welfare, buyer surplus and developer surplus are lower under separation than under integration if and only if $K > 0$, that is $u_B/u_S < R_p(\sigma)$. The factor K divides the welfare difference exactly; the sign of the cofactor, and the two surplus rankings, are verified on the reference grid at $\gamma \in \{1/2, 1, 2, 4, 8, 16\}$. On the region where the platform prefers separation, the rival's profit ranks the other way: higher under separation if and only if $K > 0$.*

Proof of Proposition B.6. Write the two welfare levels as rational functions of (u_B, u_S, σ) on the plane, with $\gamma = (\sigma + 2u_B u_S)/(1 - 2u_B u_S)$, separation at $\hat{a}$ and integration at a^{**} . Their difference has a denominator that is a product of squares and of $(\sigma + 1)(\sigma + 4)^4(3\sigma + 4)^2$, hence positive, and a numerator that K divides exactly, with cofactor $8(v + r)^2(\sigma + 1)(\sigma + 4)^2(3\sigma + 4)^2 Q(u_B, u_S, \sigma)$ for a polynomial Q of degree 15 in σ (script `rep18b_welfare_factor.py`). The sign of Q is checked at every point of the reference grid on which $\hat{a} > 0$ and $a^{**} > 0$, at $\gamma \in \{1/2, 1, 2, 4, 8, 16\}$: it is negative at all of them (847 to 2,103 points per γ), so that on that domain $W^s < W^I$ if and only if $K > 0$. Buyer surplus and developer surplus rank with total welfare at every one of those points; the rival's profit ranks against it at every point of the region where the platform prefers separation, and at 89 to 95% of the whole domain, the exceptions being strongly buyer-skewed points with $K < 0$ at which the lower separation fee $\hat{a} < a^{**}$ lifts the application stock enough for the rival to earn more under separation (`rep18_contract_vs_merger.py`, block B2). $\square$

The platform thus prefers separation, where it does, mostly on the side of the curve on which separation is the dearer copy: on 97% of its preferred region at $\gamma = 1/2$, and 97, 95, 92, 85 and 74% at $\gamma = 1$ to 16. Figure OA-5 draws the three propositions on one plane. The two-host family is left out of the comparison: its interior optimum exceeds the integration gain only at contract values at which both device prices are negative, and we read that as the linear model's signal rather than impose a price floor (Remark B.3).

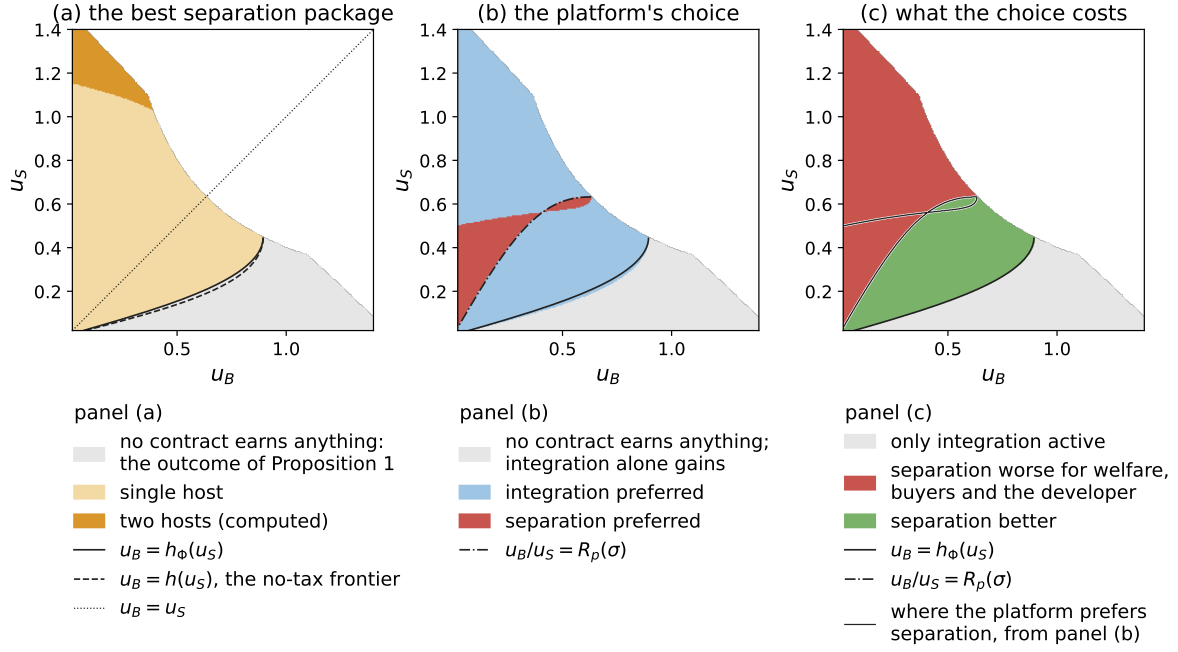


Figure OA-5: Section 4 on the plane of Figure 1, at $\gamma = 4$, $v = 2$, $r = 1/2$. (a) Proposition 6: on the grey region, 28% of the admissible set, no contract earns anything and the outcome is that of Proposition 1; the band is the sliver between h_Φ (solid) and $\underline{h}$ (dashed), 1.1% of the set (0.1% at $\gamma = 1$, 3.6% at $\gamma = 16$). Below h_Φ the optimum is a single-host package except in the darker corner, above the diagonal, where a two-host package wins. (b) Proposition B.5: the platform prefers integration to its best single-host package on the blue region and separation on the red one, 9.4% of the set; the dash-dotted curve is $u_B/u_S = R_p(\sigma)$. The red region is a cone to the left of that curve, at weak and developer-skewed network effects, plus a thin tongue to its right at strong ones. (c) Proposition B.6, on the region where both instruments are active: separation is worse than integration for welfare, buyers and the developer on the red side of the curve and better on the green side; the black contour is the red region of (b), 91% of which lies on the red side. White is inadmissible. All shares in this caption are on a step-0.002 grid over $[0.02, 1.40]^2$. Scripts: `rep3_contracts.py`, `rep9_dnr.py`, `rep18_contract_vs_merger.py` and `band_share.py`, drawn by `rep18_figure.py`.

Remark B.3 (Two hosts deep in the developer-skewed region). A two-host package taxes at its interior optimum only where network effects are strictly developer-skewed, $u_S > u_B$ (Lemma B.1), and beats the single-host package only deeper in that range: on the reference grid, at 3 to 11% of the admissible points of the unraveling region, all with $u_S \geq 0.90$ (at $\gamma = 4$, 162 of 1,692 points, all with $u_S \geq 1.04$). What it does there is not what the single host does. The platform posts two rebates generous enough that each manufacturer accepts whatever the other does, and funds them with a higher fee on the developer, 1.1 to 1.7 times $\hat{a}$ at the median and up to 3.7 times. Both manufacturers earn more than their benchmark at 76 to 97% of those points, buyers gain at the same points (both devices are cheaper), the developer loses at 75 to 100%, and total welfare falls at 35 to 72%: the family is a coalition of the three firms financed by the developer rather than a deeper distortion of the device market. Two caveats bound its reach. The

package sells devices at a negative price on the strength of the rebate (the margin $p_k + \beta^k r$ stays positive) at most of those points once $\gamma \geq 2$, 74 to 100%, which is where the linear model stops describing anything; and beyond the admissible set the optimum runs away to contract values several times the market's scale. Wherever its interior optimum exceeds the integration gain, which happens at no point for $\gamma \leq 2$, at one point at $\gamma = 4$ and at 101 and 146 points at $\gamma = 8$ and 16, both device prices are negative; we impose no price floor, and state the comparison of Propositions B.5 and B.6 on the single-host family (`rep19_remark_table.py`).

STATUS OF THE PROOFS. It is worth recording exactly which steps are analytic and which are computational. Analytic: the concavity (A.18), the frontier (9), the ordering (B.34) and $R_p = R_\Phi/(2 + \sigma) \leq 1 < R_\Phi$ (all polynomial identities), the free-riding exclusion of subsidies, the exclusion of taxes with both shares at or below one via (A.21) and Proposition 1, the closed forms $\hat{a}$, $\hat{\beta}$, $\Phi(\hat{a})$, the dominance of constraint-saturating single-host packages (last part of Lemma A.1), the second-order condition of the two-host program on $\mathcal{A}^*$ (Corollary B.3) and with it the placement of its interior optimum (Lemma B.1), the comparison (B.39) and its limit (B.38), the outcome identities (B.40), and the exact division of the welfare difference by K . Computational, on the admissible set at the reference calibrations and with the protocol of Online Appendix B.7: the exclusion of two-host packages wherever $u_B \geq u_S$, which rests on the selection rule (within the bounds of the sweep; beyond them it follows from concavity), the region where (A.23) holds, the interiority of the two-host optimum, the welfare signs at the unrestricted optimum, the sign of the cofactor Q in Proposition B.6, and the shares reported after Propositions B.5 and B.6. The three propositions should be read accordingly: exact where the objects have closed forms, and computational findings on a specified domain elsewhere.

QUANTITATIVE ILLUSTRATION. Table B.1 reports the single-host optimum, the integrated optimum and the separation benchmark at four reference points ($\gamma = 4$, $v = 2$, $r = 1/2$), chosen one in each situation of Section 4: a point where the platform prefers separation ($K > 0$, first column), one where it prefers integration with $K < 0$ (second), one where it prefers integration although $K > 0$, deep in the developer-skewed region where network effects are strong in level (third), and one close to the frontier h_Φ (fourth). The closed-form values coincide with the regime-aware game engine to 10^{-3} at all four points (the publication condition $u_{sn}^I \geq a$ holds at the relevant configurations; it starts binding only beyond the two-host frontier). Script: `rep19_remark_table.py`.

INTEGRATION OR SEPARATION. Table B.2 reports, for each γ , the share of the admissible set on which the platform prefers separation, at its best single-host package, to integration (Proposition B.5), how thin that preference is, how much of the preferred region lies below R_p , and what the two arrangements do to the other parties there (medians over the preferred region, in percent of the benchmark). The two-host family is excluded from the comparison: wherever its interior optimum exceeds the merger's gain, both device prices are negative. Script: `rep18_contract_vs_merger.py`; the exact division of the welfare difference by K is `rep18b_welfare_factor.py`.

B.8 More Manufacturers

This section supports Section 5.5 of the paper. Every step is checked against the two-manufacturer model at $N = 2$. Scripts: `rep6_manufacturers.py`, which drives the files

(u_B, u_S)	(0.10, 0.40)	(0.30, 0.30)	(0.15, 1.00)	(0.50, 0.18)
u_B/u_S against $R_p(\sigma)$	0.25 vs 0.60	1.00 vs 0.63	0.15 vs 0.66	2.78 vs 0.63
a^{**} (integration)	0.710	0.413	2.307	0.072
$\hat{a}$ (separation)	0.709	0.407	1.961	0.046
$\hat{\beta} = \hat{\beta}(\hat{a})$	1.084	1.153	1.395	1.029
separation gain $\Phi(\hat{a})$	0.5259	0.1835	4.6055	0.0023
integration gain $\Pi^I(a^{**}) - \Pi^I(0)$	0.5198	0.1893	5.3347	0.0056
ratio	101.2%	96.9%	86.3%	41.7%
rival's profit	1.214/1.174 (1.317)	1.342/1.360 (1.542)	1.354/0.831 (1.902)	1.504/1.495 (1.542)
applications q_S	0.860/0.879 (1.602)	0.863/0.849 (1.314)	2.597/2.968 (4.945)	0.738/0.706 (0.788)
buyer surplus V_B	3.851/3.946 (4.009)	4.481/4.422 (4.796)	5.202/7.157 (6.113)	4.736/4.669 (4.796)
developer surplus V_S	0.370/0.386 (1.283)	0.372/0.360 (0.863)	3.372/4.404 (12.227)	0.272/0.249 (0.311)
welfare W	7.279/7.343 (7.927)	7.921/7.874 (8.744)	16.436/19.628 (22.144)	8.056/7.961 (8.191)

Table B.1: Separation at its single-host optimum and integration at its optimum, at four reference points ($\gamma = 4$, $v = 2$, $r = 1/2$). Lower block: separation / integration (benchmark). The first column is the cone of Proposition B.5 where separation is preferred, by 1.2% of the integration gain, and where it is the dearer copy for everyone but the rival: q_S , V_B , V_S and W are all lower than under integration, and only the rival earns more. The second and fourth columns have $K < 0$ and the reverse rankings for q_S , V_B , V_S and W ; in the fourth, a strongly buyer-skewed point outside the region where separation is preferred, the rival too earns slightly more under separation, the lower fee $\hat{a} < a^{**}$ lifting the application stock. The third has $K > 0$ but strong network effects, so that integration is preferred and yet separation is still the dearer copy; there integration raises buyer surplus above the benchmark while separation lowers it. The host is held at π^{bench} by construction in every column.

γ	1/2	1	2	4	8	16
separation preferred, share of $\mathcal{A}^*$	4.9%	6.6%	8.3%	9.6%	10.2%	11.3%
$\Phi(\hat{a})/[\Pi^I(a^{**}) - \Pi^I(0)] - 1$, med.	0.11%	0.32%	0.55%	0.67%	0.52%	0.35%
maximum	0.56%	1.42%	2.54%	3.16%	2.80%	1.94%
of which below R_p (dearer copy)	97%	97%	95%	92%	85%	74%
welfare, separation / integration	-1.6/ - 1.5	-3.0/ - 2.6	-5.5/ - 4.9	-8.4/ - 7.8	-11.9/ - 11.3	-15.8/ - 15.4
buyers, separation / integration	-0.8/ - 0.4	-1.5/ - 0.6	-2.8/ - 1.4	-4.4/ - 2.7	-6.5/ - 5.0	-9.2/ - 8.3
developer, separation / integration	-58.1/ - 57.9	-62.4/ - 61.9	-65.9/ - 65.0	-68.9/ - 68.1	-71.2/ - 70.6	-72.7/ - 72.3
rival, separation / integration	-1.6/ - 1.7	-3.0/ - 3.4	-5.5/ - 6.7	-8.7/ - 11.6	-12.8/ - 17.4	-18.0/ - 23.9

Table B.2: Integration or separation on the reference grid ($v = 2$, $r = 1/2$). The lower block gives median changes from the benchmark, in percent, over the region on which the platform prefers separation: separation is the dearer copy for welfare, buyers and the developer, and the cheaper one for the rival, on the share of that region reported in the fourth row.

of $\text{ext}/$.

DEMAND AND EQUILIBRIUM. With N symmetric manufacturers and $\bar{p} = \frac{1}{N} \sum_j p_j$, buyers and the developer give

$$M_N D_k = v - (1 + \sigma_N) p_k + \sigma_N \bar{p} - u_B a, \quad M_N q_S = u_S \left(Nv - \sum_j p_j \right) - a,$$

with $\sigma_N = \gamma - Nu_B u_S (1 + \gamma)$ and $M_N = 1 - Nu_B u_S$, so that $(1 + \gamma)M_N = 1 + \sigma_N$ exactly as in the paper. The contraction condition is $Nu_B u_S < 1$. At stage 3 the integrated firm maximizes $(p_1 + r)D_1 + a q_S$ against $N - 1$ rivals each maximizing $(p + b)D$;

imposing symmetry among the rivals after differentiating, the equilibrium is in closed form (`ext/extN_eq2.py`), and at $N = 2$, $b = r$ it reproduces `rep_core.prices_int` identically. The two factors of the two-manufacturer model become $A_N = N(\sigma + 2) - \sigma$ and $B_N = 2N(\sigma + 1) - \sigma$, equal to $4 + \sigma$ and $4 + 3\sigma$ at $N = 2$.

HARM TO A NON-INTEGRATED MANUFACTURER. The envelope argument of Proposition 3 runs unchanged:

$$\frac{d\pi_{\text{rival}}}{da} = (p^I + r) \frac{dD_{\text{rival}}}{da}, \quad \frac{dD_{\text{rival}}}{da} = - \frac{[\sigma(N-1) + N] [u_B B_N + \sigma u_S]}{M_N A_N B_N} < 0$$

on the admissible set, with no ambiguity of sign at any N : a rival is harmed by a higher fee whenever its margin is positive, which interiority supplies, hence by the merger if and only if $a^{**}(N) > 0$.

THE FRONTIER. Writing $a^{**}(N)$ over positive factors, the bracket that carries its sign is linear in (u_B, u_S) , and $a^{**}(N) > 0$ if and only if $u_B/u_S < R_a^N(\sigma)$ with

$$R_a^N(\sigma) = \left(2N^3\sigma^3 + 8N^3\sigma^2 + 10N^3\sigma + 4N^3 - 5N^2\sigma^3 - 13N^2\sigma^2 - 8N^2\sigma + 4N\sigma^3 + 4N\sigma^2 - \sigma^3 + \sigma^2 \right) / \left(2[N(\sigma+1) - \sigma][2N(\sigma+1) - \sigma] \right).$$

At $N = 2$ this is R_a of (A.10). Three properties are identities. First, $R_a^N(0) = N$ exactly. Second, $R_a^N/N \rightarrow (\sigma + 2)/2$ as $N \rightarrow \infty$, the leading term of the numerator being $2N^3(\sigma + 1)^2(\sigma + 2)$ and that of the denominator $4N^2(\sigma + 1)^2$. Third, R_a^N is strictly increasing in N : writing $N = 2 + m$, $m \geq 0$, every coefficient of the expanded numerator of $\partial R_a^N / \partial N$ is positive, as is every coefficient of the denominator, which is the method of `verifI.py` (`ext/extN_frontier.py`). Numerically, R_a^N at $\sigma = 0, 1, 2, 4$ takes the values 2.00, 2.48, 2.95, 3.92 at $N = 2$ and 10.00, 14.49, 18.99, 27.98 at $N = 10$.

WELFARE. On the admissible set of the paper transposed to N (concavity together with a positive rival demand at a^{**}), and on the reference grid, both halves of Proposition 5 survive, though as grid findings rather than as theorems: $\Delta W'(0) < 0$ is not an identity at general N . Welfare rises at every subsidizing point, at every γ and every $N \in \{2, 3, 5, 10\}$ (at $\gamma = 4$: 738 of 738 points at $N = 2$ and 442, 238 and 99 of as many at $N = 3, 5, 10$); and among the taxing points whose fee lies in the lowest decile of a^{**} , which is the neighborhood the proposition speaks of, it falls at every one, again at every γ and N (177, 148, 109 and 67 points at $\gamma = 4$). The two channels move the aggregate the same way: at $\gamma = 4$ the subsidizing region shrinks from 29% of the admissible set at $N = 2$ to 13% at $N = 10$, and the taxing corner where welfare rises nonetheless recedes faster still, from 192 of 1,776 taxing points (10.8%) to 16 of 677 (2.4%); at $\gamma = 1/2$ that corner falls from 28 to 10% of the taxing points, and for $\gamma \geq 8$ it is all but empty at every N (`ext/extN_welf2.py`). The admissible counts at $N = 2$ coincide with those of $\mathcal{A}$ at every γ , which is the check that the transposition is exact.

The share of the admissible set on which the merger raises total welfare falls both as the market fragments and as product-market competition intensifies, with the admissible count in parentheses:

γ	$N = 2$	$N = 3$	$N = 5$	$N = 10$
1/2	56% (1 380)	48% (1 010)	39% (658)	30% (353)
1	52% (1 853)	44% (1 380)	36% (925)	27% (513)
2	46% (2 252)	37% (1 703)	30% (1 162)	22% (658)
4	37% (2 514)	28% (1 923)	21% (1 336)	15% (776)
8	24% (2 654)	17% (2 051)	12% (1 435)	8% (837)
16	15% (2 737)	9% (2 135)	5% (1 495)	2% (876)

The $N = 2$ column is the model of the paper. Every share in the table is a grid finding (`rep6_manufacturers.py`); the demand system, the equilibrium, the factors A_N and B_N , the generalization of Proposition 3 and the monotonicity of R_a^N in N are identities.

B.9 An Integrated Rival

This section supports Section 5.6 of the paper. Scripts: `rep7_twostores.py`.

SETUP. M_2 owns its operating system and its store. The developer writes q applications at cost $q^2/2$ and publishes on every store where $u_S n_B^i \geq a_i$, so its net profit is $q \sum_i (u_S n_B^i - a_i) - q^2/2$ and $q_S = u_S(n_B^1 + n_B^2) - a_1 - a_2$. The two fees therefore enter the developer's problem only through their sum, and the buyer-side demand system is the one of (1) at $a = a_1 + a_2$: the two stores are Cournot complements. Before the merger the platform licenses to M_1 at $\beta = 1$, $a_1 = 0$ and earns nothing, while M_2 chooses (p_2, a_2) ; after it, two symmetric integrated firms choose (p_k, a_k) . Removing M_2 's store from the problem returns the a^{**} of (A.9) exactly, which is the control on the setup (`ext/ext2_setup.py`).

THE REDUCTION TO A TOTAL FEE PRESUMES THAT THE DEVELOPER PUBLISHES ON BOTH STORES, $u_S n_B^k \geq a_k$ FOR EACH; ON THE REFERENCE GRID THIS HOLDS AT EVERY RETAINED POINT, BEFORE THE MERGER (STORE 2 ALONE CHARGES, SMALLEST SLACK 0.005 TO 0.012 ACCORDING TO γ) AND AFTER IT (BOTH STORES CHARGE, SMALLEST SLACK 0.018; `EXT/EXT2_FULL.PY`). THE FRONTIER IS INVARIANT. The bracket $\mathcal{K} = u_S(3\sigma^3 + 21\sigma^2 + 48\sigma + 32) - 2(\sigma + 2)(3\sigma + 4)u_B$ of (A.9) signs M_2 's pre-merger fee and each firm's post-merger fee as well as the integrated fee of the paper, so all three change sign on $u_B/u_S = R_a(\sigma)$ and the taxing frontier of Figure 1 is unchanged.

THE CHANGE IN THE TOTAL FEE. The difference between the post-merger and the pre-merger total fee factors, over positive denominators, as $(v + r)(\sigma + 4)\mathcal{K}L$ with

$$L = 9\sigma^3 + 60\sigma^2 + 112\sigma + 64 - 2u_B u_S (15\sigma^2 + 44\sigma + 32) - 8\sigma u_S^2(\sigma + 1).$$

$L > 0$ on the admissible set as a polynomial identity. In (σ, x) -coordinates with $x = u_S^2$ and $u_B u_S = (\gamma - \sigma)/(2(1 + \gamma))$, L is affine and decreasing in x and vanishes at

$$x_L = \frac{(3\sigma + 4)(3\gamma\sigma + 8\gamma + 8\sigma + 16)}{8\sigma(\gamma + 1)}.$$

The two signs that place x_L strictly to the right of the admissible interval, namely that the relevant denominator is negative at x_L and decreasing there, hold as polynomial identities under the substitution $\gamma = \sigma + t$, $t > 0$: every coefficient of the expanded numerator has a single sign, as does the denominator. This is the method of `verifI.py` (`ext/ext2_Lproof3.py`). Hence $L > 0$ throughout, and the sign of the change in the total fee is the sign of $\mathcal{K}$.

THE ADMISSIBLE SET HERE. Admissibility in this section is interiority in both configurations, positive device demands and a positive application stock at the benchmark and after the merger, and not the concavity screen of $\mathcal{A}$; the second-order conditions of the two fee problems, M_2 's before the merger and each store's own-fee problem after it, hold at every point retained at every γ (`ext/ext2_full.py`). The two sets coincide for $\gamma \leq 2$ and differ by 5, 7 and 7 points at $\gamma = 4, 8$ and 16 , every one of them subsidizing: hence 733 subsidizing points here where Section 5.5 counts 738 on $\mathcal{A}$. The five dropped at $\gamma = 4$ are $(u_B, u_S) = (1.32, 0.30), (1.36, 0.28), (1.38, 0.28), (1.40, 0.26)$ and $(1.40, 0.28)$, each failing only the post-merger conditions.

DOUBLE MARGINALIZATION. The counterfactual holds prices at their equilibrium behavior while the two firms choose (a_1, a_2) jointly. In the taxing region the uncoordinated total fee exceeds the coordinated one at 99.2, 98.6, 98.2, 99.0, 99.7 and 99.7% of the admissible taxing points, with median ratios 1.43, 1.46, 1.47, 1.48, 1.46 and 1.44, at $\gamma = 1/2, 1, 2, 4, 8$ and 16 (`ext/ext2_dm.py`). The ratio is hump-shaped in γ and peaks at the baseline.

SURPLUSES. The shares below are computed on the reference grid at $v = 2, r = 1/2$ (`ext/ext2_full.py`). Welfare rises at every subsidizing point at every γ (733 of 733 at $\gamma = 4$) and at 15, 14, 10, 4, 0.4 and 0.1% of the taxing ones at $\gamma = 1/2$ to 16 , which is the shape of Proposition 5. The rival's and the developer's shares lie within five grid points of each other at every γ , both being pushed the same way by the total fee.

On the same grid, with the screen above:

γ	admissible	$\Delta W > 0$	$\Delta \pi_2 > 0$	$\Delta V_B > 0$	$\Delta V_S > 0$
1/2	1 380	48%	39%	76%	39%
1	1 853	45%	37%	70%	38%
2	2 252	40%	35%	62%	35%
4	2 509	32%	30%	50%	30%
8	2 647	23%	23%	33%	23%
16	2 730	15%	15%	18%	15%

The shares converge on the paper's own as γ grows (23 against 24% at $\gamma = 8, 15$ against 15% at $\gamma = 16$): duplication matters where the two-sided logic matters, which is where product-market competition is soft. The invariance of the frontier, the factorization of the change in the total fee and the positivity of L are identities; every share here, the alignment of the rival with the developer and the double-marginalization factor are grid findings (`rep7_twostores.py`).

B.10 Welfare Objects, Micro-foundation, and the General Model

THE REPRESENTATIVE BUYER AND THE WELFARE OBJECTS. The buyer surplus used throughout the paper derives from the quadratic representative-consumer utility (Shubik–Levitan, with additively appended network benefits)

$$U_B(q_1, q_2, n_S^1, n_S^2) = q_0 + \sum_k (v q_k + u_B n_S^k q_k) - \frac{1}{4(1 + \gamma)} \left(2 \sum_k q_k^2 + \gamma \left(\sum_k q_k \right)^2 \right). \quad (\text{B.41})$$

The first-order conditions $\partial U_B / \partial q_k = p_k$ invert exactly into the demand system of Section 2 of the paper: $q_k = v - p_k - \frac{\gamma}{2}(p_k - p_\ell) + u_B n_S^k + \frac{\gamma}{2} u_B (n_S^k - n_S^\ell)$. Buyer surplus is the indirect

utility $V_B = U_B(D_1, D_2, q_S, q_S) - p_1 D_1 - p_2 D_2$ (application availability is symmetric across devices in every configuration studied, by costless porting), the developer's surplus is $V_S = q_S^2/2$ (the rent of the quadratic-cost developer under linear participation), and $W = V_B + V_S + \sum_k \pi_k + \Pi^{plat}$. These are the objects evaluated in Appendix A.1 of the paper and in the simulations.

THE SINGLE-HOMING MICRO-FOUNDATION OF SECTION 2. The same demand system arises from a population of single-homing buyers, none of whom buys two devices. Let each manufacturer k face a *captive* segment of mass m_c : consumers with reservation values ω uniform on $[0, \bar{V}]$ who consider only device k , buy iff $\omega + u_B n_S^k - p_k \geq 0$, and hence demand $m_c - \frac{m_c}{\bar{V}}(p_k - u_B n_S^k)$ in the interior. Let both manufacturers additionally compete on a fully covered Hotelling segment of mass m_h : consumers uniform on $[0, 1]$ with transportation cost t and a common gross value Ω high enough for full coverage, so that k 's Hotelling demand is $m_h [\frac{1}{2} + \frac{1}{2t}(p_\ell - p_k + u_B(n_S^k - n_S^\ell))]$. Summing the two segments and normalizing $m_c/\bar{V} = 1$:

$$Q_B^k = \underbrace{\left(m_c + \frac{m_h}{2}\right)}_{=v} - p_k - \underbrace{\frac{m_h}{2t}}_{=\gamma/2} (p_k - p_\ell) + u_B n_S^k + \frac{m_h}{2t} u_B (n_S^k - n_S^\ell),$$

which is the paper's demand system exactly, with $\gamma = m_h/t$ and $v = m_c + m_h/2$. Participation is elastic on the captive margin (this is what makes total demand price-sensitive, unlike a pure covered Hotelling model) and rivalry operates on the contested margin.

Nothing forces the captive support to start at zero. With ω uniform on $[\underline{V}, \bar{V}]$ the same matching holds, with $m_c = \bar{V} - \underline{V}$ and $v = \bar{V} + m_h/2$, and $\underline{V}$ is free: it fixes the captive mass and nothing else. The aggregation is literal at a given price vector when both segments are interior, which requires $\underline{V} \leq p_k - u_B n_S^k \leq \bar{V}$ for each k and a shared Hotelling segment, $|p_1 - p_2| \leq t$. The lower inequality costs nothing, since $\underline{V}$ can be taken as low as one likes. The other two bind m_h in opposite directions, $\gamma |p_1 - p_2| \leq m_h \leq 2(v - \max_k(p_k - u_B n_S^k))$, so a calibration exists if and only if $\gamma |p_1 - p_2| \leq 2(v - \max_k(p_k - u_B n_S^k))$. On the reference grid this holds at every admissible point, at the integrated optimum and at the separation benchmark alike, at every γ of the reference calibration; the smallest slack is 0.012 on $\mathcal{A}$, at $\gamma = 4$, and 1.55 on $\mathcal{A}^*$. Comparing the two outcomes requires one calibration serving both, a common m_h in the intersection of the two intervals: it exists at every point of $\mathcal{A}^*$ (smallest slack 0.73), and on $\mathcal{A}$ at all but 16, 23 and 4 points at $\gamma = 2, 4$ and 8, extreme points where the integrated optimum's price gap calls for a Hotelling segment larger than the separation outcome admits (`rep31_microfoundation.py`). This is a grid finding and not a proof. The analysis nonetheless takes (1) as a primitive rather than as an aggregation maintained pointwise, as is standard for linear demand systems.

The two constructions agree on welfare. In the single-homing population, aggregate buyer surplus is $V_B^{sh} = \sum_k \frac{1}{2}(\bar{V} - p_k + u_B n_S^k)^2 + (\text{Hotelling terms})$, the captive density $m_c/(\bar{V} - \underline{V})$ being one under the matching; by quasi-linearity, its gradient satisfies Roy's identity $\partial V_B^{sh}/\partial p_k = -Q_B^k$, and by the envelope theorem at the indifferent Hotelling consumer, $\partial V_B^{sh}/\partial n_S^k = u_B Q_B^k$. The indirect utility of the representative consumer (B.41) has exactly the same gradient in (p_1, p_2, n_S^1, n_S^2) . Two functions with identical gradients on a connected domain differ by a constant: all welfare *changes* computed in the paper are therefore identical under the two interpretations, which is what Section 2 asserts.

THE GENERAL MODEL: EXISTENCE, UNIQUENESS, MONOTONICITY. Appendix A.4 of the paper states (G1)–(G4) for general demand systems. The supporting lemmas are as

follows. In the last stage, given (p_1, p_2, a) , participation solves

$$n_B^1 = Q_B^1(p_1, p_2, n_S), \quad n_B^2 = Q_B^2(p_2, p_1, n_S), \quad n_S = Q_S(u_S(n_B^1 + n_B^2) - a). \quad (\text{B.42})$$

Substituting the third equation into the first two, total buyer participation $D_B = n_B^1 + n_B^2$ is a fixed point of the participation map

$$\mathcal{T}(x) = Q_B^1(p_1, p_2, Q_S(u_S x - a)) + Q_B^2(p_2, p_1, Q_S(u_S x - a)).$$

Assumption B.1. *In the relevant range, writing $\partial Q_S/\partial n_B$ for the derivative of the composite seller response $D_B \mapsto Q_S(u_S D_B - a)$, that is $u_S Q'_S(z)$ at $z = u_S D_B - a$: (a) network effects are positive, $\partial Q_S/\partial n_B \geq 0$ and $\partial Q_B^k/\partial n_S \geq 0$; (b) the feedback is contracting, $\frac{\partial Q_S}{\partial n_B} \left(\frac{\partial Q_B^1}{\partial n_S} + \frac{\partial Q_B^2}{\partial n_S} \right) < 1$; (c) some $\bar{X} > 0$ satisfies $\mathcal{T}(\bar{X}) \leq \bar{X}$ (bounded demands suffice); (d) $\mathcal{T}(0) \geq 0$ (demands are non-negative at zero participation).*

Lemma B.2. *Under Assumption B.1, system (B.42) has a unique solution with $D_B \in [0, \bar{X}]$, defining demands $D_k(p_k, p_\ell, a)$ and $D_S(p_1, p_2, a)$; and $D_B > 0$ whenever $\mathcal{T}(0) > 0$. Part (c) of the assumption is what delivers existence: without a self-map of a complete domain the pointwise feedback bound gives uniqueness conditional on existence only, and D_k and D_S need not be defined. Total buyer participation being positive does not by itself make the solution interior: $D_B < \bar{X}$, the positivity of each device demand and $D_S > 0$ are separate requirements, which the linear model of the paper verifies directly (Online Appendix B.1) and which we impose here where they are used.*

Proof. By (a), $\mathcal{T}'(x) = \frac{\partial Q_S}{\partial n_B} \left(\frac{\partial Q_B^1}{\partial n_S} + \frac{\partial Q_B^2}{\partial n_S} \right) \geq 0$, and by (b), $\mathcal{T}' < 1$: hence $0 \leq \mathcal{T}' < 1$. Write $g(x) = \mathcal{T}(x) - x$. With $\mathcal{T}(0) \geq 0$ one has $g(0) \geq 0$, and (c) gives $g(\bar{X}) \leq 0$, so g vanishes somewhere on $[0, \bar{X}]$ by continuity; and $g' = \mathcal{T}' - 1 < 0$ pointwise makes g strictly decreasing, so it vanishes once. This argument needs only a pointwise bound on $\mathcal{T}'$, where a Banach contraction would need a uniform one; (a) is part of the assumption because (b) alone bounds $\mathcal{T}'$ from above only. Then $D_S = Q_S(u_S D_B - a)$ and $D_k = Q_B^k(p_k, p_\ell, D_S)$ are uniquely defined. $\square$

Lemma B.3. *Under Assumption B.1 with (a) holding strictly, and the demand assumptions of Section 2: $\partial D_k/\partial p_k < 0$, $\partial D_S/\partial p_k < 0$, $|\partial D_k/\partial p_k| > |\partial D_k/\partial p_\ell|$ at symmetric prices, $\partial D_k/\partial a < 0$ and $\partial D_S/\partial a < 0$. At the boundary cases permitted by the weak form of (a), where the composite seller response or the supply response to the net return vanishes, the same signs hold weakly.*

Proof. Differentiating the fixed-point equation for D_B in p_1 : $\frac{\partial D_B}{\partial p_1} \left[1 - \frac{\partial Q_S}{\partial n_B} \left(\frac{\partial Q_B^1}{\partial n_S} + \frac{\partial Q_B^2}{\partial n_S} \right) \right] = \frac{\partial Q_B^1}{\partial p_1} + \frac{\partial Q_B^2}{\partial p_1}$. The bracket is positive by Assumption B.1; the right-hand side is negative because the direct price effect dominates: $\partial D_B/\partial p_1 < 0$, and likewise in p_2 . Since $\partial Q_S/\partial n_B > 0$, D_S is decreasing in both prices. Then $\partial D_1/\partial p_1 = \partial Q_B^1/\partial p_1 + (\partial D_B/\partial p_1)(\partial Q_B^1/\partial n_S)(\partial Q_S/\partial n_B) < 0$. For the reduced-form dominance of the direct price effect at symmetric prices, split it into two inequalities. In the *difference*, the feedback cancels by symmetry (a symmetric price pair moves D_B equally through either price, so $\partial D_B/\partial p_1 = \partial D_B/\partial p_2$), leaving $\partial D_1/\partial p_1 - \partial D_1/\partial p_2 = \partial Q_B^1/\partial p_1 - \partial Q_B^1/\partial p_2 < 0$ by primitive dominance. In the *sum*, the response to a uniform price increase decomposes as $\partial D_1/\partial p_1 + \partial D_1/\partial p_2 = (\partial Q_B^1/\partial p_1 + \partial Q_B^1/\partial p_2) + (\partial Q_B^1/\partial n_S)(\partial Q_S/\partial n_B)(\partial D_B/\partial p_1 + \partial D_B/\partial p_2) < 0$: the first term is negative by primitive dominance, the second is a product of nonnegative factors and the negative uniform-price response of D_B established

above. The two inequalities together give $|\partial D_1/\partial p_1| > |\partial D_1/\partial p_2|$ whatever the sign of the reduced-form cross effect. (In the linear model the sum is the primitive sum times $1/(1 - 2u_B u_S)$, while the difference is the primitive difference exactly: the feedback terms do *not* cancel inside each absolute value separately, only in the difference.) For the fee, differentiating the fixed-point equation for D_S : $\partial D_S/\partial a = \frac{\partial Q_S/\partial a}{1 - \frac{\partial Q_S}{\partial n_B}(\frac{\partial Q_B^1}{\partial n_S} + \frac{\partial Q_B^2}{\partial n_S})} < 0$, and

$$\partial D_k/\partial a = (\partial Q_B^k/\partial n_S)(\partial D_S/\partial a) < 0. \quad \square$$

In the running example, Assumption B.1 is $2u_B u_S < 1$, and all the properties of Lemma B.3 hold with the explicit expressions of Section 2; the pricing subgame is well-behaved in every configuration used in the paper (profits strictly concave in own price, best responses with slopes in $(0, 1)$ in the substitutes range and in $(-\frac{1}{2}, 0)$ in the complements range, prices decreasing in the fee at configuration-specific rates: at $2u_B/(4 + \sigma)$ in the non-integrated configuration, and at $(2u_B + u_S)/(4 + \sigma) \pm u_S/(4 + 3\sigma)$ for the integrated firm's and the rival's device respectively in the integrated one), which is the content of the verification blocks in the replication scripts.

B.11 Numerical Simulations

Unless otherwise stated: $\gamma = 4$, $v = 2$, $r = 1/2$; maps are drawn on the admissible set of Section B.1, blue marking a rise and red a fall of the object relative to the separation benchmark. The figures are drawn by `rep1_figures.py` (baseline and γ -map), `rep4_figures.py` (efficiency gains and coordination motives), from the closed-form solutions of each regime; the dominant-platform map is Figure OA-3 of Section B.5.7, drawn by `rep5_figure.py`; no simulation of the game itself is involved except where noted.

BASLINE (SECTION 3 OF THE PAPER). Figure OA-6 maps the impact of integration at the optimal fee a^{**} on the rival's profit, buyer surplus and developer surplus, and Figure OA-7 the change in total welfare. The rival's map is exactly the region split by $\underline{h}$ (Proposition 3), and the buyer and developer maps display the frontier geometry established in Section B.2, including the intermediate-region dividing curve for buyers.

THE ROLE OF PRODUCT-MARKET COMPETITION. Figure OA-8 repeats the buyer-surplus map for $\gamma \in \{2, 6, 10\}$. As γ increases, prices become more rigid and cost-based, the asymmetric price structure that harnesses network effects becomes less available, and the region where buyers gain from integration shrinks: both the strength *and* the skewness of network effects, and the intensity of competition, determine the merger's sign.

EFFICIENCY GAINS (SECTION 5.1). Figure OA-9 maps the impact of integration on the rival and on buyers at the constrained optimum of Section B.4 (regimes $\beta^{**} \in \{1, \beta^*, \underline{\beta}\}$ as applicable), for $r_0 - r \in \{0.3, 0.6\}$. Three findings, announced in the paper: the qualitative map is robust; rival harm is undone *not* in the developer-skewed range (there β^{**} is at or near one, yet the still-higher fee leaves $\Delta\pi_2 < 0$ throughout, as the left panels show at high u_S) but in the buyer-skewed, subsidizing range, where the rival's profit, pinned at its self-supply value, rises with the subsidized commons (25% of the admissible grid at $\Delta = 0.3$, the largest gains at strongly buyer-skewed points such as $(1.10, 0.36)$); and the unambiguously harmful region concentrates on weak, balanced network effects, where the merger's only bite is extraction.

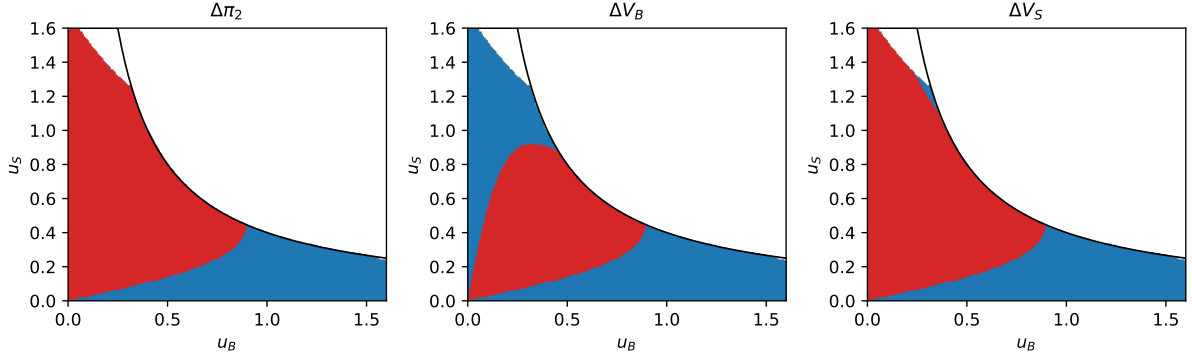


Figure OA-6: Impact of vertical integration at a^{**} ($\beta = 1$, $\gamma = 4$), on the baseline set $\mathcal{A}$: change relative to the separation benchmark in the rival's profit, buyer surplus and developer surplus (blue: rise; red: fall). White inside the $\sigma = 0$ curve, in the strongly developer-skewed corner ($u_S \gtrsim 1.27$), is where the rival's demand would not survive the integrated fee and $\mathcal{A}$ therefore stops. The thin blue band at the top of the ΔV_S panel is the region $u_S \geq \bar{h}_S(u_B)$ in which the developer gains despite a rising fee.

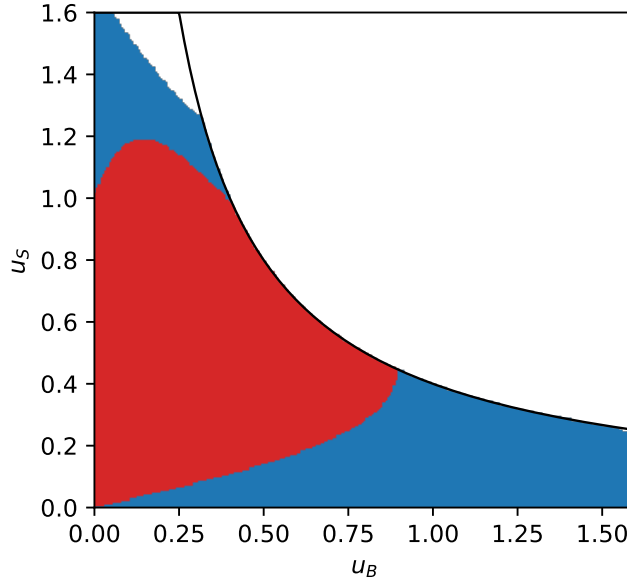


Figure OA-7: The change in total welfare induced by the merger, computed on the admissible set at $\gamma = 4$, $v = 2$, $r = 1/2$; blue is a gain, and the white area is inadmissible (`rep1_figures.py`). Welfare rises throughout the subsidizing region, as Proposition 5(i) states, and falls just inside the taxing frontier.

COORDINATION MOTIVES (SECTION 5.3). Figure OA-10 repeats the exercise for coordination motives $\alpha_B \in \{0.2, 0.4\}$, at the constrained optimum of Section B.6, against the corrected benchmark of that section, in which the independent platform already prices part of the premium, as the dominance block above does (`rep34_coordination.py`, block F for the figure and block C for the shares). Two comparisons organize the reading. Against the undistorted benchmark, the one Figure OA-9 uses for efficiency gains, harm to the rival is slightly *less* common than with efficiency gains at the calibrations shown

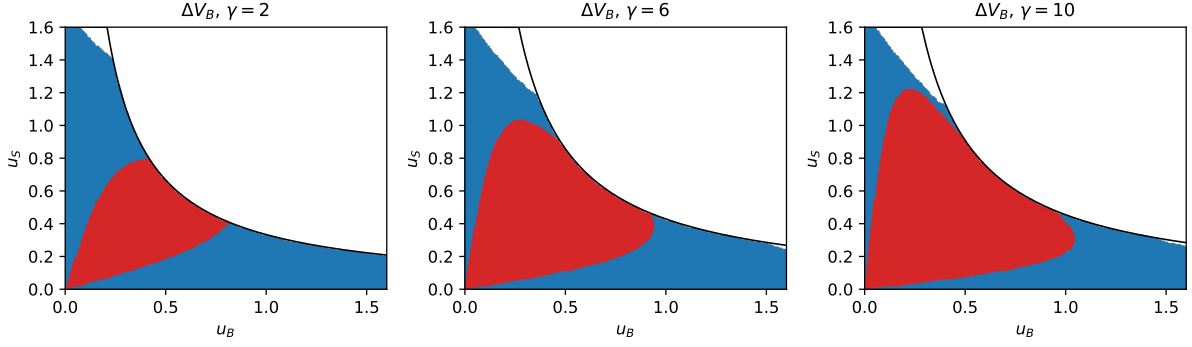


Figure OA-8: Change in buyer surplus at a^{**} for $\gamma \in \{2, 6, 10\}$ (blue: rise; red: fall).

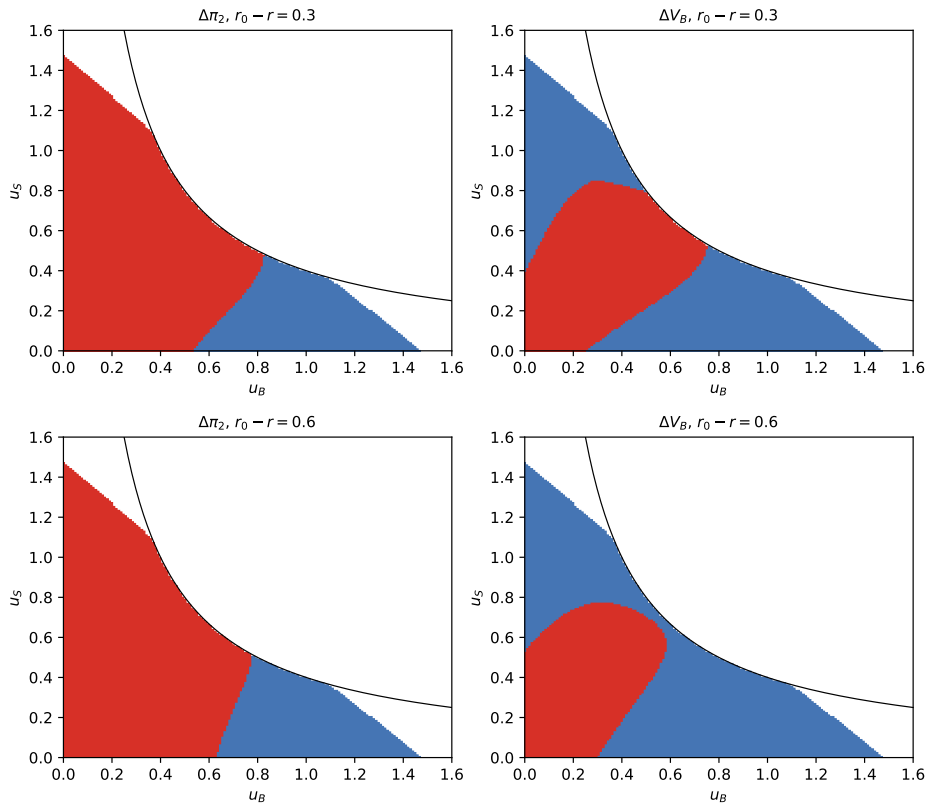


Figure OA-9: Efficiency gains: change in the rival's profit (left) and buyer surplus (right) at the constrained optimum, for $r_0 - r = 0.3$ (top) and 0.6 (bottom).

($\Delta\pi_2 < 0$ at 69% and 66% of the admissible grid at $\gamma = 4$ for $\alpha_B \in \{0.2, 0.4\}$, against 75% at both $\Delta \in \{0.3, 0.6\}$; for buyers, $\Delta V_B < 0$ at 28% and 13% against 44% and 35%; `rep4_figures.py`): the coordination benefit lifts demand on both devices once both are hosted, and the blue wedge at low u_S , absent from Figure OA-9, is its direct imprint. Against the corrected benchmark the premium is on both sides of the comparison and the maps change most for the developer: at $\gamma = 4$, $\Delta V_S < 0$ at 46% and 27% of the admissible grid instead of 68% and 64%, since the independent platform taxes the developer at most points already; the rival loses at 71% at both premia, buyers at 31% and

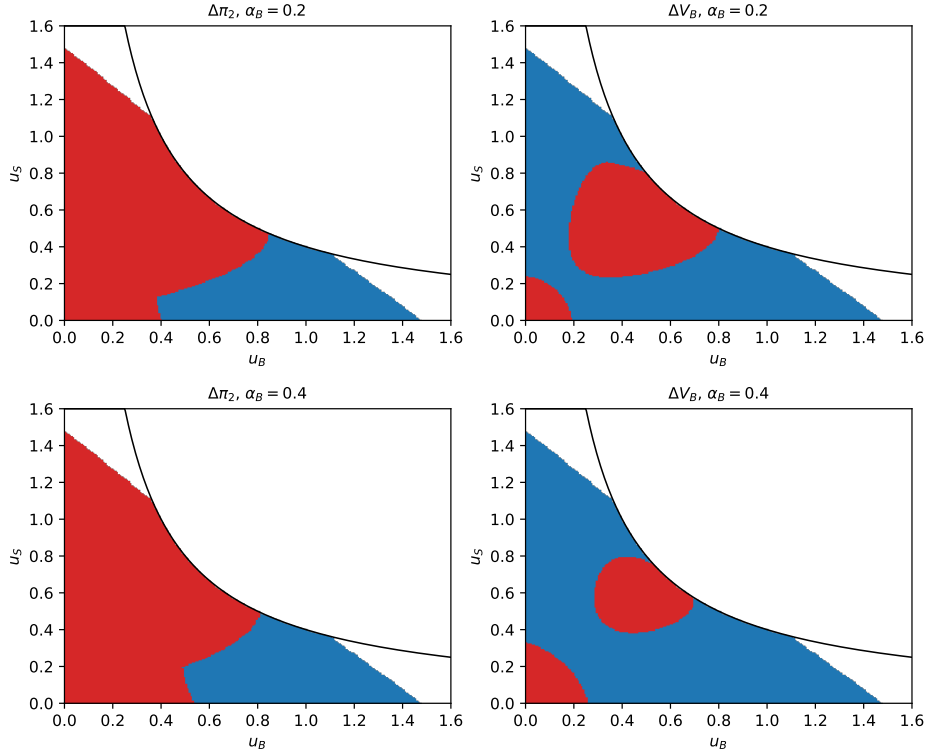


Figure OA-10: Coordination motives: change in the rival's profit (left) and buyer surplus (right) at the constrained optimum, for $\alpha_B = 0.2$ (top) and 0.4 (bottom), against the corrected benchmark of Online Appendix B.6, at $\gamma = 4$.

18%, and welfare falls at 38% and 23%. A small region of buyer losses appears near the origin, the imprint of the lower floor of the rival. Across the six values of γ the rival loses at 61 to 86% of the admissible set at $\alpha_B = 0.2$ and 58 to 85% at 0.4 , buyers at 0 to 74% and 0 to 68%, the developer at 24 to 61% and 14 to 49%, and welfare falls at 7 to 69% and 0 to 59%, every share rising with γ . In mechanism the comparison with efficiency gains runs the other way: the premium priced by $\underline{\beta} < 1$ is not a merger-created gain, so nothing is passed through, and in the developer-skewed range, where the independent platform had left the whole share and taxed, the rival suffers the fee increase, and at many of those points a lower share as well, without compensation. ('Comparable size' has no canonical definition, since α_B and Δ enter the premium through different formulas, so the percentages are read at the calibrations shown.)